

Numerical Analysis of Partial Differential Equations

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Introduction

These notes are based on the lectures delivered by Professor Paola Antonietti during the Numerical Analysis of Partial Differential Equations course at Politecnico di Milano during Semester 2025-1. Official course information can be found on the official website (*Accessed: 2024-02-15, 2024*). Some sections, particularly the practical components, were taught by Teaching Assistant Francesco Regazzoni. Further details on the practical part can be found in the provided resources. This material was prepared independently by the author(s) and has not been reviewed or approved by the professors named herein. This material may be outdated, the latest version is available in the repository. To report an error, please contact mjeon@uc.cl.

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1 The Problem

1.1 Variational Formulation

Consider the problem where u satisfies:

$$\begin{cases} \mathcal{L}u = f & \text{in } \Omega, & \text{i.e. the function } u \text{ receives values in } \Omega \\ BC & \text{in } \partial\Omega, & \text{i.e. the function } u \text{ receives values in } \partial\Omega \end{cases}$$

where:

- Ω : Open bounded domain in $\mathbb{R}^d$, $d=2,3$
- $\partial\Omega$: Boundary of Ω
- f : Forcing Term. It is given
- Boundary conditions to be prescribed according to $\mathcal{L}$
- $\mathcal{L}$: 2nd order differential equation

This is called the **variational formulation** of the problem, we will see others later.

1.2 The boundaries

We can split the boundary conditions into two parts. We notate:

$$\Gamma_D \quad \text{and} \quad \Gamma_N$$

where:

- $\Gamma_D^\circ \cap \Gamma_N^\circ = \emptyset$
- $\Gamma_D \cup \Gamma_N = \partial\Omega$

Define:

$$BC = \begin{cases} u = 0 & \text{in } \Gamma_D \\ \mu \nabla u \cdot \underline{n} = g & \text{in } \Gamma_N \end{cases}$$

There is a third type of boundary, the Robin type boundary, but we will not address it.

1.3 Weak Formulation

PDEs are a very difficult problem, so the formulation is often relaxed to find a solution. The weak formulation consists of finding a solution that is not satisfied pointwise, but on average. It does this by checking for equality under the integral, but satisfying it for every function in a defined function set. We usually choose Sobolev Spaces as a function set, which allow for functions that are not classically differentiable.

The usual steps for building a weak formulation are:

1. Take v (regular enough) and multiply the first equation by v

2. Integrate over Ω
3. Integrate it by parts (vector calculus):

$$\int_{\Omega} u \nabla \cdot V \, d\Omega = \int_{\partial\Omega} u V \cdot \hat{n} \, d(\partial\Omega) - \int_{\Omega} \nabla u \cdot V \, d\Omega$$

2 The Poisson Problem

2.1 The problem

The Poisson Problem consists of using the diffusion operator:

1. **Non-conservative form:**

$$\mathcal{L}u = -\text{div}(\mu \nabla u) + \mathbf{b} \cdot \nabla u + \sigma u$$

2. **Conservative form:**

$$\mathcal{L}u = -\text{div}(\mu \nabla u) + \text{div}(\mathbf{b} \cdot u) + \sigma u$$

Where:

- Diffusion Coefficient:

$$\mu \in L^{\infty}(\Omega), \mu(x) \geq \mu_0 > 0 \quad (\text{It makes the sup essential bounded})$$

- Advection Coefficient:

$$b \in (L^{\infty}(\Omega))^d \quad (\text{It makes the sup essential bounded})$$

- $\sigma \in L^2(\Omega)$ (It makes σ^2 integral bounded)

- $f \in L^2(\Omega)$ (It makes f^2 integral bounded)

(The first term is called the diffusion term and the second the transportation term.)

Note that if b is constant, by the chain rule, both forms are equal.

Remember that:

- Divergence:

$$\text{div } F = \nabla \cdot F = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

Where F_x is the x component of function F. F_y and F_z are defined analogously.

- Gradient:

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

2.2 Strong Formulation

$$\text{Find } u \text{ st } \begin{cases} -\nabla \cdot (\mu \nabla \underline{u}) + \nabla \cdot (\underline{b} \underline{u}) = f & \text{in } \Omega \\ \underline{u} = g_D & \text{in } \Gamma_D \\ \mu \nabla \underline{u} \cdot \underline{n} - \underline{b} \cdot \underline{n} = g_N & \text{in } \Gamma_N \end{cases}$$

Where:

- $u : \Omega \rightarrow \mathbb{R}$
- $f \in L^2(\Omega)$
- $0 < \mu_0 \leq \mu(\underline{x}) \leq \mu_1 \quad \text{a.e. } \Omega$
- $|\underline{b}(\underline{x})| \leq b_1 \quad \text{a.e. } \Omega$
- $g_D : \Gamma_D \rightarrow \mathbb{R}$ regular enough
- $g_N : \Gamma_N \rightarrow \mathbb{R}$ regular enough

Where the conditions are required to use the theorem that we will use to state the existence of the solutions (see section 4.3)

2.3 Weak formulation

Now, to simplify the analysis, we will start from this specific case, the **Generalized Poisson Problem**:

$$(MP) \begin{cases} -\text{div}(\mu \nabla u) = f & \text{in } \Omega \\ u = 0 & \text{in } \partial\Omega \end{cases}$$

1,2:

$$\int_{\Omega} -\text{div}(\mu \nabla u) \cdot v = \int_{\Omega} f \cdot v$$

3:

$$\int_{\Omega} \mu \nabla u \cdot \nabla v - \int_{\partial\Omega} \mu \nabla u \cdot \underline{n} \cdot v = \int_{\Omega} f \cdot v$$

Now, we will restrict the space conveniently, so $v = 0$. This implies that $\mu \nabla u \cdot \underline{n} \cdot v = 0$ in the second integral.

Now, we reformulate the variational problem:

$$\begin{cases} \text{Find } u \in V \\ \int_{\Omega} \mu \nabla u \cdot \nabla v = \int_{\Omega} f v \quad \forall v \in V \end{cases}$$

We define the space V as:

$$V = H_0^1(\Omega)$$

Being:

$$L^2(\Omega) = \{v : \Omega \rightarrow \mathbb{R} \text{ st } \|v\|_{L^2(\Omega)} < +\infty\} \quad , \|v\| = \left(\int_{\Omega} v^2 \right)^{1/2}$$

$$H^1(\Omega) = \{v : \Omega \rightarrow \mathbb{R}, v \in L^2(\Omega) \text{ and } \nabla v \in [L^2(\Omega)]^d\}$$

$$H_0^1(\Omega) = H_D^1 = \{v \in H^1(\Omega) \text{ st } v|_{\Gamma_D} = 0\}$$

2.4 Non-homogeneous boundary conditions

An extension over the simplest case is having non-homogeneous boundary conditions, so it is not enough to set it to 0 on the boundary.

2.4.1 Dirichlet Boundary

$$(MP) \begin{cases} -\Delta u = f & \text{in } \Omega \\ u = g & \text{on } \partial\Omega = \Gamma_D \end{cases}$$

Note that $u \in \{u \in H^1(\Omega) \quad \wedge \quad u = g \text{ on } \Gamma_D\} \equiv V_g$.

The weak form:

$$\begin{aligned} \int_{\Omega} -\Delta u \cdot v &= \int_{\Omega} f \cdot v & \forall v \in H_{\Gamma_D}^1(\Omega) \\ \int_{\Omega} \nabla u \cdot \nabla v - \int_{\Gamma_D} \nabla u \cdot n \cdot v &= \int_{\Omega} f \cdot v & \forall v \in H_{\Gamma_D}^1(\Omega) \end{aligned}$$

And since the integral is 0 in the boundary, the problem we must solve is:

$$\text{Find } u \in V_g \quad \text{st} \quad \int_{\Omega} \nabla u \cdot \nabla v = \int_{\Omega} f \cdot v \quad \forall v \in H_{\Gamma_D}^1(\Omega)$$

Anyway, this is not a good formulation, since the sum of two elements in V_g is not in V_g (In the boundary, the result would be $2g$). Then, this is an affine space, and therefore not a Hilbert space (not allowing us to use LM, see theorem 3.6).

Now, take u with the lifting operator:

$$u = \overset{\circ}{u} + Rg, \quad \text{st} \quad \overset{\circ}{u} \in H_{\Gamma_D}^1(\Omega) \quad \wedge \quad Rg \in H^1(\Omega) \text{ st } Rg|_{\Gamma_D} = g$$

Therefore, a solvable formulation is:

$$\text{Find } \overset{\circ}{u} \in H_{\Gamma_D}^1 \quad \text{st} \quad \int_{\Omega} \nabla \overset{\circ}{u} \cdot \nabla v = \int_{\Omega} f \cdot v - \int_{\Omega} \nabla Rg \cdot \nabla v \quad \forall v \in H_{\Gamma_D}^1(\Omega)$$

Note that note, as u is in the same hilbert space as v , we can apply LM (see theorem 3.6).

2.4.2 Neumann Boundary

$$(MP) \begin{cases} -\Delta u = f & \text{in } \Omega \\ \frac{\partial u}{\partial n} = \Phi & \text{on } \partial\Omega = \Gamma_N \end{cases}$$

Following the same procedure as before, the problem we must solve is:

$$\text{Find } u \in ? \quad \text{st} \quad \int_{\Omega} \nabla u \cdot \nabla v = \int_{\Omega} f \cdot v + \int_{\Gamma_N} \Phi v \quad \forall v \in ?$$

The space V is just $H^1(\Omega)$, since we can deal with the extra term in the weak formulation.

Note that in this formulation, if u is a solution, $u + \text{constant}$ will also be a solution. Then, we don't have uniqueness in our solution. We must add an extra condition to ensure this. For example:

$$\int_{\Omega} u = 0$$

Plus, if we want the solution to exist, we need a compatibility condition on the data (replace v by 1 in the problem):

$$\int_{\Omega} f = - \int_{\Gamma_N} \Phi$$

Note that in this case, V_h will be chosen as before, but boundaries will add degrees of freedom.

2.4.3 Mixed Boundary

The third case is the most general one, where we have both kinds of boundaries

$$(MP) \begin{cases} -\Delta u = f & \text{in } \Omega \\ u = g & \text{on } \partial\Omega = \Gamma_D \\ \frac{\partial u}{\partial n} = \Phi & \text{on } \partial\Omega = \Gamma_N \end{cases}$$

Exercise

3 Lax-Milgram

Lax-Milgram is a theorem that is fundamental to prove existence and uniqueness in some PDE formulations.

3.1 Bounded Lemma

Lemma 3.1

Let be $w, v \in L^2(\Omega)$

$$\implies \left| \int_{\Omega} w \cdot v \right| < +\infty$$

Taking $V = H_0^1(\Omega)$

$$\int_{\Omega} f v$$

is bounded since $f, v \in L^2(\Omega)$

and

$$\int \mu \nabla u \cdot \nabla v$$

is bounded, since $\mu \in L^\infty, \nabla u \in L^2(\Omega), \nabla v \in L^2(\Omega)$

Definition 3.2: $a(u, v)$

Define $a(u, v) : H_{\Gamma_D}^1(\Omega) \times H_{\Gamma_D}^1(\Omega) \rightarrow \mathbb{R}$ in the Generalized Poisson Problem:

$$a(u, v) := \int_{\Omega} \mu \nabla u \nabla v$$

Definition 3.3: $F(v)$

Define $F(v) : H_{\Gamma_D}^1(\Omega) \rightarrow \mathbb{R}$ in the Generalized Poisson Problem:

$$F(v) := \int_{\Omega} f v$$

Then, (MP) is finding:

$$\mu \in H_{\Gamma_D}^1(\Omega) \quad \text{st} \quad a(u, v) = F(v) \quad \forall v \in H_{\Gamma_D}^1(\Omega)$$

This is called abstract model problem (AMP), since a and F can change. We want to find out if this problem is well posed.

3.2 Poincaré**Theorem 3.4: Poincaré Inequality**

Suppose $|\Gamma_D| > 0$, then

$$\exists C_p > 0 \quad \text{st} \quad \|v\|_{L^2(\Omega)} \leq C_p \|\nabla v\|_{L^2(\Omega)} \quad \forall v \in H_{\Gamma_D}^1(\Omega)$$

Corollary 3.5

Under the Poincaré assumptions:

$$\|v\|_V^2 = \|v\|_{L^2(\Omega)}^2 + \|\nabla v\|_{L^2(\Omega)}^2 \leq (1 + C_p^2) \|\nabla v\|_{L^2(\Omega)}^2$$

Therefore

$$\|\nabla v\|_{L^2(\Omega)}^2 \geq (1 + C_p^2)^{-1} \|v\|_V^2$$

3.3 Lax-Milgram

Theorem 3.6: Lax-Milgram

Assume:

1. V is an hilbert space, with norm $\|\cdot\|_V$

2. $F \in V'$:

$$|F(v)| \leq \|F\|_{V'} \|v\|_V \quad \forall v \in V$$

3. a is continuous:

$$\exists M > 0 \quad st \quad |a(u, v)| \leq M \|u\|_V \|v\|_V \quad \forall u, v \in V$$

4. a is coercive:

$$\exists \alpha > 0 \quad |a(v, v)| \geq \alpha \|v\|_V^2 \quad \forall v \in V$$

Then, the following problem:

$$(AMP) = \text{Find } u \in V \quad st \quad a(u, v) = F(v) \quad \forall v \in V$$

is well posed, i.e.:

$$\exists! u \in V \quad st \quad \|u\|_V \leq \frac{\|F\|_{V'}}{\alpha}$$

Remark.

This implies continuous dependence on data, i.e., stability

Our diffusion model problem satisfies LN hypotheses

- On $H^1(\Omega)$, define:

$$\|v\|_{H^1(\Omega)}^2 = \|v\|_{L^2(\Omega)}^2 + \|\nabla v\|_{L^2(\Omega)}^2$$

- On $L^2(\Omega)$:

$$\|v\|_{L^2(\Omega)}^2 = \int_{\Omega} v^2$$

Note that:

1. Our V is a Hilbert space

2.

$$|F(v)| = \left| \int_{\Omega} f \cdot v \right| \leq \int_{\Omega} |f \cdot v| \leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \leq \|f\|_{L^2(\Omega)} \|v\|_V$$

3.

$$\begin{aligned}
|a(u, v)| &= \left| \int \mu \nabla u \nabla v \right| \leq \|\mu\|_{L^\infty(\Omega)} \|\nabla u\|_{L^2(\Omega)} \|\nabla v\|_{L^2(\Omega)} \\
&\leq \|\mu\|_{L^\infty(\Omega)} \|u\|_V \|v\|_V \quad \text{By definition of norm in } H^1(\Omega) \\
&= M \|u\|_V \|v\|_V
\end{aligned}$$

4.

$$|a(u, u)| = \int_{\Omega} \mu \nabla u \nabla u$$

And assuming $\mu(x) \geq \mu_0 > 0$ Ω a.e.

$$\begin{aligned}
|a(u, u)| &= \left| \int_{\Omega} \mu \nabla u \nabla u \right| \geq \mu_0 \int_{\Omega} (\nabla u)^2 = \mu_0 \|\nabla u\|_{L^2(\Omega)}^2 \\
&\geq \frac{\mu_0}{1 + C_p^2} \|v\|_V^2 \quad (\text{Using Poincaré, see section 3.2}) \\
&= \alpha \|v\|_V^2
\end{aligned}$$

Theorem 3.7

Assuming regularity on the data:

$$\exists \mu_0 \text{ st } \mu(x) \geq \mu_0 > 0 \text{ a.e. in } \Omega$$

$\implies$ (MP) is well posed, and

$$\|u\|_V \leq \frac{M}{\alpha} \sup_{v \in V \setminus \{0\}} \frac{|F(v)|}{\|v\|_V}$$

Remark.

The previous supremum is the definition of the dual norm

4 Garlekin Paradigm

The Garlekin Paradigm is a way to reformulate the (AMP) (see section 3.1) in order to be able to find a solution.

4.1 Galerkin Method

Definition 4.1: Galerkin Method

Find

$$u_h \in V_h \quad st \quad a(u_h, v_h) = F(v_h) \quad \forall v_h \in V_h \quad (G)$$

where:

- $\{V_h, h > 0\}$ are the finite-dimensional subspaces of V
- $\dim V_h = N_h < \infty$ with $N_h > N_{h+1}$

This is called Galerkin Formulation.

Theorem 4.2

(G) is well posed

Proof. The proof can be done with two different approaches, seen in sections 4.2 and 4.3 $\square$

4.2 Equivalence to Linear System

We can show that (G) is a linear system of N_h equations and N_h unknowns.

Since V_h has finite dimensions, $\exists \varphi_i \in V_h \quad \wedge \quad \exists u_i \in \mathbb{R} \quad st$

$$V_h = span\{\varphi_1, \varphi_2, \dots, \varphi_{N_h}\} \quad \wedge \quad u = \sum_{i=1}^{N_h} u_i \varphi_i \quad \forall u$$

Using the expansion, (G) becomes:

$$\begin{aligned} \text{find } \{u_j\}_{j=1}^{N_h} \quad st \quad & a\left(\sum_{j=1}^{N_h} u_j \varphi_j, v_h\right) = F(v_h) \quad \forall v_h \in V_h \\ st \quad & \sum_{j=1}^{N_h} u_j \cdot a(\varphi_j, v_h) = F(v_h) \quad \forall v_h \in V_h \quad (\text{Using bilinearity of } a) \\ st \quad & \sum_{j=1}^{N_h} u_j \cdot a(\varphi_j, \varphi_i) = F(\varphi_i) \quad \forall i = 1, \dots, N_h \end{aligned}$$

The last condition is equivalent since φ_j generates V_h , so any $v_h \in V_h$ can be written as a linear combination of φ_j s, and each condition is needed separately since a is bilinear and F is linear (by definition of integral).

Finally, this expression is a linear system $A\underline{u} = \underline{F}$, where:

- $\underline{u} = [u_1, u_2, \dots, u_{N_h}]^T \in \mathbb{R}^{N_h}$
- $\underline{F} = [F(\varphi_1), F(\varphi_2), \dots, F(\varphi_{N_h})]^T \in \mathbb{R}^{N_h}$

- $A \in \mathbb{R}^{N_h \times N_h}$, $(A)_{ij} = a(\varphi_j, \varphi_i)$

This is called The Algebraic Formulation.

By the stability of the linear system theorem, we know that the solution of a linear system depends on semi-definite positive property on A .

In our diffusion problem, A :

$$(A)_{ij} = a(\varphi_j, \varphi_i) = \int_{\Omega} \mu \nabla \varphi_j \nabla \varphi_i$$

Prove that A is SDP (exercise)

Then A is invertible, and $A\underline{u} = \underline{F}$ admits a unique solution

4.3 Well-Posedness and Uniqueness

We came from

$$\text{Find } u \in V \quad \text{st} \quad a(u, v) = F(v) \quad \forall v \in V \quad (AMP)$$

to the problem (G) with the Galerkin method.

We can use that:

$$V_h \subset V$$

Therefore every property held on V is true in V_h , so LM hypotheses trivially hold.

(G) is trivially well-posed, since it has a unique solution.

Remark.

Note that an eventual solution in (C) has to, particularly, satisfy:

$$a(u, v_h) = F(v_h) \quad \forall v_h \in V_h \quad (*)$$

Using linearity on $(*)$ -(G):

$$\begin{aligned} a(u, v_h) - a(u_h, v_h) &= F(v_h) - F(v_h) & \forall v_h \in V_h \\ \implies \underbrace{a(u - u_h, v_h)}_{\text{ERROR}} &= 0 & \forall v_h \in V_h \end{aligned}$$

This last expression is called Galerkin Orthogonality

4.4 Céa Lemma

Lemma 4.3: Céa

Assume the LM assumptions are satisfied, then:

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V$$

Proof.

$$\begin{aligned} \alpha \|u - u_h\|_V^2 &\leq a(u - u_h, u - u_h) && \text{(By coercivity)} \\ &= a(u - u_h, u - v_h) + \underbrace{a(u - u_h, v_h - u_h)}_{=0 \text{ (Galerkin orthogonality)}} && \text{(Using linearity)} \\ &\leq M \|u - u_h\|_V \|u - v_h\|_V \quad \forall v_h \in V_h && \text{(Continuity)} \end{aligned}$$

Then,

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \|u - v_h\|_V \quad \forall v_h \in V_h$$

Finally,

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V$$

□

Basically, the Galerkin error is bounded by the distance of u to V_h .

For our Poisson Model problem, Cea Lemma trivially holds (exercise) with:

$$\begin{aligned} M &= \|\mu\|_{L^\infty(\Omega)} \\ \alpha &= \frac{\mu_0}{1 + C_p^2} \end{aligned}$$

Note that $\|\mu\|_{L^\infty(\Omega)}$ is the maximum and μ_0 is the minimum μ .

Plus, in common domains, the Poincaré constant is not important, so:

$$\frac{M}{\alpha} \approx \frac{\|\mu\|_{L^\infty(\Omega)}}{\mu_0}$$

If μ values are piecewise constant, this is easy to find.

Finally, if we assume space saturation:

$$\forall v \in V \quad \lim_{h \rightarrow 0} \inf_{v_h \in V_h} \|v - v_h\|_V = 0$$

Then:

$$\lim_{h \rightarrow 0} \|u - u_h\| = 0$$

5 The Finite Element Method (FEM)

The Finite Element Method (FEM) is a way to choose the sequence V_h such that:

$$\forall v \in V \quad \lim_{h \rightarrow 0} \inf_{v_h \in V_h} \|v - v_h\|_V = 0$$

So we can use Lemma Céa to state convergence:

$$\lim_{h \rightarrow 0} \|u - u_h\|_V = 0$$

5.1 Mesh

We want to define a mesh of Ω , so we have finite elements that divide it. For this, we use a polyhedrization of Ω :

$$\mathcal{T}_h = \{K_1, \dots, K_{N_h}\} \quad st \quad \Omega_h = \bigcup K \quad \wedge \quad K_i \cap K_j = \emptyset \quad \forall i \neq j$$

Ω_h is an approximation of Ω , that does not consider the smooth non-polyhedral parts of the boundary domain that are not considered in the $\{K\}$.

K 's are called tiles.

Over these polyhedrons, and a finite space with $r \geq 1$:

$$V_h = \{v_h \in C^0(\bar{\Omega}) \quad st \quad v_h|_K \in \mathbb{P}^r(K) \quad \forall K \in \mathcal{T}_h \quad st \quad v_h|_{\Gamma_D} = 0\}$$

where $\mathbb{P}^r(K)$ is the space of polynomials on total degree $r \geq 1$ on K .

To have desired properties, we must have a mesh that satisfies:

- Conforming Mesh:

$$\partial K_i \cap \partial K_j = \begin{cases} \emptyset \\ \text{Vertex} \\ \text{Full edge} \end{cases} \quad \forall K_i, K_j \in \mathcal{T}_h$$

- Shape Regularity: Also called Minimum Angle Condition. It assures that the polyhedrons' angles are not too small or too big but close to a regular polyhedron.

5.2 Polynomial Typical Spaces

- 1D ($\mathbb{P}^r$):

$$p(x) = \sum_{k=0}^r a_k x^k$$

- **2D ($\mathbb{P}^r$):**

$$p(x_1, x_2) = \sum_{\substack{k=0, \dots, r \\ m=0, \dots, r \\ k+m \leq r}} a_{km} x_1^k x_2^m$$

- **3D tetrahedra ($\mathbb{P}^r$):**

$$p(x_1, x_2, x_3) = \sum_{\substack{k=0, \dots, r \\ m=0, \dots, r \\ n=0, \dots, r \\ k+m \leq r}} a_{kmn} x_1^k x_2^m x_3^n$$

- **3D hexahedra($\mathbb{Q}^r$):**

$$p(x_1, x_2, x_3) = \sum_{\substack{k=0, \dots, r \\ m=0, \dots, r \\ n=0, \dots, r}} a_{kmn} x_1^k x_2^m x_3^n$$

5.3 Lagrangian Basis Functions

The most natural basis that comes up to build polynomials is monomials. Anyway, this is not the best option for the problem we want to solve. As we must state continuity between tiles, other shape functions φ_i allow us to build the continuity more easily.

These functions are called lagrangian basis functions, and like any basis satisfy:

$$V_h = \text{span}\{\varphi_1, \dots, \varphi_{N_h}\}$$

We do this by imposing values on vertices $\{x_i\}_i$, doing the polynomials continuous over all the edges. Specifically, we have to set exterior vertices according to the boundary conditions. In the simplest case, we only have to set to 0 the exterior vertices.

If we have more degrees of freedom than vertices for each tile, we can set new points on the side to completely define the polynomial in the tile.

The vertices set: $\underline{v} = (v_1, v_2, \dots, v_{N_h})$ are the values of the function in the respective points, i.e.: $(v_h(x_1), v_h(x_2), \dots, v_h(x_{N_h}))$

On the other side, $\varphi_j(\underline{x})$ is a function that satisfies:

$$\begin{cases} 1 & \text{if } \underline{x} \equiv x_j \\ 0 & \text{if } \underline{x} \neq x_i, \quad i \neq j \end{cases}, \quad \text{if } \underline{x} \text{ belongs to the vertices set.}$$

Note that there is no restriction on the values the function can take on other points, but it must be continuous so it belongs to the space.

Then, any point $v_h \in V_h$ can be written as:

$$v_h(\underline{x}) = \sum_{j=1}^{N_h} v_j \varphi_j(\underline{x})$$

If we use triangles (tetrahedra) as polynomials, we can build the $\mathbb{P}^r$ spaces. If we use

quadrilaterals (hexahedra) instead, we can build the $\mathbb{Q}^r$ space.

The reason is that using polyhedral with more vertices requires more degrees of freedom to fix the points in the function. In the case of $p=2$, possible bases are:

$$\begin{aligned}\mathbb{P}_1(x_1, x_2) &= \text{span}\{1, x_1, x_2\} \\ \mathbb{Q}_1(x_1, x_2) &= \text{span}\{1, x_1, x_2, x_1x_2\}\end{aligned}$$

5.4 Interpolation Estimates

Let be Π_h^r the finite element (lagrangian) interpolant operator. Given u (smooth enough), we construct $\Pi_h^r u$ and estimate:

$$\|u - \Pi_h^r\|_V$$

1D: The interpolant consists of evaluating the function on each point in the mesh, and taking the line between each point.

Note that in this case:

$$\Pi_h^r \in V_h \quad \text{st} \quad v(x_i) = \Pi_h^r(x_i) \quad \forall i$$

Theorem 5.1: Interpolation

If $v \in H^{r+1}(\Omega)$, then $\exists C > 0$ st:

$$\begin{aligned}\|v - \Pi_h^r v\|_{L^2(\Omega)} &\leq C \left(\sum_{k \in \mathcal{T}_n} h_k^{2(r+1)} |v|_{H^{r+1}(k)}^2 \right)^{1/2} \quad (* \approx h^{r+1} |v|_{H^{r+1}(\Omega)}) \\ \|v - \Pi_h^r v\|_{H^1(\Omega)} &\leq C \left(\sum_{k \in \mathcal{T}_n} h_k^{2r} |v|_{H^{r+1}(k)}^2 \right)^{1/2} \quad (* \approx h^r |v|_{H^{r+1}(\Omega)})\end{aligned}$$

(*) if $\mathcal{T}_h$ is quasi-uniform

If $h \rightarrow 0$, it converges to 0. If we take higher degree polynomials, this convergence is faster.

Theorem 5.2: General Interpolation

Let $v \in H^{r+1}(\Omega)$. Then $\exists C > 0$ st:

$$\|v - \Pi_h^r v\|_{H^m(\Omega)} \leq C \left(\sum_{k \in \mathcal{T}_n} h_k^{2(r+1-m)} |v|_{H^{r+1}(k)}^2 \right)^{1/2} \approx h^r |v|_{H^{r+1}(\Omega)}$$

Remark.

Anyway, this works for small r 's (1 or 2, maybe 3). On the contrary, we can suffer from numerical oscillations, since the interpolant won't be numerically stable. For higher orders, a different approach is needed.

5.5 Convergence

Theorem 5.3: Convergence

Let $u \in V$ be the solution of (MP). Assume that $u \in H^{r+1}(\Omega)$, $r \geq 1$. Then $u_h \in V_h$ is the finite element approximation of order r such that:

$$\|u - u_h\|_V \lesssim \left(\sum_{k \in \mathcal{T}_n} h_k^{2r} |v|_{H^{r+1}(k)}^2 \right)^{1/2}$$

Plus, if the model is quasi-uniform:

$$\|u - u_h\|_V \lesssim h^r |u|_{H^{r+1}(K)}$$

Remark.

$\lesssim$ is a $\leq$, but with a constant that is not shown.

Proof.

$$\begin{aligned} \|u - u_h\|_V &= \|u - u_h\|_{H^1(\Omega)} && \text{(Definition)} \\ &\leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_{H^1(\Omega)} && \text{(Cea lemma)} \\ &\leq \frac{M}{\alpha} \|u - \Pi_h^r u\|_{H^1(\Omega)} && \text{(Taking } v_h = \Pi_h^r u) \\ &\leq C \frac{M}{\alpha} \left(\sum_{k \in \mathcal{T}_n} h_k^{2r} |v|_{H^{r+1}(k)}^2 \right)^{1/2} && \text{(Interpolation estimates)} \\ &\lesssim h^r |u|_{H^{r+1}(\Omega)} && \text{(If } \mathcal{T}_h \text{ is quasi-uniform)} \end{aligned}$$

□

Theorem 5.4: Convergence in L_2 norm

Let $u \in V$ be the solution of (MP), and assume that $u \in H^{r+1}(\Omega)$ $r \geq 1$. Let $u_h \in V_h$ be its finite element approximation of order r on a quasi-uniform mesh.

$$\implies \|u - u_h\|_{L^2(\Omega)} \lesssim h^{r+1} \|u\|_{H^{r+1}(\Omega)}$$

Proof. Consider the following auxiliary problem:

$$\begin{cases} -\Delta \Phi = u - u_h & \text{in } \Omega \\ \Phi = 0 & \text{on } \partial\Omega \end{cases}$$

Then, by elliptic regularity, it can be proved that:

$$\exists! \Phi \in V \cap H^2(\Omega) \text{ and } \|\Phi\|_{H^2(\Omega)} \leq C_1 \|u - u_h\|_{L^2(\Omega)}$$

On the other hand, the weak form of the problem is:

$$\int_{\Omega} \nabla \cdot \nabla v = \int_{\Omega} (u - u_h) v \quad \forall v \in V$$

So:

$$\begin{aligned} \|u - u_h\|_{L^2(\Omega)}^2 &= \int_{\Omega} (u - u_h)^2 \\ &= \int_{\Omega} \nabla \Phi \cdot \nabla (u - u_h) && \text{(Taking in the weak form } v = u - u_h) \\ &= a(\Phi, u - u_h) = a(u - u_h, \Phi) && \text{(By symmetry)} \\ &= a(u - u_h, \Phi - \Phi_h) && \text{(By Galerkin orthogonality)} \\ &\leq M \|u - u_h\|_V \|\Phi - \Phi_h\|_V && \text{(Continuity)} \end{aligned}$$

So we have shown:

$$\|u - u_h\|_{L^2(\Omega)}^2 \lesssim \|u - u_h\|_V \|\Phi - \Phi_h\|_V$$

Now, focus on this term:

$$\begin{aligned} \|\Phi - \Phi_h\|_V &= \|\Phi - \Pi_h^1 \Phi\|_V \lesssim h \|\Phi\|_{H^2(\Omega)} \\ &\leq h \|\Phi\|_{H^2(\Omega)} \\ &\lesssim h \|u - u_h\|_{L^2} && \text{(By elliptic regularity)} \end{aligned}$$

So we have now:

$$\begin{aligned} \|u - u_h\|_{L^2(\Omega)}^2 &\lesssim h \|u - u_h\|_V \\ &\lesssim h h^r \|u\|_{H^{r+1}(\Omega)} && \text{(By convergence theorem (see 5.3))} \\ &= h^{r+1} \|u\|_{H^{r+1}(\Omega)} \end{aligned}$$

□

5.6 Remarks on the algebraic problem

Proposition 5.5

Let $A_{ij} = a(\Phi_j, \Phi_i)$. If $a(\cdot, \cdot)$ is symmetric and coercive, then A is spd

Proof. On slides

□

Fact 5.6

if A is spd, then: $K_2(A) = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)}$

From now on, we will talk about P1, as the space of continuous functions piecewise linear that satisfy the condition boundaries.

Let be:

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

From discretizing with P1 FEM: $A\underline{u} = \underline{F}$. where:

$$\underline{u} = \begin{bmatrix} u_1 & u_2 & \dots & u_{N_h} \end{bmatrix}^T$$

$$A \in \mathbb{R}^{N_h \times N_h}, \quad (A)_{ij} = a(\varphi_j, \varphi_i) = \int_{\Omega} \nabla \varphi_j \cdot \nabla \varphi_i \quad \forall i, j = 1, \dots, N_h$$

$$\underline{F} \in \mathbb{R}^{N_h}, \quad (\underline{F})_i = \int_{\Omega} f \varphi_i \quad \forall i = 1, \dots, N_h$$

Remark.

It can be shown that

$$K_2(A) \leq C(r)h^{-2}$$

Remark.

It can be shown that:

$$\left(\frac{K_2(A) - 1}{K_2(A) + 1} \right)^k < 10^{-1} \iff k > \frac{\log 10}{\log \left(\frac{K_2(A) - 1}{K_2(A) + 1} \right)}$$

6 FEM Implementation

6.1 The problem

Consider the problem (MP) with $\mu = 1$ and its weak formulation (see section 2.3).

The construction will follow:

1. Mesh: Find a triangulation of Ω with shape regularity and conforming mesh.
2. Consider the space V_h of linear continuous functions within each element K in the mesh.
3. Denote by N_h the dimension of V_h . For coding purposes, it's more convenient to include vertices in the boundary condition.

4. Use a nodal basis defined by:

$$\phi_i \in V_h, \quad \phi_i(v_j) = \delta_{ij} \quad 1 \leq i, j \leq N_h$$

5. Note that the solution is a linear combination of the elements in the base, with weights $\{u_j\}_j$.

6.2 Reference Elements

The triangles in the mesh should be obtained from a linear transformation from a reference triangle $\hat{K}$.

In 2D, (ξ, η) reference system, the reference triangle has vertices $\hat{v}_1 = (0, 0)$, $\hat{v}_2 = (1, 0)$, $\hat{v}_3 = (0, 1)$. It is important to keep the order of the vertices. For stability reasons, usually an anti-clockwise order is followed.

Each vertex i has a linear function $\hat{\Phi}_i(\xi, \eta) = C_1 + C_2\xi + C_3\eta$ with 3 degrees of freedom C_1, C_2, C_3 (for each triangle in the mesh). Note that each piece linear function will keep piece linear properties under linear transformations.

On $\hat{K}$, any linear polynomial has the form $C_1 + C_2\xi + C_3\eta$.

To uniquely identify one linear polynomial on $\hat{K}$, we take as dof the three vertices with the shape functions:

$$\begin{cases} \hat{\phi}_1(\xi, \eta) = 1 - \xi - \eta \\ \hat{\phi}_2(\xi, \eta) = \xi \\ \hat{\phi}_3(\xi, \eta) = \eta \end{cases}$$

with:

$$\begin{aligned} \hat{\nabla} \hat{\phi}_1(\xi, \eta) &= (-1, -1) \\ \hat{\nabla} \hat{\phi}_2(\xi, \eta) &= (1, 0) \\ \hat{\nabla} \hat{\phi}_3(\xi, \eta) &= (0, 1) \end{aligned}$$

In 3D, (ξ, η, ζ) reference system, the reference triangle has vertices $\hat{v}_1 = (0, 0, 0)$, $\hat{v}_2 = (1, 0, 0)$, $\hat{v}_3 = (0, 1, 0)$, $\hat{v}_4 = (0, 0, 1)$. Plus we have to add the points $(0.5, 0, 0)$; $(0, 0.5, 0)$; $(0, 0, 0.5)$; $(0.5, 0.5, 0)$; $(0, 0.5, 0.5)$; $(0.5, 0.5, 0.5)$ to have the same reference points as degrees of freedom (dof).

Exercise: Write the shape functions for $r=1, 2$.

6.3 Affine Mapping

After finding the reference triangle (polynomial), we must linear-transform this into any triangle (polynomial) in the space.

In d=2, the affine mapping F_K of any (ξ, η) onto (x, y) is defined by:

$$\begin{pmatrix} x \\ y \end{pmatrix} = B_K \begin{pmatrix} \xi \\ \eta \end{pmatrix} + b_K$$

where:

$$B_k = \begin{bmatrix} x_2 - x_1 & x_3 - x_1 \\ y_2 - y_1 & y_3 - y_1 \end{bmatrix} \quad b_K = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

After mapping the reference cell, the shape functions change in (x,y):

$$\hat{\phi}_j = \phi_j \circ F_K \quad \phi_j = \hat{\phi}_j \circ F_K^{-1}$$

And also the gradients:

$$\hat{\nabla} \hat{\phi}_j = B_k^T \nabla \phi_j \quad \nabla \phi_j = B_k^{-T} \hat{\nabla} \hat{\phi}_j$$

6.4 Representation of Mesh

The representation of a normal mesh consists of a list of vertices coordinates:

$$\begin{bmatrix} v_1 & v_2 & \dots & v_{N_h} \end{bmatrix}^T$$

And a connectivity matrix with the index of each polytope:

$$\begin{bmatrix} pol_{x_1}^1 & pol_{x_1}^2 & \dots & pol_{x_1}^{N_h} \\ pol_{x_2}^1 & pol_{x_2}^2 & \dots & pol_{x_2}^{N_h} \\ \dots & \dots & \dots & \dots \\ pol_{x_d}^1 & pol_{x_d}^2 & \dots & pol_{x_d}^{N_h} \end{bmatrix}$$

It can also have an extra row indicating which elements belong to the boundary in each polynomial, or store this information on a separate matrix

6.5 Calculating A and F

Our objective is to solve $Au = F$.

Note that:

$$\begin{aligned} (A)_{ij} &= \int_{\Omega} \nabla \phi_j \cdot \nabla \phi_i = \sum_{\substack{K \in \mathcal{T}_h \\ K \subset \text{supp}(\phi_i) \cap \text{supp}(\phi_j)}} \int_K \nabla \phi_j \cdot \nabla \phi_i \\ (\underline{F})_i &= \int_{\Omega} f \phi_i = \sum_{\substack{K \in \mathcal{T}_h \\ K \subset \text{supp}(\phi_i)}} \int_K f \phi_i \end{aligned}$$

where $\text{supp}(f)$ is the subset where f is not 0 (adding this restriction allows us to save computation).

To calculate F, from the transformations we saw on section 6.2, we need to calculate, for

every element $K \in \mathcal{T}_h$, and for every ϕ_i, ϕ_j shape functions with non-null support over K :

$$b_K(i) = \int_K f(x, y) \phi_i(x, y) dx dy = \det(B_K) \int_{\hat{K}} f(F_K(\xi, \eta)) \hat{\phi}_i(\xi, \eta) d\xi d\eta$$

However, we are not able to calculate this exact integral efficiently. Instead, we will use the quadrature rule to calculate integrals:

$$\int_K g dx \approx \sum_{\ell=1}^{nqn} g(x_\ell) w_\ell$$

- nqn stands for the number of quadrature nodes.
- w_ℓ are the weights of each node ℓ

We will pin some quadrature nodes over the bounds of the reference triangle so we use them to calculate the integral as follows:

$$b_K(i) = \det(B_K) \int_{\hat{K}} f(F_K(\xi, \eta)) \hat{\phi}_i(\xi, \eta) d\xi d\eta \approx \det(B_K) \sum_{\ell=1}^{nqn} f(F_K(\xi_\ell, \eta_\ell)) \hat{\phi}_i(\xi_\ell, \eta_\ell) \cdot \hat{w}_\ell$$

Analogously, to calculate A :

$$A_K(i, j) = \int_K \nabla \phi_j(x, y) \cdot \nabla \phi_i(x, y) dx dy = \det(B_K) \int_{\hat{K}} \left(B_K^{-T} \hat{\nabla} \hat{\phi}_j(\xi, \eta) \right) \cdot \left(B_K^{-T} \hat{\nabla} \hat{\phi}_i(\xi, \eta) \right) d\xi d\eta$$

And then the approximation is:

$$\begin{aligned} A_K(i, j) &= \det(B_K) \int_{\hat{K}} \left(B_K^{-T} \hat{\nabla} \hat{\phi}_j(\xi, \eta) \right) \cdot \left(B_K^{-T} \hat{\nabla} \hat{\phi}_i(\xi, \eta) \right) d\xi d\eta \\ &\approx \det(B_K) \sum_{\ell=1}^{nqn} \left(B_K^{-T} \hat{\nabla} \hat{\phi}_j(\xi_\ell, \eta_\ell) \right) \cdot \left(B_K^{-T} \hat{\nabla} \hat{\phi}_i(\xi_\ell, \eta_\ell) \right) \cdot \hat{w}_\ell \end{aligned}$$

6.6 Loading

Besides we perform all the calculations over a reference polygon, we need our final result to be global. This is called loading, and consists of taking the calculations obtained for each K and put them on the final result vector (see figures 1 and 2). As we are considering every ϕ_i separately for each K (ϕ_i, ϕ_j in the case of A), the load function should sum and not overwrite what is already in the final F . Every integral will consider only a specific base of ϕ for each K , the final result has to consider the sum of all of them.

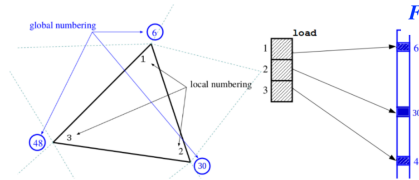


Figure 1: Loading F from local to global vector: Example

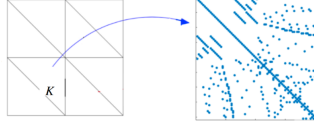


Figure 2: Loading A from local to global vector: Example

6.7 A further example

Suppose the diffusion problem:

$$\begin{cases} -\Delta u + c = f & \text{on } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

So the weak problem is finding $u \in H_0^1(\Omega)$ st

$$\int_{\Omega} \nabla u \cdot \nabla v + \int_{\Omega} uv = \int_{\Omega} fv \quad \forall v \in H_0^1(\Omega)$$

And the FEM for $r=1$ is finding $u_h \in V_h$ st

$$\int_{\Omega} \nabla u_h \cdot \nabla v_h + \underbrace{\int_{\Omega} u_h v_h}_M = \underbrace{\int_{\Omega} f v_h}_F \quad \forall v_h \in V_h$$

So we can write it as:

$$(A + M)\underline{u} = \underline{F}$$

Now, to assemble the matrix M to the model we can follow a similar approach.

For any element $K \in \mathcal{T}_n$, and for any ϕ_i, ϕ_j with not null support on K:

$$M_{loc} = \int_K \phi_j(x, y) \cdot \phi_i(x, y) \, dxdy = \det(B_K) \int_{\hat{K}} \hat{\phi}_j(\xi, \eta) \hat{\phi}_i(\xi, \eta) \, d\xi d\eta$$

Now, it is approximately:

$$\det(B_K) \sum_{\ell=1}^{nqn} \hat{\phi}_j(\xi_{\ell}, \eta_{\ell}) \hat{\phi}_i(\xi_{\ell}, \eta_{\ell}) \cdot \hat{w}_{\ell}$$

Note that both ϕ_i and ϕ_j are linear, so $\phi_i \phi_j$ is quadratic. It is needed to use as many points in the quadrature in order to calculate the integral.

Now, we can update M with M_{loc} .

6.8 Non-homogeneous Dirichlet BC

Now, take a problem with a non-homogeneous Dirichlet boundary condition:

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ u = g & \text{on } \Gamma_D \\ \frac{\partial u}{\partial n} = g_N & \text{on } \Gamma_N \end{cases}$$

Let $V = H_{\Gamma_D}^1$. Let $V_g = \{v \in H^1 \text{ st } v = g \text{ on } \Gamma_D\}$. Then, as seen in the section 2.4, we need to redefine u as $\overset{\circ}{u}$, and follow the same reformulation.

Now, to assemble this to the FEM, we can follow different approaches:

- For every $x_i \in \partial\Omega$, change the i th equation of the system:
 - Set the i th row of A to e_i .
 - Set the right-hand side F_i to $g(x_i)$

Note that with this approach, A loses its symmetry. Then, the numerical methods for solving the system are much slower. Anyway, it is a good approach if the system is not symmetric from the beginning.

- This consists of:
 1. Define the vector $u_g \in \mathbb{R}^{N_h}$ such that

$$(u_g)_i = \begin{cases} g(x_i) & \text{for any } x_i \in \partial\Omega \\ 0 & \text{for any } x_i \notin \partial\Omega \end{cases}$$

2. Compute the vector $F_g = F - Au_g$
3. If $x_i \in \partial\Omega$
 - Set the i th row of A to e_i^T
 - Set the i th column of A to e_i
 - Set $F_g(i) = 0$
4. Solve the system $A\overset{\circ}{u} = F_g$
5. Set $u = \overset{\circ}{u} + u_g$

This is equivalent to introducing the lifting operator

7 The discontinuous Galerkin Method (DG)

7.1 Limitations of Continuous Method

Until now, we have been focused on $\Delta u = f$ problems. In this case, we have used FEM with the Galerkin Method. This has limitations:

- Non-conforming Meshes: We cannot create a grid where a node converges with a side of a polygon, due to continuity. This doesn't allow us to have, for example, grids that have more details in one part than another.

- High order approximations: Due to numerical stability, we can only have low order approximations with $r = 1, 2, 3$.

The solution to this is to relax conformity:

$$V_h \not\subseteq V$$

7.2 The idea

The Discontinuous Galerkin (DG) methods are a family of finite element methods for approximating partial differential equations.

The discrete solution is sought in a discrete space made of polynomials that are completely discontinuous across mesh elements. Therefore, $V_h \not\subseteq V$.

The DG methods have several advantages:

- Wide range of PDEs treated within the same unified framework.
- Weak approximation of boundary conditions.
- Flexibility in mesh design.
- Flexibility in polynomial degree distribution.

Otherwise, they have some disadvantages:

- As different tiles do not share vertices, there are a high number of degrees of freedom.
- Larger algebraic linear systems to be solved.

DG methods were introduced in the 70s for purely hyperbolic problems. Then, they were extended to elliptic PDEs.

During the 80s, the methods were abandoned, due to the larger number of degrees of freedom, without having much computational power.

In the late 90s, they were used to solve a wide range of problems: Linear/non-linear PDEs, time-dependent/stationary problems, spectrum approximation, and multi-physics problems.

Some fundamental books regarding this topic are:

- Discontinuous Galerkin Methods for Solving Elliptic and Parabolic Equations: Theory and Implementation, by B. Rivière.
- Discontinuous Galerkin Methods. Algorithms, Analysis, and Applications, by J.S. Hesthaven and T. Warburton.
- High-order Discontinuous Galerkin Methods for the Maxwell Equations, by H. Fahs.
- Mathematical Aspects of Discontinuous Galerkin Methods, by D. A. Di Pietro.
- Discontinuous Galerkin Methods on Polygonal and Polyhedral Meshes, by A. Cangiani, Z. Dong, E.H. Geourgoulis, and P. Houston.

- Review of discontinuous Galerkin Finite Element Methods for Partial Differential Equations on Complicated Domains, by P.F. Antonietti, A. Cangiani, J. Collis, Z. Dong, E.H. Georgoulis, S.Giani, and P. Houston.

7.3 Definitions

Definition 7.1: $\{\{v\}\}$ and $[[v]]$

We define two elements given a function v :

$\{\{v\}\} :=$ The average of v across the edge

$[[v]] :=$ The jump of v across an edge between two neighboring elements

Definition 7.2: $\mathcal{F}_h, \mathcal{F}_h^I, \mathcal{F}_h^B$

Given a mesh of Ω . Define:

$\mathcal{F}_h :=$ Set of all edges/faces on the mesh

$\mathcal{F}_h^I :=$ Set of all interior edges/faces on the mesh

$\mathcal{F}_h^B :=$ Set of all boundary edges/faces on the mesh

Definition 7.3

Let v be scalar and $\underline{v}$ vector functions.

Let f^+ be the function evaluated in k^+ and f^- be the function evaluated in k^- ($F = \partial K^+ \cap \partial K^-$). For any $F \in \mathcal{F}_h^I$, shared by tiles K^+ and K^- :

$$\bullet \{\{v\}\} := \frac{(v^+ + v^-)}{2} \quad \text{and} \quad [[v]] = v^+ \underline{n}^+ + v^- \underline{n}^-$$

$$\bullet \{\{\underline{v}\}\} := \frac{(\underline{v}^+ + \underline{v}^-)}{2} \quad \text{and} \quad [[\underline{v}]] = \underline{v}^+ \cdot \underline{n}^+ + \underline{v}^- \cdot \underline{n}^-$$

where:

- $\underline{n}^+$ is the normal vector pointing to K^+

- $\underline{n}^-$ is the normal vector pointing to K^-

Otherwise, for $F \in \mathcal{F}_h^B$

$$\bullet \{\{v\}\} = v \quad \text{and} \quad [[v]] = v \underline{n}$$

$$\bullet \{\{\underline{v}\}\} = \underline{v} \quad \text{and} \quad [[\underline{v}]] = \underline{v} \cdot \underline{n}$$

Note that usually $\underline{n}^+ = -\underline{n}^-$, so the jump formulas make sense in regular tiles.

7.4 Development of Homogeneous Dirichlet Boundary Condition Setting

7.4.1 DG Formulation for the Poisson Problem

We will consider the case for a model with the Dirichlet condition on 0 only:

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

We then have to introduce a finite element polygonization $\mathcal{T}_h$. We are going to deal with the simplest case, triangles in 2D and tetrahedra in 3D. We are going to continue calling K a generic mesh element.

Then, we can multiply by v and integrate:

$$\int_K -\Delta v = \int_K f v \quad \forall v \text{ smooth enough}$$

The next step is to integrate by parts:

$$\sum_{K \in \mathcal{T}_h} \int_K \nabla u \cdot \nabla v - \int_{\partial K} \nabla u \cdot \underline{n}_K \cdot v = \sum_{K \in \mathcal{T}_h} \int_K f v \quad \forall v \text{ smooth enough}$$

We don't know how to deal with the integral on ∂K , since we don't have the same function values in contiguous K's, since in this formulation we don't enforce continuity.

7.4.2 Magic Formula

Theorem 7.4: Magic Formula(Arnold, 1982)

$\forall \tau$ vector function, $\forall v$ scalar function, both piecewise regular

$$\sum_{K \in \mathcal{T}_h} \int_{\partial K} \tau \cdot \underline{n}_K v = \sum_{F \in \mathcal{F}_h} \{ \{ \tau \} \} \cdot [[v]] + \sum_{F \in \mathcal{F}_h^I} [[\tau]] \{ \{ v \} \}$$

Proof.

$$\underbrace{\sum_{K \in \mathcal{T}_h} \int_{\partial K} \tau \cdot \underline{n}_K v}_{(A)} = \underbrace{\sum_{F \in \mathcal{F}_h} \{ \{ \tau \} \} \cdot [[v]] + \sum_{F \in \mathcal{F}_h^I} [[\tau]] \{ \{ v \} \}}_{(B)}$$

The proof is not necessary for the exam.

Rewrite:

$$\begin{aligned}
(A) &= \underbrace{\sum_{F \in \mathcal{F}_h^I} \int_F (\underline{\tau}^+ \cdot \underline{n}^+ v^+ + \underline{\tau}^- \cdot \underline{n}^- v^-)}_{A_1} + \sum_{F \in \mathcal{F}_h^B} \int_F \underline{\tau} \cdot \underline{n} v \\
(B) &= \underbrace{\sum_{F \in \mathcal{F}_h} \int_F \frac{1}{2} (\underline{\tau}^+ + \underline{\tau}^-) \cdot (v^+ \underline{n}^+ + v^- \underline{n}^-)}_{C_1} + \sum_{F \in \mathcal{F}_h^B} \int_F \underline{\tau} \cdot v \underline{n} + \underbrace{\sum_{F \in \mathcal{F}_h^I} \int_F (\underline{\tau}^+ \cdot \underline{n}^+ + \underline{\tau}^- \cdot \underline{n}^-) \frac{1}{2} (v^+ + v^-)}_{C_2}
\end{aligned}$$

Now, note that the boundary part is the same in both, so we have to show:

$$A_1 = C_1 + C_2$$

Now, note that:

$$\begin{aligned}
C_1 + C_2 &= \sum_{F \in \mathcal{F}_h} \left(\int_F \frac{1}{2} (\underline{\tau}^+ + \underline{\tau}^-) \cdot (v^+ \underline{n}^+ + v^- \underline{n}^-) + \int_F (\underline{\tau}^+ \cdot \underline{n}^+ + \underline{\tau}^- \cdot \underline{n}^-) \frac{1}{2} (v^+ + v^-) \right) \\
&= \sum_{F \in \mathcal{F}_h} \left(\int_F (\underline{\tau}^+ \cdot \underline{n}^+ v^+ + \underline{\tau}^- \cdot \underline{n}^- v^- + \frac{1}{2} \underline{\tau}^+ \cdot \underline{n}^- v^+ + \frac{1}{2} \underline{\tau}^- \cdot \underline{n}^+ v^- + \frac{1}{2} \underline{\tau}^+ \cdot \underline{n}^+ v^+ + \frac{1}{2} \underline{\tau}^- \cdot \underline{n}^- v^-) \right) \\
&= \sum_{F \in \mathcal{F}_h} \left(\int_F (\underline{\tau}^+ \cdot \underline{n}^+ v^+ + \underline{\tau}^- \cdot \underline{n}^- v^-) \right) \quad (\text{Since } \underline{n}^+ = -\underline{n}^-) \\
&= A_1
\end{aligned}$$

□

7.4.3 Reformulation using magic formula

Definition 7.5: Elliptic Regularity

We say that Ω satisfies elliptic regularity if $\forall g \in L^2(\Omega)$ the solution z of the problem:

$$\begin{cases} -\Delta z = g & \text{in } \Omega \\ z = 1 & \text{on } \partial\Omega \end{cases}$$

satisfies $z \in H^2(\Omega)$ and $\|z\|_{H^2(\Omega)} \lesssim \|g\|_{L^2(\Omega)}$

Remark.

A sufficient condition for this property to be true is that Ω is a convex polygonal/polyhedral domain.

Using the magic formula, our problem, up to what we saw in section 7.4.1:

$$\sum_{K \in \mathcal{T}_h} \left(\int_K \nabla u \cdot \nabla v - \int_{\partial K} \nabla u \cdot \underline{n}_K \cdot v \right) = \sum_{K \in \mathcal{T}_h} \int_K f v \quad \forall v \text{ smooth enough}$$

Turns into:

$$\sum_{K \in \mathcal{T}_h} \int_K \nabla u \cdot \nabla v - \sum_{F \in \mathcal{F}_h} \{ \{ \nabla u \} \} \cdot [[v]] - \sum_{F \in \mathcal{F}_h^I} [[\nabla u]] \{ \{ v \} \} = \sum_{K \in \mathcal{T}_h} \int_K f v \quad \forall v \text{ smooth enough}$$

Plus, assuming Ω is convex, with elliptic regularity, the solution u of this problem:

$$u \in H^2(\Omega) \cap H_0^1(\Omega) \quad (\text{By theoretical PDEs})$$

By this, if $u \in H^2(\Omega)$

$$\implies \nabla u \in [H^1(\Omega)]^d$$

Plus, a function in H^1 is continuous along the normal direction (In $\mathbb{R}^1, \mathbb{R}^2$, but in $\mathbb{R}^3$ we need elliptic regularity), so $\nabla u^+ = \nabla u^-$, and therefore:

$$\implies [[\nabla u]] = \nabla u^+ \cdot \underline{n}^+ + \nabla u^- \cdot \underline{n}^- = 0 \quad \forall F \in \mathcal{F}_h^I$$

Since this, the third integral in the problem is 0, and the problem turns into:

$$\sum_{K \in \mathcal{T}_h} \int_K \nabla u \cdot \nabla v - \sum_{F \in \mathcal{F}_h} \int_F \{ \{ \nabla u \} \} \cdot [[v]] = \sum_{K \in \mathcal{T}_h} \int_K f v \quad \forall v \in H^2(\Omega) \cap H_0^1(\Omega)$$

On the other hand, note that as $u \in H^2(\Omega)$, $[[u]] = 0 \quad \forall F \in \mathcal{F}_h^I$ as well. And, by our problem, u is 0 in the boundary, so $[[u]] = 0 \quad \forall F \in \mathcal{F}_h^B$.

Finally, as we have lost symmetry in the, we can recover it by adding a term that is 0:

$$\sum_{K \in \mathcal{T}_h} \int_K \nabla u \cdot \nabla v - \sum_{F \in \mathcal{F}_h} \{ \{ \nabla u \} \} \cdot [[v]] - \sum_{F \in \mathcal{F}_h} \int_F [[u]] \cdot \{ \{ \nabla v \} \} = \sum_{K \in \mathcal{T}_h} \int_K f v \quad \forall v \in H^2(\Omega) \cap H_0^1(\Omega)$$

And we will add another 0 term (and symmetric). This term is called stabilization term, and controls the jumps:

$$\sum_{K \in \mathcal{T}_h} \int_K \nabla u \cdot \nabla v - \sum_{F \in \mathcal{F}_h} \{ \{ \nabla u \} \} \cdot [[v]] - \sum_{F \in \mathcal{F}_h} \int_F [[u]] \cdot \{ \{ \nabla v \} \} + \sum_{F \in \mathcal{F}_h} \int_F \gamma [[u]] [[v]] = \sum_{K \in \mathcal{T}_h} \int_K f v \quad \forall v \in H^2(\Omega) \cap H_0^1(\Omega)$$

Where γ is a stabilization function (that might depend on the discretization parameters).

7.4.4 Discrete Setting

Let

$$V_h^r = \{v_h \in L^2(\Omega) \quad st \quad v_h|_K \in \mathbb{P}^r(K) \quad \forall K \in \mathcal{T}_h\} \not\subseteq H_0^1(\Omega)$$

Note that this space does not contain the boundary conditions. In contrast to the section 5.1, that $v_n \in C^0(\bar{\Omega})$

So our DG formulation is as follows:

$$\text{Find } u_h \in V_h^r \quad st \quad A(u_h, v_h) = \int_{\Omega} f v_h \quad \forall v_h \in V_h^r$$

Where $A(\cdot, \cdot)$ is defined as:

$$A(u_h, v_h) = \sum_{K \in \mathcal{T}_h} \int_{\Omega} \nabla_h u_h \cdot \nabla_h v_h - \sum_{F \in \mathcal{F}_h} \{ \{ \nabla_h u_h \} \} \cdot [[v_h]] - \sum_{F \in \mathcal{F}_h} \int_F [[u_h]] \cdot \{ \{ \nabla_h v_h \} \} + \sum_{F \in \mathcal{F}_h} \int_F \gamma [[u_h]] [[v_h]]$$

Where $\nabla_h \cdot \rightarrow$ is the broken gradient (The operator is applied element by element K , without continuity between them).

Anyway, more generally, we can formulate the problem in different scenarios:

$$A(u_h, v_h) = \underbrace{\sum_{K \in \mathcal{T}_h} \int_{\Omega} \nabla_h u_h \cdot \nabla_h v_h}_V - \underbrace{\sum_{F \in \mathcal{F}_h} \{ \{ \nabla_h u_h \} \} \cdot [[v_h]]}_I - \theta \underbrace{\sum_{F \in \mathcal{F}_h} \int_F [[u_h]] \cdot \{ \{ \nabla_h v_h \} \}}_{I^T} + \underbrace{\sum_{F \in \mathcal{F}_h} \int_F \gamma [[u_h]] [[v_h]]}_S$$

where: θ is called the interior penalty.

- $\theta = 1$: Symmetric Interior Penalty (SIP)
- $\theta = -1$: Non-symmetric Interior Penalty (NIP)
- $\theta = 0$: Incomplete Interior Penalty (IIP)

Note that:

$$A = V - I - \theta I^T + S$$

where V is called volume term, I is called consistency term, and S is called stability term

7.5 Extension to Different Boundaries

The above formulation applies to the homogeneous Dirichlet boundary conditions.

- In the case of non-homogeneous Dirichlet boundary conditions, the right-hand side (rhs) changes:

$$\int_{\Omega} f v - \theta \sum_{F \in \mathcal{F}_h^B} \int_F \nabla_h v_h \cdot \underline{n} g_D + \sum_{F \in \mathcal{F}_h^B} \int_F \gamma v_h g_D$$

- In the case of Neumann boundary conditions ($\nabla u \cdot \underline{n} = g_N$ on $\partial\Omega$), the rhs changes to:

$$\int_{\Omega} f v + \sum_{F \in \mathcal{F}_h^B} \int_F v_h g_N$$

and the lhs changes to (note that the sum is over the interior edges):

$$A(u_h, v_h) = \int_{\Omega} \nabla_h u_h \cdot \nabla_h v_h - \sum_{F \in \mathcal{F}_h^I} \int_F \{ \{ \nabla_h u_h \} \} \cdot [[v_h]] - \theta \sum_{F \in \mathcal{F}_h^I} \int_F [[u_h]] \cdot \{ \{ \nabla_h v_h \} \} + \sum_{F \in \mathcal{F}_h^I} \gamma [[u_h]] [[v_h]]$$

7.6 The stabilization function γ

Define $\gamma : L^\infty(\mathcal{F}_h) \rightarrow \mathbb{R}$, where $\mathcal{F}_h$ is the set of the edges:

$$\forall F \in \mathcal{F}_h \quad st \quad \gamma|_F = \alpha \frac{\bar{r}^2}{\bar{h}}$$

where:

- $\alpha \in \mathbb{R}^{++}$
- Polynomial Degree: $\bar{r} = \begin{cases} \max\{r^+, r^-\} & \text{on } F \in \mathcal{F}_h^I \\ r & \text{on } F \in \mathcal{F}_h^B \end{cases}$
- Mesh Size: $\bar{h} = \begin{cases} \max\{h_k^+, h_k^-\} & \text{on } F \in \mathcal{F}_h^I \\ h_k & \text{on } F \in \mathcal{F}_h^B \end{cases}$

If r is the same for all $K \in \mathcal{T}_h$, and if the mesh is quasi-uniform ($h_K \simeq h \quad \forall K \in \mathcal{T}_h$)

$$\implies \gamma \cong \alpha \frac{r^2}{h}$$

7.7 Space and norms

Now, we will introduce some spaces:

Definition 7.6: Broken Sobolev Space

$$H^s(\mathcal{T}_h) = \{v \in L^2(\Omega) \quad st \quad v|_K \in H^s(K) \quad \forall K \in \mathcal{T}_h\}$$

Definition 7.7: Broken L^2 Space

$$L^2(\mathcal{F}_h) = \{v \in L^2(F) \quad \forall F \in \mathcal{F}_h\}$$

And its norms:

Definition 7.8: $\|\cdot\|_{H^s(\mathcal{T}_h)}$

$$\|v_h\|_{H^s(\mathcal{T}_h)}^2 = \sum_{K \in \mathcal{T}_h} \|v_k\|_{H^s(K)}^2$$

Definition 7.9: $\|\cdot\|_{L^2(\mathcal{F}_h)}$

$$\|v_h\|_{L^2(\mathcal{F}_h)}^2 = \sum_{F \in \mathcal{F}_h} \|v_k\|_{L^2(F)}^2$$

Now, the norm in the DG space:

Definition 7.10: $\|\cdot\|_{DG}$

$$\|v_h\|_{DG}^2 = \|\nabla_h v_h\|_{L^2(\Omega)}^2 + \|\gamma^{1/2}[[v_h]]\|_{L^2(\mathcal{F}_h)}^2 \quad \forall v \in H^2(\mathcal{T}_h)$$

Proof. The difficult part to prove is showing that norm 0 implies the 0 element. The idea is that if the jumps are always 0, taking into account the boundary conditions, the unique possible element is 0. $\square$

Plus, we need a second norm:

Definition 7.11: $\|\!\!\|\cdot\|\!\!\|$

$$\|\!\!\|v\|\!\!\|_{DG}^2 = \|v\|_{DG}^2 + \|\gamma^{-1/2}\{\{\nabla_h v\}\}\|_{L^2(\mathcal{F}_h)}^2 \quad \forall v \in H^2(\mathcal{T}_h)$$

7.8 Equivalence of $\|\cdot\|$ and $\|\!\!\|\cdot\|\!\!\|$

Theorem 7.12: Trace-Inverse Inequality

$$\|\eta\|_{L^2(F)}^2 \leq C_I \frac{r}{h_K^{1/k}} \|\nabla \eta\|_{L^2(K)} \quad \forall F \subseteq \partial K \quad \forall \eta \in \mathbb{P}^2(K)$$

Proposition 7.13

It can be shown that $\forall v \in H^2(\mathcal{T}_h)$:

$$\|v\|_{DG} \leq \|\!\!\|v\|\!\!\|_{DG} \lesssim \|v\|_{DG}^2$$

Anyway, $\forall v_h \in V_h^r$:

$$\|v_h\|_{DG} \leq \|\!\!\|v_h\|\!\!\|_{DG} \lesssim \|v_h\|_{DG}^2$$

Proof. The first inequality is trivial. The no bound in $H^2(\mathcal{T}_h)$ won't be proven. On the other hand, we will prove $\|\!\!\|v_h\|\!\!\|_{DG} \lesssim \|v_h\|_{DG}^2$ using the trace/inverse estimate:

$$\begin{aligned} \|\gamma^{-1/2}\{\{\nabla_h v_h\}\}\|_{L^2(\mathcal{F}_h)}^2 &= \frac{h}{r^2 \alpha} \sum_{F \in \mathcal{F}_h} \|\{\{\nabla_h v_h\}\}\|_{L^2(F)}^2 \\ &\lesssim \frac{h}{r^2 \alpha} \frac{r^2}{h} C_I \sum_{K \in \mathcal{T}_h} \|\nabla_h v_h\|_{L^2(K)}^2 = \frac{1}{\alpha} C_I \|\nabla_h v_h\|_{L^2(\Omega)}^2 \end{aligned}$$

$\square$

7.9 Properties of A

The needed properties for A are:

- Continuity on $H^2(\mathcal{T}_h) \times V_h^r$.

$$A(v, w_h) \lesssim \|v\|_{DG} \|w_h\|_{DG} \quad \forall v \in H^2(\mathcal{T}_h) \quad \forall w_h \in V_h^r$$

It does not work with $\|\cdot\|_{DG}$

- Coercivity on $V_h^r \times V_h^r$:

$$A(v_h, v_h) \gtrsim \|v_h\|_{DG}^2 \quad \forall v_h \in V_h^r$$

For SIP and IIP, the penalty parameter α has to be chosen large enough. For NIP $\alpha = 0$ is fine, given $p \geq 2$.

- Strong-Consistency:

$$A(u, v_h) = \int_{\Omega} f v_h \quad \forall v_h \in V_h^r \quad \implies \quad A(u - u_h, v_h) = 0 \quad \forall v_h \in V_h^r$$

(Galerkin Orthogonality)

- Approximation:

Let $\Pi_h^r u \in V_H^r$ be a suitable approximation of u , then the approximation error is:

$$\|u - \Pi_h^r u\|_{DG}$$

We need to control this.

7.10 Error bound

The error:

$$\|u - u_h\|_{DG} \leq \|u - \Pi_h^r u\|_{DG} + \|\Pi_h^r u - u_h\|_{DG}$$

And the second term squared, assuming the properties stated in section 7.9:

$$\begin{aligned} \|\Pi_h^r u - u_h\|_{DG}^2 &\lesssim A(\Pi_h^r u - u_h, \Pi_h^r u - u_h) && \text{(Coercivity)} \\ &\lesssim A(\Pi_h^r u - u, \Pi_h^r u - u_h) + \underbrace{A(u - u_h, \Pi_h^r u - u_h)}_{=0 \text{ Galerkin Orthogonality}} && \text{(Triangular ineq.)} \\ &\lesssim \|\Pi_h^r u - u\|_{DG} \|\Pi_h^r u - u_h\|_{DG} && \text{(Continuity)} \end{aligned}$$

So:

$$\|\Pi_h^r u - u_h\|_{DG} \lesssim \|\Pi_h^r u - u\|_{DG}$$

Finally, the error turns into:

$$\begin{aligned} \|u - u_h\|_{DG} &\lesssim \|u - \Pi_h^r u\|_{DG} + \|\Pi_h^r u - u_h\|_{DG} \\ &\lesssim \|u - \Pi_h^r u\|_{DG} \end{aligned}$$

The final bound is found in section 7.11.

7.11 Interpolation Error

Now, we have to find a bound for $\|u - \Pi_h^r u\|_{DG}$. Define $e_\pi = u - \Pi_h^r u$, then:

$$\|e_\pi\|_{DG}^2 = \underbrace{\|\nabla_h e_\pi\|_{L^2(\Omega)}^2}_I + \underbrace{\sum_{F \in \mathcal{F}_h} \|\gamma^{1/2}[[e_\pi]]\|_{0,\mathcal{F}_h}^2}_{II} + \underbrace{\sum_{F \in \mathcal{F}_h} \|\gamma^{-1/2}\{\{\nabla_h e_\pi\}\}\|_{0,\mathcal{F}_h}^2}_{III} \quad (1)$$

Now, we need two extra results:

Theorem 7.14: Babuska & Suri, 1987

Let $K \in \mathcal{T}_h$. Let $v \in H^{s+1}(K)$, $s \geq 0$. Then, there exists a sequence of operators $\Pi_h^r v : H^{s+1}(K) \rightarrow \mathbb{P}^r(K)$, $r = 1, 2, \dots$ such that:

$$\|v - \Pi_h^r v\|_{H^m(K)} \lesssim \frac{h^{\min(r,s)+1-m}}{r^{r+1-m}} \|v\|_{H^{s+1}(K)} \quad 0 \leq m \leq s+1$$

The hidden constant is independent of v , h , and r , but depends on the shape-regularity of K in s .

Proof. Find it in Babuska & Suri, 1987. □

Theorem 7.15: Multiplicative Trace Inequality for Polynomial Spaces

$$\|\eta\|_{L^2(\partial K)}^2 \lesssim \|\eta\|_{L^2(K)} \|\nabla \eta\|_{L^2(K)} + h^{-1} \|\eta\|_{L^2(K)}^2 \quad \forall \eta \in H^1(K)$$

Where the hidden constant depends on the shape regularity of K , and dimension d .

Now, we need to bound I, II, and III

- With Babuska & Suri inequality, taking $m=1$.

On $K \in \mathcal{T}_h$ and $u \in H^{s+1}(K)$, the Babuska-Suri interpolate satisfies:

$$\begin{aligned} \|\nabla_h e_\pi\|_{L^2(\Omega)} &\leq \|e_\pi\|_{H^1(K)} \quad \text{By eq 1 and DG norm definition} \\ &\lesssim \frac{h^{\min(r,s)}}{r^2} \|u\|_{H^{s+1}(K)} \end{aligned}$$

- Now, bound II:

$$\begin{aligned} \sum_{F \in \mathcal{F}_h} \|\gamma^{1/2}[[e_\pi]]\|_{L^2(\mathcal{T}_h)}^2 &\lesssim \frac{r^2}{h} \sum_{K \in \mathcal{T}_h} \|e_\pi\|_{L^2(\partial K)}^2 \quad (\text{Constant by the definition of } \gamma) \\ &\lesssim \frac{r^2}{h} \sum_{K \in \mathcal{T}_h} \left[\|e_\pi\|_{L^2(K)} \|\nabla e_\pi\|_{L^2(K)} + h^{-1} \|\eta\|_{L^2(K)}^2 \right] \quad (\text{Mult. trace ineq.}) \\ &\lesssim \sum_{K \in \mathcal{T}_h} \frac{h^{2\min(r,s)}}{r^{2s-1}} \|u\|_{H^{s+1}(K)}^2 \end{aligned}$$

- Now, bound III, by a similar process:

$$\sum_{F \in \mathcal{F}_h} \|\gamma^{-1/2} \{\{\nabla_h e_\pi\}\}\|_{0,F}^2 \lesssim \frac{h}{r^2} \sum_{K \in \mathcal{T}_h} \|\nabla e_\pi\|_{L^2(\partial K)}^2 \lesssim \frac{h^{2 \min(r,s)}}{r^{2s+1}} \|u\|_{H^{s+1}(K)}^2$$

Gathering all together:

Theorem 7.16: Interpolation Bounds

If the exact solution is regular enough, then:

$$\|u - \Pi_h^r u\|_{DG} \lesssim \frac{h^{\min(r,s)}}{r^{s-1/2}} \|u\|_{H^{s+1}(\Omega)}$$

Moreover, if $r \geq s$, the estimate becomes:

$$\|u - \Pi_h^r u\|_{DG} \lesssim \left(\frac{h}{r}\right)^s r^{1/2} \|u\|_{H^{s+1}(\Omega)}$$

And as a consequence:

Theorem 7.17: Convergence Theorem

If the exact solution is smooth enough, and continuity, coercivity, and galerkin orthogonality hold, then:

$$\|u - u_h\|_{DG} \lesssim \frac{h^{\min(s,r)}}{r^{s-1/2}} \|u\|_{H^{s+1}(\Omega)}$$

Remark.

For the SIP and IIP methods, the above result holds if the penalty constant α is chosen sufficiently large (this ensures coercivity).

Remark.

- The bound is optimal in h and suboptimal in r by a factor $r^{1/2}$. See, for example (Houston, Schwab, Suli, 2001), (Riviere, Wheeler, Girault, 1999), (Perugia, Schotzau, 2001).
- Optimal error estimate with respect to r can be shown using the projector of (Georgoulis & Suli, 2005), provided the solution belongs to a suitable augmented space, or whenever a continuous interpolant can be built (Stamm & Wihler, 2010).

7.12 L^2 -norm error

An estimate for the L^2 -error can be obtained by using a duality argument.

1. If the exact solution $u \in H^s(\Omega)$, $s \geq 2$ and if u_h is the solution obtained with the SIP method ($\theta = 1$).

Plus, if the following adjoint consistency property is satisfied: $A(v_h, z) = \int_{\Omega} f v_h \quad \forall v_h \in V_h^r$ and α is sufficiently large. Then, it holds:

$$\|u - u_h\|_{L^2(\Omega)} \lesssim \frac{h^{\min(r,s)+1}}{r^{s+1/2}} \|H\|_{H^{s+1}(\Omega)}$$

2. On the other hand, for NIP and IIP formulations, if the corresponding bilinear form is non-symmetric and does not satisfy the adjoint consistency property, it holds:

$$\|u - u_h\|_{L^2(\Omega)} \lesssim \frac{h^{\min(r,s)+1}}{r^{s+1/2}} \|H\|_{H^{s+1}(\Omega)}$$

Remark.

It is numerically observed that for NIP formulation:

$$\|u - u_h\|_{L^2(\Omega)} \lesssim \begin{cases} h^{r+1} & \text{if } r \text{ is odd} \\ h^r & \text{if } r \text{ is even} \end{cases}$$

7.13 Proof of A properties

- **Coercivity:**

Take $u_h = v_h$ to obtain:

$$A(v_h, v_h) = \|\nabla_h v_h\|_{L^2(\Omega)}^2 - (1 + \theta) \underbrace{\sum_{F \in \mathcal{F}_h} \int_F \{\{\nabla_h v_h\}\} \cdot [[v_h]]}_{(1)} + \|\gamma^{1/2} [[v_h]]\|_{0, \mathcal{F}_h}^2$$

First, if $\theta = -1$, we are ok:

$$A(v_h, v_h) = \|\nabla_h v_h\|_{DG}^2$$

If not, note that:

$$\begin{aligned} (1) &= \int_{\mathcal{F}_h} \gamma^{-1/2} \{\{\nabla_h v_h\}\} \gamma^{1/2} [[v_h]] \stackrel{C-S}{\leq} \|\gamma^{-1/2} \{\{\nabla_h v_h\}\}\|_{L^2(\mathcal{F}_h)} \|\gamma^{1/2} [[v_h]]\|_{0, \mathcal{F}_h}^2 \\ &\leq \frac{\epsilon}{2} \|\gamma^{-1/2} \{\{\nabla_h v_h\}\}\|_{L^2(\mathcal{F}_h)}^2 + \frac{1}{2\epsilon} \|\gamma^{1/2} [[v_h]]\|_{0, \mathcal{F}_h}^2 \quad (ab \leq \frac{\epsilon}{2} a^2 + \frac{1}{2\epsilon} b^2) \end{aligned}$$

Therefore:

$$-|(1)| \geq -\frac{\epsilon}{2} \|\gamma^{-1/2} \{\{\nabla_h v_h\}\}\|_{L^2(\mathcal{F}_h)}^2 - \frac{1}{2\epsilon} \|\gamma^{1/2} [[v_h]]\|_{0, \mathcal{F}_h}^2$$

Now,

$$A(v_h, v_h) \geq \|\nabla_h v_h\|_{L^2(\Omega)}^2 - \frac{\epsilon(1+\theta)}{2} \|\gamma^{-1/2} \{\{\nabla_h v_h\}\}\|_{L^2(\mathcal{F}_h)}^2 + \left(1 - \frac{(1+\theta)}{2\epsilon}\right) \|\gamma^{1/2} [[v_h]]\|_{0, \mathcal{F}_h}^2$$

Now, using the trace inequality in the second term, it can be proven it is got:

$$A(v_h, v_h) \geq \left(1 - \frac{C_I \epsilon(1+\theta)}{2\alpha}\right) \|\nabla_h v_h\|_{L^2(\mathcal{F}_h)}^2 + \left(1 - \frac{(1+\theta)}{2\epsilon}\right) \|\gamma^{1/2} [[v_h]]\|_{0, \mathcal{F}_h}^2$$

Now, for example, we can choose $\epsilon = 1 + \theta$:

$$A(v_h, v_h) \geq \left(1 - \frac{C_I(1+\theta)^2}{2\alpha}\right) \|\nabla_h v_h\|_{L^2(\mathcal{F}_h)}^2 + \frac{1}{2} \|\gamma^{1/2} [[v_h]]\|_{0, \mathcal{F}_h}^2$$

And now, assuming:

$$1 - \frac{C_I(1+\theta)^2}{2\alpha} \geq \frac{1}{2} \quad \Longleftrightarrow \quad \alpha \geq C_I(1+\theta)^2 \equiv \alpha^*$$

$$A(v_h, v_h) \geq \frac{1}{2} \|\nabla_h v_h\|_{L^2(\mathcal{F}_h)}^2 + \frac{1}{2} \|\gamma^{1/2} [[v_h]]\|_{0, \mathcal{F}_h}^2 = \frac{1}{2} \|v_h\|_{DG}^2$$

- Continuity: PD: $|A(u - u_h, w_h)| \lesssim \|u - u_h\|_{DG} \|w_h\|_{DG} \quad \forall w_h \in V_h^r$.

Lemma 7.18

Let $\tilde{V}_h = V_h^r + V$, then:

$$|A(u, w)| \leq \|v\|_{DG} \|w\|_{DG} \quad \forall v, w \in \tilde{V}_h^r$$

$$|A(u, w_h)| \leq \|v\|_{DG} \|w_h\|_{DG} \quad \forall v \in \tilde{V}_h^r \quad \forall w_h \in V_h^r$$

Proof. It is trivial that (1) $\implies$ (2), since, on V_h^r , the norms are equivalent:

$$\|w_h\|_{DG} \lesssim \|w_h\|_{DG} \lesssim \|w_h\|_{DG} \quad \forall w_h \in V_h^r$$

Now, we prove (1): Use Cauchy Schwarz and the definition of A:

$$|A(v, w)| \lesssim \underbrace{\left| \sum_{k \in \mathcal{T}_h} \int_K \nabla u \nabla w \right|}_I + \underbrace{\left| \sum_{F \in \mathcal{T}_h} \int_F \{\{\nabla v\}\} \cdot [[w]] \right|}_{II} + \underbrace{\left| \sum_{F \in \mathcal{T}_h} \int_F \{\{\nabla w\}\} \cdot [[v]] \right|}_{III} + \underbrace{\left| \sum_{F \in \mathcal{T}_h} \int_F [[v]] \cdot [[w]] \right|}_{IV}$$

Now, note, by C-S:

$$\begin{aligned}
(I) &= \left(\sum_{K \in \mathcal{T}_h} \int_K |\nabla v|^2 \right)^{1/2} \left(\sum_{K \in \mathcal{T}_h} \int_K |\nabla w|^2 \right)^{1/2} = \|\nabla_h v\|_{L^2(\Omega)} \|\nabla_h w\|_{L^2(\Omega)} \\
&\leq \|v\|_{DG} \|w\|_{DG} \\
(II) &= \left| \int_{\mathcal{F}_h} |\gamma^{-1/2} \{\{\nabla_h v\}\} \cdot \gamma^{1/2} [[w]]| \right| \leq \|\gamma^{-1/2} \{\{\nabla_h v\}\}\|_{L^2(\mathcal{T}_h)} \|\gamma^{1/2} [[w]]\|_{L^2(\mathcal{T}_h)} \stackrel{\text{def norm}}{\leq} \|v\|_{DG} \|w\|_{DG} \\
(III) &= \text{The same as (II)} \\
(IV) &\leq \|\gamma^{1/2} [[v]]\|_{L^2(\mathcal{T}_h)} \|\gamma^{1/2} [[w]]\|_{L^2(\mathcal{T}_h)} \stackrel{\text{def norm}}{\leq} \|v\|_{DG} \|w\|_{DG}
\end{aligned}$$

The thesis concludes by summarizing the key points. $\square$

7.14 Remarks on Implementation

1. There is a lot of freedom in setting up the DOFs and the basis of polynomials, since everything is local, and there are no shared points.
2. We can use Lagrangian Shape Functions, as usual. Anyway, Legendre-type polynomials can be used, with models as Coefficients Expansion.
3. The volume V , since the basis functions have support in the correspondent K only, is:

$$V_{K_i} = \int_{K_i} \nabla \phi_j \nabla \phi_i \in \mathbb{R}^{3 \times 3}$$

So:

$$V = \begin{bmatrix} V_{K_1} & 0 & \dots & 0 \\ 0 & V_{K_2} & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & V_{K_{N_h/3}} \end{bmatrix}$$

4. The stability term S in the element K :

$$S_K = \sum_{\substack{F \in \mathcal{T}_h \\ F \subseteq \partial K}} \int_F \gamma^{1/2} [[\phi_j]] [[\phi_i]] \in \mathbb{R}^{3 \times 3}$$

But in this case, we also have some terms that are calculated between neighbors. Then, S will have non-zero values not only in the diagonals, but in the terms (i,j) , where i and j are neighbors. Note that we are considering each element as a 3×3 matrix.

5. The consistency term I in the element K :

$$I_K = \sum_{\substack{F \in \mathcal{T}_h \\ F \subseteq \partial K}} \int_F \gamma^{1/2} \{\{\nabla \phi_j\}\} [[\phi_i]] \in \mathbb{R}^{3 \times 3}$$

As S , this matrix has non-zero values between neighbors, but in this case, I is not symmetric

8 Spectral Element Methods (SEM)

8.1 Introduction

Spectral Element Methods are a class of models that compute PDEs in conforming meshes

$$V_h \subseteq V$$

(where continuity is needed). These methods are designed with quads (quadrilaterals). Although, under some assumptions, it has been extended to triangles.

This method is designed to solve high-order numerical methods $r > 1$ with a convergence of $\left(\frac{h}{r}\right)^s$, without stability problems, unlike the continuous Galerkin method.

SEM methods are also known as either spectral/hp or hp-FEM. h stands for high, and p for the degrees. In the literature, p is often used instead of r to denote the polynomial degrees.

8.2 Gauss-Legendre-Lobatto nodes

Suppose $K \in \mathcal{T}_h$ is a quad. In the Galerkin method, N_h increases a lot for each unit r (or p) increases (see section 5.3). For example, for $\mathbb{Q}_1$, 4 nodes are needed for each tile. for $\mathbb{Q}_2$ 9, $\mathbb{Q}_3$ 16, and so on. This is not numerically stable for high r's, due to Runge's phenomenon.

In the previous case, the problem is that the points were equally spaced on the mesh. Spectral methods have a higher density of points near the boundaries and more space between points further, improving the accuracy of high-degree polynomials. We can achieve this with the Gauss-Legendre-Lobatto (GLL) method, where the points are placed as follows:

1. Set $\xi_0 = -1$ and $\xi_p = 1$
2. Define $\mathbf{L}_p(\xi)$ the standard Legendre Polynomial of p-th degree:
 - (a) Starting from polynomial 1
 - (b) P_n is the polynomial of weight 1 such that

$$\int_{-1}^1 P_m P_n \quad \forall m < n$$

3. ξ_ℓ , $\ell = 1, \dots, p-1$ is defined as the roots of

$$\frac{\partial \mathcal{L}_p(\xi)}{\partial \xi}$$

Remark.

■ All the Legendre Polynomials are orthogonal.

Now, note that in 1D, we have the GLL nodes:

$$\begin{aligned}\xi_0, \xi_1, \dots, \xi_p & \quad \text{1D GLL nodes in } \xi \text{ axis.} \\ \eta_0, \eta_1, \dots, \eta_p & \quad \text{1D GLL nodes in } \eta \text{ axis.}\end{aligned}$$

Now, the tensor product of 1D GLL nodes gives the 2D GLL nodes:

$$(\xi_i, \eta_j), \quad i, j = 0, \dots, p$$

Remark.

■ The GLL nodes are symmetric with respect to 0

This method takes 2 discretization parameters:

- p (or N): Polynomial Degree
- h: Mesh size (maximum of elements' diameter)

Anyway there are 3 types:

- h-FEM: Fix p and refine h
- p-FEM: fix h and refine p
- hp-FEM: Refine h and p

8.3 The problem

We go back to the conforming methods, so the strong form is:

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

And the weak formulation is given $f \in L^2(\Omega)$, look for $u \in V = H_0^1(\Omega)$ st:

$$a(u, v) = F(v) \quad \forall v \in V$$

where:

$$a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v \quad F(v) = \int_{\Omega} f v$$

8.4 Mesh

Given $K \in \Omega$, the local SEM space in 2D and 3D is:

$$\mathbb{Q}^p = \{v_h = \sum_{i,j=0}^p v_{ij} x^i y^j, \quad v_{ij} \in \mathbb{R}\}$$

$$\mathbb{Q}^p = \{v_h = \sum_{i,j,k=0}^p v_{ijk} x^i y^j z^k, \quad v_{ijk} \in \mathbb{R}\}$$

And the DOFs are $\dim(\mathbb{Q}^p) = (p+1)^d$.

8.5 Mapping from Reference to Real Space F_K

Now, the difficulty is to find the function F_k from the reference element $\hat{K} = (-1, 1)^d$ to K . If F_k is linear, tiles can be only parallelograms. If we want to have general quadrilaterals, we have to relax the restriction of F_k linear to bilinear:

Call $\hat{\Gamma}_1, \hat{\Gamma}_2, \hat{\Gamma}_3, \hat{\Gamma}_4$ the reference quad edges and $\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4$. Then, define contraction/dilatation maps:

$$\Pi_\ell : [-1, 1] \rightarrow \Gamma_\ell, \quad \ell = 1, \dots, 4$$

Now, define $F_K : \hat{K} \rightarrow K$ in 2D as:

$$\begin{aligned} \begin{pmatrix} x \\ y \end{pmatrix} &= F_K \left(\begin{pmatrix} \xi \\ \eta \end{pmatrix} \right) \\ &= \frac{1-\eta}{2} \Pi_1(\xi) + \frac{1+\eta}{2} \Pi_3(\xi) \\ &\quad + \frac{1-\xi}{2} \left[\Pi_4(\eta) - \frac{1+\eta}{2} \Pi_4(1) - \frac{1-\eta}{2} \Pi_4(-1) \right] \\ &\quad + \frac{1+\xi}{2} \left[\Pi_2(\eta) - \frac{1+\eta}{2} \Pi_2(1) - \frac{1-\eta}{2} \Pi_2(-1) \right] \end{aligned}$$

Note that:

$$F_K \left(\begin{pmatrix} -1 \\ -1 \end{pmatrix} \right) = \Pi_1(-1), \quad F_K \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix} \right) = \Pi_3(1), \quad F_K \left(\begin{pmatrix} -1 \\ 1 \end{pmatrix} \right) = \Pi_3(-1), \quad F_K \left(\begin{pmatrix} 1 \\ -1 \end{pmatrix} \right) = \Pi_1(1)$$

Then, the SEM space will be:

$$V_h^p = \{v_h \in C^0(\bar{\Omega}) \quad st \quad v_h|_K = \hat{v}_0 \circ F_K^{-1}, \quad \hat{v} \in \mathbb{Q}^p(\hat{K}), \quad \forall K \in \mathcal{T}_h \quad v_h = 0 \text{ on } \partial\Omega\}$$

We choose the degree of the polynomials in the reference space. Anyway, note that as the map is not linear anymore, the approximation functions in the real space will not coincide with the reference ones (Unless we use parallelograms and another transformation).

8.6 Basis construction on $\hat{K}$

Define $\phi_0(\xi), \dots, \phi_p(\xi)$ the lagrangian functions associated with the GLL nodes $\xi_0, \dots, \xi_p$ respectively, i.e.:

$$\phi_j(\xi_i) \quad \forall i = 0, \dots, p \quad \forall j = 0, \dots, p$$

This is called Nodal Basis.

Now, the shape functions in 2D are defined as:

$$\phi_s(\xi, \eta) = \phi_i(\xi) \phi_j(\eta), \quad s = 0, \dots, (1+p)^d$$

Note that we can generate all the space:

$$\mathbb{Q}^p = \text{span}\{\phi_s(\xi, \eta) = \phi_i(\xi)\phi_j(\eta), \quad i, j = 0, \dots, p\}, \quad \text{where } \phi_i(x) = \begin{cases} 1 & \text{if } x = x_i \\ 0 & \text{if } x \neq x_i \end{cases} \quad i = 0, \dots, p$$

8.7 Problem Reconstruction

Finally, the discrete problem is:

$$\text{Find } u_h \in V_h^p \quad \text{st} \quad a(u_h, v_h) = F(v_h) \quad \forall v_h \in V_h^p$$

And note that $V_h^p \subseteq V = H_0^1(\Omega)$, so well-posedness is trivial since LM holds.

Now, as we have seen in FEM (section 4.2), the problem is equivalent to solve a linear system:

$$A \underline{u}_h = F$$

where:

- $A \in \mathbb{R}^{N_h \times N_h}$:

$$A_{s\ell} = a(\phi_s, \phi_\ell) = \int_{\Omega} \nabla \phi_s(x, y) \nabla \phi_\ell(x, y), \quad s, \ell = 1, \dots, N_h$$

- $F \in \mathbb{R}^{N_h}$:

$$F_s = \int_{\Omega} f(x, y) \phi_s(x, y)$$

8.8 Error Analysis

The SEM is a conforming method $V_h^p \subseteq V$, so well-posedness is inherited by the continuous problem.

Plus, from C  a Lemma, we also have:

$$\|u - u_h\|_V \lesssim \inf_{v_h \in V_h^p} \|u - v_h\|_V$$

So, to bound the error we only need a suitable interpolation estimation (Taking $v_h = \Pi_h^p u$ above, where Π_h^p is the Lagrangian interpolant).

Theorem 8.1: Interpolation Bounds

Let be $v \in H^{s+1}(\Omega)$, $s \geq 0$, then $\exists \Pi_h^p v$ an interpolant such that:

$$\|v - \Pi_h^p v\|_{H^1(\Omega)} \lesssim \frac{h^{\min(p, s)}}{p^s} \|v\|_{H^{s+1}(\Omega)}$$

$$\|v - \Pi_h^p v\|_{L^2(\Omega)} \lesssim \frac{h^{\min(p, s)+1}}{p^{s+1}} \|v\|_{H^{s+1}(\Omega)}$$

Remark.

Note that if $s \leq p$:

$$\|v - \Pi_h^p v\|_{H^1(\Omega)} \lesssim \left(\frac{h}{p}\right)^s \|v\|_{H^{s+1}(\Omega)}$$

and similarly with $L^2(\Omega)$

Convergence of SEM Let u be the exact solution of the problem. Assume that $u \in V \cap H^{s+1}(\Omega)$ for $s \geq 0$.

Let $u_h \in V_h^p$ its approximation with SEM, then:

$$\|u - u_h\|_V \lesssim \frac{h^{\min(s,p)}}{p^s} \|u\|_{H^{s+1}(\Omega)}$$

$$\|u - u_h\|_{L^2(\Omega)} \lesssim \frac{h^{\min(s,p)+1}}{p^{s+1}} \|u\|_{H^{s+1}(\Omega)}$$

Moreover, if u is analytic, then, for a constant γ dependent on u :

$$\|u - u_h\| \lesssim \exp(-\gamma p)$$

8.9 Spectral Element Methods with Numerical Integration (SEM-NI)

The idea of SEM-NI is to use the GLL nodes as quadrature points to compute the integrals of A and F :

$$\begin{aligned} a_h(\phi_j, \phi_i) &= (\nabla \phi_j, \nabla \phi_i)_h \\ F_h(\phi_i) &= (f, \phi_i)_h \end{aligned}$$

In practice,

- $a_h(v_h, z_h) = \sum_{q,r=0}^p \nabla v_h(x_q, y_r) \nabla z_h(x_q, y_r) w_q w_r$
- $F_h(v_h) = \sum_{q,r=0}^p f(x_q, y_r) v_h(x_q, y_r) w_q w_r$

where $w_q w_r$ is the product of 1D GLL weights.

Proposition 8.2

The GLL node weights are:

$$\begin{aligned} \hat{w}_\ell &= \frac{2}{p(p+1)} \frac{1}{[\mathcal{L}_p(\xi_\ell)]^2} & \ell = 1, \dots, p-1 \\ \hat{w}_\ell &= \frac{2}{p(p+1)} & \ell = 0, p \end{aligned}$$

Proposition 8.3

The degree of exactness of a 1D GLL quadrature rule with $p+1$ nodes is $2p-1$.

Proof. Proofs as exercise in both □

We can calculate specifically the mass-matrix M :

$$(M)_{ij} = (\phi_j, \phi_i)_h = \sum_{q,r=0}^p \nabla \phi_j(x_q, y_r) \nabla \phi_i(x_q, y_r) w_q w_r$$

And we note that this matrix is diagonal, since the values are 0 when $i \neq j$.

Remark.

As $\hat{\phi}_i, \hat{\phi}_j \in \mathbb{P}^p([-1, 1])$, $\hat{\phi}_i \hat{\phi}_j \in \mathbb{P}^{2p}$, and it can be proved that the error in the following integral is small:

$$\int_{-1}^1 \hat{\phi}_i(\xi) \hat{\phi}_j(\xi) d\xi \approx \sum_{\ell=0}^p \hat{\phi}_i(\xi) \hat{\phi}_j(\xi) \hat{w}_\ell$$

Plus, $\hat{\phi}'_i(\xi) \hat{\phi}'_j(\xi) \in \mathbb{P}^{2p-2}([-1, 1])$, and the following integral is computed exactly:

$$\int_{-1}^1 \hat{\phi}'_i(\xi) \hat{\phi}'_j(\xi) d\xi \approx \sum_{\ell=0}^p \hat{\phi}'_i(\xi) \hat{\phi}'_j(\xi) \hat{w}_\ell$$

Therefore, note that the stiffness matrix has a very small error, and the mass matrix is calculated exactly.

On the other hand:

Proposition 8.4

Let u be the solution of the weak formulation and u_h the approximate solution with the SEM-NI. Assuming $u \in H^{s+1}(\Omega)$, then:

$$\|u - u_h\|_V \leq C_s \frac{h^{\min(s,p)}}{p^s} (\|f\|_{H^s(\Omega)} + \|u\|_{H^{s+1}(\Omega)})$$

8.10 Strang's Lemma

Let $u \in V$ and $u_h \in V_h$ be the solutions of:

$$\begin{aligned} u \in V \quad st \quad a(u, v) &= F(v) & \forall v \in V \\ u_h \in V_h \quad st \quad a(u_h, v_h) &= F(v_h) & \forall v_h \in V_h^p \end{aligned}$$

Note that for the first problem, Lax-Milgram conditions hold. For the second one, it can be proven that the conditions hold as well.

Definition 8.5: Uniformly Coercivity

Let $\{a_h\}_h$ be a family of bilinear forms. $\{a_h\}_h$ is uniformly coercive in V_h^p if:

$$\exists \tilde{a} \quad \text{st} \quad \tilde{a}(v, v) \geq \tilde{a} \|v\|_V^2 \quad \forall v \in V_h^p$$

with $\tilde{a}$ independent of h and p

Lemma 8.6: Strang's Lemma

Suppose that the SEM problem satisfies the LM theorem (see theorem 3.6). Plus, suppose $\{a_h(\cdot, \cdot)\}_h$ is uniformly coercive in V_h^p with constant $\tilde{a}$. Then $\implies \|u - u_h\|_V \leq$:

$$\inf_{v_h \in V_h} \left\{ \left(1 + \frac{M}{\tilde{a}} \right) \|u - v_h\|_V + \frac{1}{\tilde{a}} \sup_{w_h \in V_h^p \setminus \{0\}} \frac{|a(v_h, w_h) - a_h(v_h, w_h)|}{\|w_h\|_V} \right\} +$$

$$\frac{1}{\tilde{a}} \sup_{w_h \in V_h^p \setminus \{0\}} \frac{F(w_h) - F_h(w_h)}{\|w_h\|_V}$$

This is a generalization of C a Lemma

8.11 SEM on Triangular Grids

The ideas of SEM-NI were constructed in quads. Anyway, constructing conforming meshes with triangles is much more complicated than with quads, especially in 3D.

If we want to use SEM on triangular grids, it is not possible to preserve simultaneously nodal basis functions (Lagrangian type) and tensorial structure, so, we can:

1. Preserve Nodal Basis and lose tensorization
2. Preserve tensorization and lose nodal basis

8.11.1 Triangular SEM with Nodal Basis

Given $p \geq 1$, choose a suitable set of interpolation points $\{(\xi_i, \eta_i)\}_i$ for the space $\mathbb{P}^p(\hat{T})$. As those points must be placed in the boundaries of the reference triangle with the 1D GLL, it is not as easy as taking the tensorial product between two sides and having the degrees of freedom. Now, we must find another method to place the points.

There are two known alternatives, which are not known explicitly, but are computable by suitable algorithms:

- Electrostatic Points (1998, Hesthaven)
- Fekete Points (1983, Bos), (1995, Chen & Babuska), (2000, Taylor et.al.)

It can be proven that Lagrange Polynomials have an explicit form in $[-1,1]$, allowing us to compute derivatives efficiently. Anyway, there's no closed-form expression for the Lagrange Polynomials associated with an arbitrary set of points in $\hat{T}$.

To cope with this, we can express the Lagrange polynomials in terms of another polynomial basis (e.g., the orthogonal modal basis or Dubiner basis), and apply a change of basis. The basis change matrix V is known as the generalized Vandermonde Matrix.

8.11.2 Triangular SEM with Modal Basis

Another option could be to preserve tensorization and use a modal basis.

Dubiner came up with a method to introduce high-degree polynomial bases on triangles, later extended to 3D case by Karniadakis and Sherwin, but being much more costly.

The Dubiner Shape functions consists of building a “collapse function” F from the reference quad to the reference triangle $F : \hat{Q} \rightarrow \hat{T}$, being:

$$\begin{aligned}\hat{Q} &= \{(\hat{\xi}, \hat{\eta}) \quad st \quad -1 \leq \hat{\xi}, \hat{\eta} \leq 1\} \\ \hat{T} &= \{(\hat{\xi}, \hat{\eta}) \quad st \quad -1 \leq \hat{\xi}, \hat{\eta} \leq 0 \quad \wedge \quad \xi + \eta \leq 0\}\end{aligned}$$

The map is:

$$F \left(\begin{bmatrix} \hat{\xi} \\ \hat{\eta} \end{bmatrix} \right) = \begin{bmatrix} \frac{(1+\hat{\xi})(1+\hat{\eta})}{2} - 1 \\ \hat{\eta} \end{bmatrix} = \begin{bmatrix} \xi \\ \eta \end{bmatrix}$$

9 Numerical Methods for the Heat Equation

9.1 The Problem

The modeled problem describes the evolution of the temperature $u(x,t)$ at point x and time t in a heat-conducting medium.

The thermal conductivity is set to 1 in the simplest formulation, and the temperature on the boundary is kept at a constant zero temperature. u_0 describes the initial temperature, while f is the volumetric heat source.

The problem of the Heat Equation is:

$$\text{Find } u = u(x, t) \quad st \quad \begin{cases} \frac{\partial u}{\partial t} - \Delta u = f(x, t) & \forall x \in \Omega, \quad \forall t \in (0, T) \\ u(x, t) = g_D(x, t) & x \in \Gamma_D, \quad \forall t > 0 \\ \frac{\partial u(x, t)}{\partial n} = g_N(x, t) & x \in \Gamma_N, \quad \forall t > 0 \\ u(x, 0) = u_0(x) & \text{at } t = 0 \quad \forall x \in \Omega \end{cases}$$

- The first condition is the heat equation.
- The second and third ones are the boundary conditions, with g_D and g_N regular enough.
- The fourth one is the initial conditions.

Definition 9.1: Q_T

We call $Q_T = \Omega \times (0, T)$.

If $T = +\infty \implies Q = \{(x, t) \mid x \in \Omega, t > 0\}$

The heat equation, with its variations, is bounded to diverse domains of science and applied mathematics:

- Model heat transfer and diffusion processes.
- In probability theory, it is connected with the study of random walks and Brownian motion through the Fokker-Planck equation.
- In finance, the Black-Scholes equation is a slight adaptation of the heat equation.
- In quantum mechanics, the Schrödinger equation can be interpreted as a heat equation.

9.2 Weak Formulation

Setting $V = H_{\Gamma_D}^1(\Omega)$, and assuming f is a function of only t . We seek $u(t) \in V$ such that for each $t > 0$:

$$\int_{\Omega} \frac{\partial u}{\partial t} v - \int_{\Omega} \Delta u v = \int_{\Omega} f v \quad \forall v \in V$$

Integrating by parts and assuming $u = 0$ on $\partial\Omega$

$$\int_{\Omega} \frac{\partial u}{\partial t} v \, d\Omega + a(u(t), v) = \int_{\Omega} f(t) v \, d\Omega \quad \forall v \in V \quad (WFH)$$

We must solve this, taking into account also the initial conditions:

$$u(x, 0) = u_0(x) \quad \forall x \in \Omega$$

9.3 Well-Posedness**Definition 9.2: Weakly Coercive**

A bilinear form $a(\cdot, \cdot)$ is said weakly coercive if:

$$\exists \lambda \geq 0, \quad \exists \alpha > 0 \quad \text{st} \quad a(v, v) + \lambda \|v\|_{L^2(\Omega)}^2 \geq \alpha \|v\|_V^2 \quad \forall v \in V$$

Remark.

If $\lambda = 0$, it becomes the standard definition of coercivity.

Theorem 9.3

Let be $f \in L^2(Q_T)$, $u_0 \in L^2(\Omega)$. If the bilinear form $a(\cdot, \cdot)$ is continuous and weakly coercive, then (WFH) is well posed.

9.4 Discretization

The procedure to solve this problem consists of two steps:

1. Semi-discrete Formulation: Discretize in the space.
2. Fully-Discrete Formulation: Discretize in time.

9.5 Semidiscrete Discretization

Let $\mathcal{T}_h$ be a mesh (e.g. made by triangles). Let

$$V_h = \{v_h \in C^0(\bar{\Omega}) \quad st \quad v_h|_K \in \mathbb{P}^r(K) \quad \forall K \in \mathcal{T}_h, \quad v_h = 0 \text{ on } \partial\Omega\} \subseteq V$$

So the weak formulation can be approximated with: Find $u_h(t) \in V_h$ such that:

$$\int_{\Omega} \frac{\partial u_h}{\partial t} v_h \, d\Omega + a(u_h, v_h) = \int_{\Omega} f(t) v_h \, d\Omega \quad \forall v_h \in V \quad (SDF)$$

Subject to $u_h(0) = u_{0,h}$, a suitable approximation of the initial condition $u(0) = u_0$.

Remark.

- Any method we have seen can be employed. E.g. SEM or DG

Lemma 9.4

(SDF) is a system of Ordinary Differential Equations (ODEs)

Proof. Let N_h be the dimension of V_h , and let $\{\phi_1(x), \dots, \phi_{N_h}(x)\}$ a basis for the discrete space V_h , i.e.:

$$V_h = \text{span}\{\phi_1(x), \dots, \phi_{N_h}(x)\}$$

Then, there exist $u_1(t), \dots, u_{N_h}(t)$ such that:

$$u_h(x, t) = \sum_{j=1}^{N_h} u_j(t) \phi_j(x)$$

Then, we can substitute this into (SDF), to get:

$$\int_{\Omega} \sum_{j=1}^{N_h} \dot{u}_j(t) \phi_j(x) v_h + a\left(\sum_{j=1}^{N_h} u_j(t) \phi_j(x), v_h\right) = F(v_h) \quad \forall v_h \in V_h$$

where $\dot{u}_j(t)$ is the partial derivative of u_j wrt t .

Then, using the linearity:

$$\sum_{j=1}^{N_h} \dot{u}_j(t) \int_{\Omega} \phi_j(x) v_h + \sum_{j=1}^{N_h} u_j(t) a(\phi_j(x), v_h) = F(v_h) \quad \forall v_h \in V_h$$

Finally, it is only necessary to test for any ϕ_i to satisfy the above equation, i.e.:

$$\sum_{j=1}^{N_h} \dot{u}_j(t) \int_{\Omega} \phi_j(x) \phi_i(x) + \sum_{j=1}^{N_h} u_j(t) a(\phi_j(x), \phi_i(x)) = F(\phi_i(x)) \quad \forall i, j = 1, \dots, N_h$$

Now, define:

- The mass matrix $M \in \mathbb{R}^{N_h \times N_h}$ as:

$$\{M\}_{ij} = m_{ij} = \int_{\Omega} \phi_j(x) \phi_i(x) \quad \forall i, j = 1, \dots, N_h$$

- $A \in \mathbb{R}^{N_h \times N_h}$ as:

$$\{A\}_{ij} = a(\phi_j, \phi_i) = \int_{\Omega} \nabla \phi_j \cdot \nabla \phi_i$$

- $\underline{u}(t) \in \mathbb{R}^{N_1 \times 1}$ as:

$$\{\underline{u}(t)\}_j = u_j(t) \quad \forall j = 1, \dots, N_h$$

- $\underline{F}(t) \in \mathbb{R}^{N_1 \times 1}$ as:

$$\{\underline{F}(t)\}_j = \int_{\Omega} f(x, t) \phi_i(x) \quad \forall i = 1, \dots, N_h$$

And with this notation, the problem becomes: $\forall t \in (0, T)$, find $\underline{u}(t) \in \mathbb{R}^{N_h \times 1}$ such that:

$$\begin{cases} M \dot{\underline{u}}(t) + A \underline{u}(t) = \underline{F}(t) \\ \underline{u}(0) = \underline{u}_{0h} \end{cases} \quad (P3)$$

Where $\underline{u}_{0h} \in \mathbb{R}^{N_h}$ that contains the expansion coefficients of the initial datum in the chosen basis. $\square$

Remark.

(P3) is a coupled system of N_h ODEs. This is hard to solve.

- If we choose SEN-IN, M is diagonal, so ODEs are decoupled.
- If we choose DG, M is block diagonal, so ODEs are partially decoupled.
- If we choose FEM, M is fully coupled.

AS MORE DECOUPLE, IT IS MORE EASILY PARALLELIZABLE, WHICH IS A BIG COMPUTATIONAL ADVANTAGE.

9.6 Fully Discrete Formulation

We have already discretized the problem across the space. It is missing the discretization in time, that is done as follows:

Fix $\Delta t > 0$, and define snapshots as:

$$t^k = t_0 + k \cdot \Delta t \quad k = 0, 1, 2, \dots$$

Now, for any k , compute:

$$\underline{u}^k \simeq \underline{u}(t^k) \quad \forall k = 1, 2, 3, \dots$$

with $\underline{u}^0$ known by the initial datum.

Now, to obtain the fully discrete approximation, the θ -methods can be used:

For $\theta \in [0, 1]$, and given $\underline{u}^{(0)}$, for any $k = 1, 2, \dots$:

$$M \left[\frac{\underline{u}^{k+1} - \underline{u}^k}{\Delta t} \right] + A[\theta \underline{u}^{k+1} + (1 - \theta) \underline{u}^k] = \theta \underline{F}^{k+1} + (1 - \theta) \underline{F}^k$$

where $\underline{F}^k = \underline{F}(t^k)$. Anyway, some specific versions are the most used:

- Let be $\theta = 0$, the method becomes the explicit euler:

$$\frac{M}{\Delta t} \underline{u}^{k+1} + \left(A - \frac{M}{\Delta t} \right) \underline{u}^k = \underline{F}^k \quad \forall k = 1, 2, 3, \dots$$

This is the cheapest method. It is conditionally stable, the delta time must be sufficiently small to be stable ($\Delta t \leq \text{constant} \cdot h^2$). It converges in order 1 (Δt).

- Let be $\theta = 1$, the method becomes the implicit euler:

$$\left(\frac{M}{\Delta t} + A \right) \underline{u}^{k+1} - \frac{M}{\Delta t} \underline{u}^k = \underline{F}^k \quad \forall k = 1, 2, 3, \dots$$

It is an expensive method. This is unconditionally stable. It converges in order 1 (Δt).

- Let be $\theta = 1/2$, the method becomes the Crank-Nicolson method:

$$\left(\frac{M}{\Delta t} + \frac{1}{2} A \right) \underline{u}^{k+1} - \left(\frac{1}{2} A - \frac{M}{\Delta t} \right) \underline{u}^k = \frac{1}{2} (\underline{F}^{k+1} + \underline{F}^k) \quad \forall k = 1, 2, 3, \dots$$

It is an expensive method. This is unconditionally stable. It converges in order 2 ($(\Delta t)^2$).

Remark.

The accuracy of the methods is on the same order, what changes is the computation cost and stability.

9.7 Stability of Continuous Problem

The weak formulation is:

$$\forall t \in (0, T) \quad \text{find } u = u(x, t) \in V \quad \text{st} \quad \begin{cases} \int_{\Omega} \frac{\partial u}{\partial t} v + a(u, v) = \int_{\Omega} f(x, t) v & \forall v \in V \equiv H_0^1(\Omega) \\ u(x, 0) = u_0(x) & \forall x \in \Omega, \quad t = 0 \end{cases}$$

Theorem 9.5

Suppose that the data $(f(x, t), u_0(x))$ are regular enough. Then:

$$\|u(t)\|_{L^2(\Omega)}^2 + \alpha \int_0^t \|\nabla u(s)\|_{L^2(\Omega)}^2 ds \leq \|u_0\|_{L^2(\Omega)}^2 + \frac{C_{\Omega}^2}{\alpha} \int_0^t \|f(s)\|_{L^2(\Omega)}^2 ds$$

Where C_{Ω} is the poincarè constant, and α is the coercivity constant of $a(\cdot, \cdot)$

Proof. Fix $t \in (0, T)$ and take $v = u$ in the weak formulation:

$$\underbrace{\int_{\Omega} \frac{\partial u}{\partial t} u}_{(A)} + \underbrace{a(u, u)}_{(B)} = \underbrace{\int_{\Omega} f u}_{(C)}$$

Note that:

$$\begin{aligned} (A) &= \frac{1}{2} \frac{\partial}{\partial t} \left(\int_{\Omega} |u|^2 \right) = \frac{1}{2} \frac{\partial}{\partial t} \left(\|u\|_{L^2(\Omega)}^2 \right) \\ (B) &\geq \alpha \|u\|_{L^2(\Omega)}^2 && \text{(By Coercivity)} \\ (C) &\leq \|f\|_{L^2(\Omega)} \|u\|_{L^2(\Omega)} && \text{(Cauchy-Schwarz)} \\ &\leq C_{\Omega} \|f\|_{L^2(\Omega)} \|\nabla u\|_{L^2(\Omega)} && \text{(By Poincarè)} \\ &\leq \frac{C_{\Omega}^2}{\epsilon} \|f\|_{L^2(\Omega)}^2 + \frac{\epsilon}{4} \|\nabla u\|_{L^2(\Omega)}^2 && \text{(By Young's inequality)} \end{aligned}$$

Young's ineq: $ab \leq \epsilon a^2 + \frac{1}{4\epsilon} b^2$

So this implies:

$$\begin{aligned} \frac{1}{2} \frac{\partial}{\partial t} \left(\|u\|_{L^2(\Omega)}^2 \right) + \alpha \|\nabla u\|_{L^2(\Omega)}^2 &\leq \frac{C_{\Omega}^2}{4\epsilon} \|f\|_{L^2(\Omega)}^2 + \epsilon \|\nabla u\|_{L^2(\Omega)}^2 && \forall t \\ \frac{1}{2} \frac{\partial}{\partial t} \left(\|u\|_{L^2(\Omega)}^2 \right) + (\alpha - \epsilon) \|\nabla u\|_{L^2(\Omega)}^2 &\leq \frac{C_{\Omega}^2}{4\epsilon} \|f\|_{L^2(\Omega)}^2 && \forall t \end{aligned}$$

And taking $\epsilon = \alpha/2$:

$$\frac{1}{2} \frac{\partial}{\partial t} \left(\|u\|_{L^2(\Omega)}^2 \right) + \frac{\alpha}{2} \|\nabla u\|_{L^2(\Omega)}^2 \leq \frac{C_{\Omega}^2}{2\alpha} \|f\|_{L^2(\Omega)}^2 \quad \forall t$$

Now, integrating in time (noticing that the elements are all positive):

$$\|u(t)\|_{L^2(\Omega)}^2 - \|u_0\|_{L^2(\Omega)}^2 + 2 \frac{\alpha}{2} \int_0^t \|\nabla u(s)\|_{L^2(\Omega)}^2 ds \leq \int_0^t \frac{C_{\Omega}^2}{\alpha} \|f(s)\|_{L^2(\Omega)}^2 ds$$

□

Plus, it can be proven:

$$\|u(t)\|_{L^2(\Omega)} \leq \|u_0(t)\|_{L^2(\Omega)} + \int_0^t \|f(s)\|_{L^2(\Omega)} ds$$

9.8 Semidiscrete Problem: Stability and Convergence Analysis

Now, assume the data is regular enough, and let u_h be the solution of the semidiscrete problem. Then, similarly to what was done in the continuous setting (see section 9.7), it can be proven that:

$$\|u_h(t)\|_{L^2(\Omega)}^2 + \alpha \int_0^t \|\nabla u_h(s)\|_{L^2(\Omega)}^2 ds \leq \|u_{0h}\|_{L^2(\Omega)}^2 + \frac{C_\Omega^2}{\alpha} \int_0^t \|f(s)\|_{L^2(\Omega)}^2 ds \quad \forall t$$

where the initial data is $u_h(0) = u_{0h}$.

And then:

$$\|u_h(t)\|_{L^2(\Omega)} \leq \|u_{0h}(t)\|_{L^2(\Omega)} + \frac{C_\Omega^2}{\alpha} \int_0^t \|f(s)\|_{L^2(\Omega)} ds$$

On the other hand:

Theorem 9.6: Convergence of Semidiscrete Scheme

Suppose that the exact solution of the problem is regular enough. Let $u_h(t)$ be the semidiscrete solution. Then, for any time t :

$$\left(\|u(t) - u_h(t)\|_{L^2(\Omega)}^2 + \int_0^t \|\nabla u(s) - \nabla u_h(s)\|_{L^2(\Omega)}^2 ds \right)^{1/2} \lesssim h^r$$

Or more specifically:

$$\begin{aligned} & \left(\|u(t) - u_h(t)\|_{L^2(\Omega)}^2 + \alpha \int_0^t \|\nabla u(s) - \nabla u_h(s)\|_{L^2(\Omega)}^2 ds \right)^{1/2} \\ & \leq Ch^r \left(\|u_0\|_{H^r(\Omega)}^2 + \int_0^t \|u(s)\|_{H^{s+1}(\Omega)}^2 ds + \int_0^t \left\| \frac{\partial u(s)}{\partial s} \right\|_{H^{r+1}(\Omega)}^2 ds \right)^{1/2} \end{aligned}$$

Where r is the polynomial degree in the FE approximation.

9.9 Stability for Fully Discrete Scheme

We can use the θ -method, applied to the (SDF) problem (with $f \equiv 0$ for simplicity). We get:

$$\int_\Omega \left(\frac{u_h^{k+1} - u_h^k}{\Delta t} \right) v_h + a(\theta u_h^{k+1} + (1-\theta)u_h^k, v_h) = 0 \quad \forall v_h \in V_h$$

Take, for example, $\theta = 1$, so we have the implicit euler problem:

$$\int_{\Omega} \left(\frac{u_h^{k+1} - u_h^k}{\Delta t} \right) v_h + a(u_h^{k+1}, v_h) = 0 \quad \forall v_h \in V_h$$

To prove stability with $\theta = 1$, take $v_h = u_h^{k+1}$, obtaining:

$$\frac{1}{\Delta t} \|u_h^{k+1}\|_{L^2(\Omega)}^2 - \frac{1}{\Delta t} \int_{\Omega} u_h^k u_h^{k+1} + a(u_h^{k+1}, u_h^{k+1}) = 0$$

Then:

$$\|u_h^{k+1}\|_{L^2(\Omega)}^2 + \underbrace{\Delta t \cdot a(u_h^{k+1}, u_h^{k+1})}_{(A)} = \underbrace{\int_{\Omega} u_h^k u_h^{k+1}}_{(B)}$$

Note that:

$$(A) \geq \Delta t \alpha \|u_h^{k+1}\|_V^2 \quad (\text{Coercivity})$$

$$(B) \leq \frac{1}{2} \|u_h^k\|_{L^2(\Omega)}^2 + \frac{1}{2} \|u_h^{k+1}\|_{L^2(\Omega)}^2$$

Then:

$$\|u_h^{k+1}\|_{L^2(\Omega)}^2 + \Delta t \alpha \|u_h^{k+1}\|_V^2 \leq \frac{1}{2} \|u_h^k\|_{L^2(\Omega)}^2 + \frac{1}{2} \|u_h^{k+1}\|_{L^2(\Omega)}^2$$

So:

$$\|u_h^{k+1}\|_{L^2(\Omega)}^2 + 2\Delta t \alpha \|u_h^{k+1}\|_V^2 \leq \|u_h^k\|_{L^2(\Omega)}^2 \quad \forall k = 0, 1, 2, \dots$$

And as $\|u_h^{k+1}\|_V \geq \|u_h^{k+1}\|_{L^2(\Omega)}$:

$$(1 + 2\alpha \cdot \Delta t) \|u_h^{k+1}\|_{L^2(\Omega)}^2 \leq \|u_h^k\|_{L^2(\Omega)}^2 \quad \forall k = 0, 1, 2, \dots$$

And:

$$\|u_h^{k+1}\|_{L^2(\Omega)}^2 \leq \frac{1}{1 + 2\alpha \cdot \Delta t} \|u_h^k\|_{L^2(\Omega)}^2 \quad \forall k = 0, 1, 2, \dots$$

So, inductively:

$$\|u_h^k\|_{L^2(\Omega)}^2 \leq \left(\frac{1}{1 + 2\alpha \cdot \Delta t} \right)^k \|u_{0h}\|_{L^2(\Omega)}^2 \quad \forall k = 0, 1, 2, \dots$$

Therefore, u_h^k is unconditionally absolutely stable for every Δt :

$$\lim_{k \rightarrow \infty} \|u_h^k\|_{L^2(\Omega)} = 0$$

More generally, it can be proven that:

- If $\theta \geq 1/2$, the θ -method is unconditionally absolutely stable.
- If $\theta < 1/2$, the θ -method is absolutely stable only for:

$$\Delta t \leq \frac{2}{(1 - 2\theta)\lambda_h^{N_h}}$$

Where $\lambda_h^{N_h}$ is the largest eigenvalue of the generalized eigenvalue problem: $Ax = \lambda Mx$.

It can be shown that $\lambda_h^{N_h} \lesssim h^{-2}$.

Then, if $\theta < 1/2$, Δt is bound to the choice of h .

9.10 Error of Fully Discrete Model

Theorem 9.7: Error Bounds for the Fully Discrete Scheme

Assume that u_0 , f , and the exact solution u are regular enough, then:

$$\|u(t^n) - u_h^n\|_{L^2(\Omega)}^2 + 2\alpha \Delta t \sum_{k=1}^n \|u(t^k) - u_h^k\|_V^2 \leq C(u_0, f, u)(h^{2r} + \Delta t^{p(\theta)}) \quad \forall n = 0, 1, \dots$$

Where:

$$p(\theta) = \begin{cases} 4 & \theta = \frac{1}{2} \\ 2 & \theta \neq \frac{1}{2} \end{cases}$$

Remark.

What is seen is that at the beginning the method converges in h^{2r} , but as the method continues, this term becomes small, and the convergence is dominated by $\Delta t^{p(\theta)}$

10 Advection-Diffusion-Reaction (ADR) problems

10.1 The Problem

The ADR problems have two versions. The conservative form:

$$\text{Find } u(x) \quad \text{st} \quad \begin{cases} -\text{div}(\mu \nabla u) + \text{div}(\underline{b}u) + \sigma u = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

And non-conservative form:

$$\text{Find } u(x) \quad \text{st} \quad \begin{cases} -\text{div}(\mu \nabla u) + \underline{b} \cdot \nabla u + \sigma u = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

Under the assumptions:

- $\mu \in L^\infty(\Omega)$, $\mu(x) \geq \mu_0 > 0$ a.e.
- $\underline{b} \in [L^\infty]^d$
- $\sigma \in L^\infty(\Omega)$
- $f \in L^\infty(\Omega)$

Remark.

If $\underline{b}$ is constant, the two forms are equivalent.

The first term is called the diffusion term, the second is the advection term, and the third is the reaction term.

Remark.

If the advection term dominates, the formal PDE is a second-order equation but behaves like a first-order one.

The typical problems solved with these kinds of equations have sharp interior shapes.

10.2 Weak Formulation

Find $u \in H_0^1(\Omega)$ st $a(u, v) = F(v) \quad \forall v \in V$ Where:

- $F(v) = \int_{\Omega} f v$
- $a(u, v) = \begin{cases} \int_{\Omega} (\mu \nabla u - \underline{b} u) \cdot \nabla v + \int_{\Omega} \sigma u v & \text{If conservative form} \\ \int_{\Omega} \mu \nabla u \nabla v + \int_{\Omega} \underline{b} \cdot \nabla u \cdot v + \int_{\Omega} \sigma u v & \text{If non-conservative form} \end{cases}$

10.3 Lax-Milgram Hypotheses

It can be proven that coercivity holds:

- Conservative case:

$$\sigma + \frac{1}{2} \operatorname{div}(\underline{b}) \geq 0 \quad \text{in } \Omega$$

- Non-conservative case:

$$\sigma - \frac{1}{2} \operatorname{div}(\underline{b}) \geq 0 \quad \text{in } \Omega$$

In both cases, $a(u, u) \geq \mu_0 \|\nabla u\|^2$, i.e., coercivity constant $\alpha \simeq \mu_0$.

On the other hand, the continuity constant:

$$M \simeq \|\mu\|_{L^\infty} + \|\underline{b}\|_{L^\infty} + \|\sigma\|_{L^\infty}$$

Remark.

As the continuity constant may be higher, it could lead to problems in Cea Lemma, paying a high price ($\frac{M}{\alpha}$) to bound the solution.

10.4 Problematic Example

Consider the following ADR equation:

$$\begin{cases} -\mu\Delta u + \underline{b} \cdot \nabla u = f & \text{in } \Omega = (0, 1) \times (0, 1) \\ u = g & \text{on } \partial\Omega \end{cases}$$

with $\underline{b} = (-1, 1)^T$.

Now, we take an f that forces the exact solution u to be:

$$u(x, y) = x + y(1 - x) + \frac{e^{-1/\mu - e^{-(1-x)(1-y)/\mu}}}{1 - e^{-1/\mu}}$$

Note that for small values of viscosity μ , this solution features a boundary layer (region where the solution changes very rapidly) near the edges $x = 1, y = 1$. This leads to numerical instability, which, in theory, is allowed by a large C  a Lemma constant; however, the approximations in these cases are very poor. (see figure 3).

We can note here that when the advection term dominates the equation, the Galerkin Method becomes useless.

10.5 Source of Oscillations

$$\|u - u_h\|_V \stackrel{Cea}{\leq} \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V \stackrel{\text{Interpolation error}}{\leq} C \frac{M}{\alpha} h^r |u|_{H^{r+1}(\Omega)}$$

Theoretically, the error goes to 0, but in real life, there's no real $h \rightarrow 0$, but fixed.

So, if the equation is convection or reaction dominated, then $M/\alpha \gg 1$, and then there is a trade-off between M/α and h^r . Specifically, we can define:

Definition 10.1: Numerical Peclet Number

$$\mathbb{P}_{e_h} = h \frac{M}{\alpha}$$

And $\mathbb{P}_{e_h}$ should always be less than 1 for stability.

10.6 Stabilize FEM

To stabilize the finite element method in these cases, some diffusion terms can be added to avoid the oscillations. The idea in the 1D case, when we have:

$$-\mu u'' + \beta u'$$

is to add a little diffusion term, so it becomes:

$$-\mu u'' + \beta u' - \mu_\ell u''$$

where μ_ℓ is called artificial viscosity. This method will help maintain stability without altering the converged solution.

There are two ways to perform this stabilization:

- The artificial diffusion, where we add:

$$c(h) \int_{\Omega} \nabla u \cdot \nabla v$$

So we replace $a(\cdot, \cdot)$ with:

$$\begin{aligned} a_h(u_h, v_h) &= \int_{\Omega} \mu \nabla u_h \nabla v_h + \int_{\Omega} \underline{b} \nabla u_h v_h + \int_{\Omega} \sigma u_h v_h + c(h) \int_{\Omega} \nabla u_h \nabla v_h \\ &= \int_{\Omega} (\mu + c(h)) \nabla u_h \nabla v_h + \int_{\Omega} \underline{b} \nabla u_h v_h + \int_{\Omega} \sigma u_h v_h \end{aligned}$$

The problem now becomes in choosing $c(h)$ correctly. It must tend to 0 without affecting convergence.

- The Streamline Diffusion consists of adding a diffusion term, but only in the $\underline{b}$ direction:

$$\frac{c(h)}{\|\underline{b}\|_{L^\infty}} \int_{\Omega} (\underline{b} \cdot u_h)(\underline{b} \cdot v_h)$$

So A becomes such that:

$$a_h(u_h, v_h) = \int_{\Omega} \mu \nabla u_h \nabla v_h + \int_{\Omega} \underline{b} \nabla u_h v_h + \int_{\Omega} \sigma u_h v_h + \frac{c(h)}{\|\underline{b}\|_{L^\infty}} \int_{\Omega} (\underline{b} \cdot u_h)(\underline{b} \cdot v_h)$$

10.7 Choosing $c(h)$

The original problem has changed to:

$$\text{Find } u_h \in V_h \quad \text{st} \quad a(u_h, v_h) + \mathcal{L}(u_h, f; v_h) = F(v_h) \quad \forall v_h \in V_h$$

where we would like $\mathcal{L}(u_h, f; v_h)$ to depend on the residual, ideally:

$$\mathcal{L}(u_h, f; v_h) = \int_{\Omega} (\mathcal{L}u_h - f)v_h$$

where $\mathcal{L}$ is the differential operator.

But in practice, as we have smoothness only inside K:

$$\mathcal{L}(u_h, f; v_h) = \sum_{K \in \mathcal{T}_h} \int_K (\mathcal{L}u_h - f) \tau_K \phi(v_h)$$

where ϕ can be chosen and τ_K is a parameter acting as scaling factor.

Specifically:

$$\forall K \in \mathcal{T}_h \quad \tau_K = \delta \frac{h_K}{\|\underline{b}(x)\|_{L^\infty}}, \quad h_k = \text{diam}(K)$$

where $\delta > 0$ and usually $\delta = \delta(p)$

10.8 Choosing $\phi(\cdot)$

To choose $\phi(v_h)$, the other term in artificial diffusion there are different possibilities:

- Galerkin Least Squares (GLS): Applying differential operator on v_h

$$\phi(v_h) = \mathcal{L}v_h$$

- Streamline Upwind Petrov Galerkin (SUPG): Taking only skew-symmetric part of $\mathcal{L}$

$$\phi(v_h) = \mathcal{L}_{ss}v_h$$

i.e., for a matrix A :

$$A = A_s + A_{ss}$$

where:

$$A_s = \frac{1}{2}(A + A^T) \quad \wedge \quad A_{ss} = \frac{1}{2}(A - A^T)$$

Example.

Let $\mathcal{L}u = -\mu\Delta u + \underline{b} \cdot \nabla u + \sigma u$. Write $\mathcal{L}_s$ and $\mathcal{L}_{ss}$.

It can be proven that

$$\begin{aligned} \mathcal{L} &= \left[\mu\Delta u + \sigma u - \frac{1}{2} \operatorname{div}(\underline{b})u \right] + \left[\underline{b} \cdot \nabla u + \frac{1}{2} \operatorname{div}(\underline{b}u) \right] \\ &= \mathcal{L}_s + \mathcal{L}_{ss} \end{aligned}$$

Exercise: Prove this, and write the laplacian for $\mathcal{L}u = -\mu\Delta u + \operatorname{div}(\underline{b}u) + \sigma u$

10.9 GLS formulation

If we introduce GLS, the formulation of the problem becomes:

$$\text{Find } u_h \in V_h \quad \text{st} \quad \underbrace{a(u_h, v_h) + \sum_{K \in \mathcal{T}_h} \int_K \tau_K \mathcal{L}u_h \mathcal{L}v_h}_{a_h(u_h, v_h)} = \underbrace{\int_{\Omega} f v_h + \sum_{K \in \mathcal{T}_h} \int_K \tau_K f \mathcal{L}v_h}_{F_h(v_h)} \quad \forall v_h \in V_h$$

Example.

If $r = 1$, $\Delta v_h = 0$, so:

$$\mathcal{L}v_h = \underline{b} \cdot \nabla v_h$$

Then:

$$\sum_{K \in \mathcal{T}_h} \int_K \tau_K \mathcal{L}u_h \mathcal{L}v_h = \sum_{K \in \mathcal{T}_h} \int_K \tau_K (\underline{b} \cdot \nabla u_h) (\underline{b} \cdot \nabla v_h)$$

10.10 GLS stability

Theorem 10.2: GLS stability for conservative form

Suppose that $\gamma := \sigma + \frac{1}{2}\text{div}(\underline{b})$. assume that:

$$\exists \gamma_1, \gamma_2 \quad \text{st} \quad 0 < \gamma_1 \leq \gamma \leq \gamma_2$$

Then, the solution u_h of the GLS formulation satisfies:

$$\|u_h\|_{GLS} \leq \|f\|_{L^2(\Omega)}$$

where $\|\cdot\|_{GLS}$ to be defined (as usual with reverse engineering).

Now, we must construct $\|\cdot\|_{GLS}$. Note that:

$$a_h(u_h, u_h) = \int_{\Omega} \mu |\nabla u_h|^2 + \int_{\Omega} \sigma u_h^2 + \int_{\Omega} \text{div}(\underline{b}u_h)u_h + \sum_{K \in \mathcal{T}_h} \int_K \tau_K (\mathcal{L}u_h)^2$$

A priori, the third term may not be positive, so it could be problematic for defining the norm.

Now, observe that:

$$\begin{aligned} \int_{\Omega} \text{div}(\underline{b}u_h)u_h &= \int_{\Omega} \text{div}(\underline{b})u_h^2 + \int_{\Omega} \underline{b} \nabla u_h \cdot u_h \\ &= \int_{\Omega} \text{div}(\underline{b})u_h^2 + \frac{1}{2} \int_{\Omega} \underline{b} \nabla (u_h^2) \end{aligned}$$

So:

$$\begin{aligned} a_h(u_h, u_h) &= \int_{\Omega} \mu |\nabla u_h|^2 + \int_{\Omega} \sigma u_h^2 + \int_{\Omega} \text{div}(\underline{b})u_h^2 + \frac{1}{2} \int_{\Omega} \underline{b} \nabla (u_h^2) + \sum_{K \in \mathcal{T}_h} \int_K \tau_K (\mathcal{L}u_h)^2 \\ &= \int_{\Omega} \mu |\nabla u_h|^2 + \int_{\Omega} \left(\sigma + \frac{1}{2} \text{div}(\underline{b}) \right) u_h^2 + \sum_{K \in \mathcal{T}_h} \int_K \tau_K (\mathcal{L}u_h)^2 := \|u_h\|_{GLS}^2 \end{aligned}$$

NOTE: There is a mistake missing somewhere in this procedure.

10.11 Stabilization of parameter τ_K

We considered:

$$\tau_K = \delta \frac{h_K}{\|\underline{b}\|}$$

Anyway, it could be chosen another option:

Definition 10.3: Local Peclet Number

$$\mathbb{P}_{e,K} = \frac{\|\underline{b}\|}{2\mu r} h_K$$

$$\tau_K = \frac{h_K}{2r\|b\|_\infty} \xi(\mathbb{P}_{e,K})$$

where ξ is a proper function.

ξ is often chosen as:

$$\xi(\theta) = \coth(\theta) - \frac{1}{\theta}$$

Note that $\xi(0) = 0$. Plus, as $\lim_{\theta \rightarrow \infty} \xi(\theta) = 1$, if $\mathbb{P}_{e,K} \gg 1$, in the limit τ_K is reduced to the first case with $\delta = 1/2$

10.12 GLS convergence

Theorem 10.4: GLS Convergence

Let $u \in V \cap H^{r+1}(\Omega)$ be the exact solution and $u_h \in V_h^r$ the approximate obtained with GLS. Moreover, suppose that $K \in \mathcal{T}_h$. Then, the local Peclet number of K satisfies:

$$\mathbb{P}_{e,K}(\underline{x}) = \frac{|b(\underline{x})| h_K}{2\mu r} > 1 \quad \forall \underline{x} \in K$$

Now, suppose that the stabilization parameter satisfies the relation $0 < \delta \leq \delta^*$, where δ^* depends on the inverse inequality constant. Then, the following super-optimal estimate holds:

$$\|u_h - u\|_{GLS} \leq Ch^{r+1/2} |u|_{H^{r+1}(\Omega)}$$

10.13 Comparing Results

Use GLS only when the asymptotic regime is not present i.e. $\mathbb{P}_{e_K} \geq 1$. Anyway in those cases, the results improve a lot, as seen in figure 3.

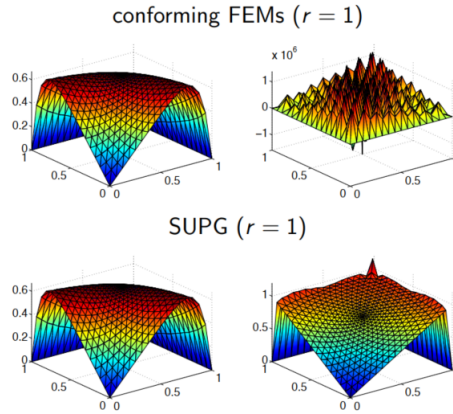


Figure 3: Approximate solutions for $h=16$. Comparing $\mu = 10^{-1}$ and $\mu = 10^{-9}$ in both, non-stabilized and stabilized galerkin.

10.14 DG methods

Let be the problem:

$$\begin{cases} -\operatorname{div}(\mu \nabla u + \underline{b}u) + \sigma u = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

Recall:

$$V_h^p = \{v_h \in L^2(\Omega) \quad \text{st} \quad v_h|_K \in \mathbb{P}^p(K) \quad \forall K \in \mathcal{T}_h\}$$

The DG formulation is:

$$\text{Find } u_h \in V_h^p \quad \text{st} \quad \underbrace{A(u_h, v_h)}_{\text{"Diffusion"}} + \underbrace{B_s(u_h, v_h)}_{\text{"Advection"}} + \underbrace{C(u_h, v_h)}_{\text{"Reaction"}} = \int_{\Omega} f v_h \quad \forall v_h \in V_h^p$$

where:

$$\bullet \quad A(u_h, v_h) = \sum_{K \in \mathcal{T}_h} \int_K \mu \nabla u_h \cdot \nabla v_h - \int_{\mathcal{F}_h} \{\mu \nabla_h u_h\} [[v_h]] - \theta \int_{\mathcal{F}_h} \{\mu \nabla_h v_h\} [[u_h]] + \int_{\mathcal{F}_h} \gamma [[u_h]] \cdot [[v_h]]$$

In this case:

$$\gamma|_F = \frac{\alpha p^2}{h} \{\{\mu\}\}$$

$$\bullet \quad C(u_h, v_h) = \sum_{K \in \mathcal{T}_h} \sum_K \sigma u_h v_h$$

$$\bullet \quad B(u_h, v_h) = \sum_{K \in \mathcal{T}_h} \int_K \underline{b} u_h \cdot \nabla v_h - \int_{\mathcal{F}_h^*} \{\underline{b} u_h\}_{\underline{b}} \cdot [[v_h]]$$

Remark.

The edge mean $\{\cdot\}$ before defined was considering equal contribution of both elements along the edge. In the case of the advection, this does not work, since there is a direction $\underline{b}$ where the flow goes. This means that an edge typically has a side from where the flow comes in, and the other side is where the flow goes out.

Definition 10.5: $\{\underline{b}u\}_{\underline{b}}$

$$\{\underline{b}u\}_{\underline{b}} = \begin{cases} \underline{b}u^+ & \underline{b} \cdot \underline{n}^+ > 0 \\ \underline{b}u^- & \underline{b} \cdot \underline{n}^+ < 0 \\ \underline{b}\{\{u\}\} & \underline{b} \cdot \underline{n}^+ = 0 \end{cases}$$

Definition 10.6: $\mathcal{F}_h^{B, \text{inflow}}$

$\mathcal{F}_h^{B, \text{inflow}}$ is the set of boundary faces that are in Γ_{IN} , the inflow part of $\partial\Omega$

Definition 10.7: $\mathcal{F}_h^*$

$$\mathcal{F}_h^* = \mathcal{F}_h^I \cup \mathcal{F}_h^{B, \text{inflow}}$$

Lemma 10.8: Magic

It holds for any stabilized function B_s :

$$B_s(u_h, v_h) = B(u_h, v_h) + \int_{\mathcal{F}_h^I} C_F[[u_h]][[v_h]] \quad \forall u_h, v_h \subseteq V_h^p$$

where:

$$B(u_h, v_h) = \sum_{K \in \mathcal{T}_h} \int_K \underline{b} u_h \cdot \nabla v_h - \int_{\mathcal{F}_h^*} \{ \underline{b} u_h \} \cdot [[v_h]] \quad \wedge \quad C_F = \frac{|\underline{b} \cdot \underline{n}|}{2} \quad \forall F \in \mathcal{F}_h^I$$

Proof. It is just enough to show that:

$$\{ \{ \underline{b} u_h \} \} \underline{b} \cdot \underline{n} = \left(\{ \underline{b} u_h \} + \frac{|\underline{b} \cdot \underline{n}|}{2} [[u_h]] \right) \cdot \underline{n} \quad F \in \mathcal{F}_h^I$$

□

Remark.

In ADR equation in non-conservative form:

$$\underline{b} \cdot \nabla u = \text{div}(\underline{b} u) - \text{div}(\underline{b}) u$$

Everything works in the same way, but the term $C(u_h, v_h)$:

$$\sum_{K \in \mathcal{T}_h} \int_K \sigma u_h v_h$$

Changes to:

$$\sum_{K \in \mathcal{T}_h} \int_K \eta u_h v_h$$

where $\eta = \sigma - \text{div}(\underline{b})$.

Now, to keep LM assumptions (see theorem 3.6), we need:

$$\exists \eta_0 > 0 \quad \text{st} \quad \eta(\underline{x}) = \sigma(\underline{x}) - \text{div}(\underline{b}(\underline{x})) \geq \eta_0 > 0 \quad \text{a.e. in } \Omega$$

Remark.

This method “ignores” the boundary conditions in the outflow sides, since it represents the boundaries as the inflow part. Specifically, it does not impose the boundary conditions on the outflow boundaries.

10.15 Error Bounds

The diffusion part has exactly the same boundary.

On the other side, the advection-reaction part changes.

Suppose that the conditions on $\underline{b}, \sigma, f$ are satisfied. Let the exact solution be regular enough. Then:

$$\|u - u_h\| \lesssim h^{p+1/2} \|u\|_{H^{p+1}(\Omega)}$$

where:

$$\|w_h\|^2 = \|u - u_h\|_{L^2(\Omega)}^2 + \|S_F^{1/2}[[w_h]]\|_{L^2(\mathcal{F}_h^*)}^2 \quad \forall w_h \in V_h^p$$

where: $S_F = \alpha \frac{|\underline{b} \cdot \underline{n}|}{2} \quad \forall F \in \mathcal{F}_h^I$

11 Methods for Navier-Stokes (NSa) Equations

11.1 The problem

11.1.1 Introduction

Navier-Stokes equations describe the motion of a fluid with constant density ρ in a domain $\Omega \subseteq \mathbb{R}^d$, with $d=2$ or 3 . The equations are as follows:

$$\begin{cases} \frac{\partial \underline{u}}{\partial t} - \text{div} [\nu(\nabla \underline{u} + \nabla \underline{u}^T)] + (\underline{u} \cdot \nabla) \underline{u} + \nabla p = f & x \in \Omega, t > 0 \\ \text{div}(\underline{u}) = 0 & x \in \partial\Omega, t > 0 \end{cases}$$

where:

- $\underline{u} = \underline{u}(x)$ is the fluid's velocity.
- $p = p(x)$ is the pressure divided by the density, which will be simply called “pressure”.
- $\nu = \nu(x)$ is the kinematic viscosity.
- $f \in L^2(\mathbb{R}^+; [L^2(\Omega)]^d)$ is a forcing term per unit of mass.

The first equation indicates the conservation of linear momentum, and the second one is the conservation of mass, which is also called the continuity equation.

Specifically, the term $(\underline{u} \cdot \nabla) \underline{u}$ describes the process of convective transport. On the other hand, $-\text{div} [\nu(\nabla \underline{u} + \nabla \underline{u}^T)]$ describes the process of molecular diffusion.

Now, if we assume ν constant and $\underline{u}$ incompressible ($\nabla \cdot \underline{u} = \text{div}(\underline{u}) = 0$):

$$\text{div} [\nu(\nabla \underline{u} + \nabla \underline{u}^T)] = \nu \text{div}(\nabla \underline{u}) + \nu \text{div}(\nabla \underline{u}^T) = \nu \Delta \underline{u} + \nu \nabla \text{div}(\underline{u}) = \nu \Delta \underline{u}$$

Then the problem becomes the incompressible Navier-Stokes Equations:

$$(INSP) \begin{cases} \frac{\partial \underline{u}}{\partial t} - \nu \Delta \underline{u} + (\underline{u} \cdot \nabla) \underline{u} + \nabla p = f & x \in \Omega, t > 0 \\ \text{div}(\underline{u}) = 0 & x \in \Omega, t > 0 \end{cases}$$

Specifically, fluids satisfying the incompressibility condition $\text{div}(\underline{u}) = 0$ are said to be incompressible.

11.1.2 Boundary and initial conditions

If we want (INSP) to be well posed, it is necessary to assign the initial condition:

$$\underline{u}(x, 0) = \underline{u}_0(x) \quad \forall x \in \Omega$$

where u_0 is a given divergence-free vector field.

On the other hand, suitable boundary conditions could be:

$$\begin{cases} \underline{u}(x, t) = \varphi(x, t) & \forall x \in \Gamma_D \\ \left(\nu \frac{\partial \underline{u}}{\partial n} - p \underline{n} \right)(x, t) = \psi(x, t) & \forall x \in \Gamma_N \end{cases} \quad \forall t > 0$$

where:

- φ and ψ are given vector functions
- $\underline{n}$ is the outward unit normal vector to $\partial\Omega$

11.1.3 Newtonian Fluids example

Taking an incompressible viscous flow in which advective inertial forces are small compared with viscous forces, the convective term can be neglected (when $Re \ll 1$). Also, the inertial terms of the momentum balance in NS equations can be eliminated.

For example, for incompressible newtonian fluids with dirichlet boundary conditions only, the Stokes equations read:

$$\text{Find } (\underline{u}, p) \quad \text{st} \quad \begin{cases} -\nu \Delta \underline{u} + \nabla p = f & \text{in } \Omega \\ \text{div}(\underline{u}) = 0 & \text{in } \Omega \\ \underline{u} = 0 & \text{on } \partial\Omega \end{cases}$$

Remark.

When only Dirichlet boundary conditions are imposed, the pressure appears as a term of its gradient only. In this case, if $(\underline{u}, p)$ is a solution, $(\underline{u}, p+c)$ with c as any constant it is as well. This, because $\nabla(p+c) = \nabla p$. To avoid this, an extra condition on p must be defined.

Remark.

When non-null Dirichlet boundary conditions are imposed, the defined dirichlet data $\underline{\varphi}$ must be compatible with the compressibility condition:

$$\int_{\partial\Omega} \underline{\varphi} \cdot \underline{n} = \int_{\Omega} \text{div}(\underline{u}) = 0$$

11.2 Weak form

Definition 11.1: Frobenius Inner Product or Double Contraction

Let A and B be two tensors, we call Frobenius Inner Product or Double Contraction to:

$$A : B = \sum_{i,j=1}^d a_{ij} b_{ij}$$

It is also used $A \dot{ : } B$

Let $V = [H_0^1(\Omega)]^d$ and $Q = L_0^2(\Omega)$:

$$L_0^2(\Omega) = \left\{ q \in L^2(\Omega) \quad st \quad \int_{\Omega} q = 0 \right\}$$

The weak formulation of the problem is as follows:

$$\text{Find } \underline{u} \in V, p \in Q \quad st \quad \begin{cases} \nu \nabla \int_{\Omega} \underline{u} : \nabla v \, d\Omega - \int_{\Omega} p \cdot \text{div}(v) \, d\Omega = \int_{\Omega} \underline{f} \cdot \underline{v} \, d\Omega & \forall \underline{v} \in V \\ \int_{\Omega} q \cdot \text{div}(\underline{u}) \, d\Omega = 0 & \forall q \in Q \end{cases}$$

Specifically, define the following bilinear forms:

$$\begin{aligned} a(\underline{u}, \underline{v}) &= \int_{\Omega} \nu \nabla \underline{u} : \nabla \underline{v} \, d\Omega \\ b(\underline{u}, q) &= - \int_{\Omega} q \cdot \text{div}(\underline{u}) \, d\Omega \end{aligned}$$

and define the functional:

$$F(\underline{v}) = \int_{\Omega} \underline{f} \cdot \underline{v} \, d\Omega$$

With this notation the weak problem is find $(u, p) \in V \times Q$ st :

$$\begin{cases} a(\underline{u}, \underline{v}) + b(\underline{v}, p) = F(\underline{v}) & \forall \underline{v} \in V \\ b(\underline{u}, q) = 0 & \forall q \in Q \end{cases} \quad (\text{WSE})$$

11.3 Galerkin Approximation

Let $\{V_h \subseteq V\}$ and $\{Q_h \subseteq Q\}$ be two families of conforming finite-dimensional subspaces depending on a real discretization parameter h . Then, the galerkin approximation of the weak problem is:

$$\text{Find } (u_h, q_h) \in V_h \times Q_h \quad st \quad \begin{cases} a(\underline{u}_h, \underline{v}_h) + b(\underline{v}_h, p_h) = F(\underline{v}_h) & \forall \underline{v}_h \in V_h \\ b(\underline{u}_h, q_h) = 0 & \forall q_h \in Q_h \end{cases}$$

11.4 Abstract saddle-point problems

Consider now the following abstract problem, a generalization of the Stokes problem:

Definition 11.2: Abstract Saddle-Point Problem

$$\text{Find } (u, \eta) \in X \times M \quad \text{st} \quad \begin{cases} a(\underline{u}, \underline{v}) + b(\underline{v}, \eta) = \langle l, v \rangle & \forall \underline{v} \in V \\ b(\underline{u}, \mu) = \langle \sigma, \mu \rangle & \forall \mu \in M \end{cases}$$

where:

- X and M are two Hilbert spaces, with their own norms $\|\cdot\|_X, \|\cdot\|_M$.
- $\langle \cdot, \cdot \rangle$ is the pairing between the elements and their dual spaces X' and M' .
- $l \in X', \sigma \in M'$
- The bilinear forms are continuous, i.e. $\exists \gamma, \delta > 0$ such that:

$$\begin{aligned} |a(w, v)| &\leq \gamma \|w\|_X \|v\|_X & w, v \in X \\ |b(w, \mu)| &\leq \gamma \|w\|_X \|\mu\|_M & w \in X, \mu \in M \end{aligned}$$

To analyze the abstract saddle-point problems, we will use further notation:

Definition 11.3: The kernel of $b(\cdot, \cdot)$: X^0

$$X^0 = \{v \in X \quad \text{st} \quad b(v, \mu) = 0 \quad \forall \mu \in M\}$$

Definition 11.4: Affine manifold X^σ

$$X^\sigma = \{v \in X \quad \text{st} \quad b(v, \mu) = \langle \sigma, \mu \rangle \quad \forall \mu \in M\}$$

X^0 is a closed subspace of X .

With this notation, the problem is:

$$(SSPP) \quad \text{Find } u \in X^\sigma \quad \text{st} \quad a(u, v) = \langle l, v \rangle \quad \forall v \in X^0$$

Specifically, we will look for $\sigma = 0$.

11.5 Well-posedness of Continuous Abstract Saddle-Point Problems

Now, to make sure this is well posed, it is enough to use Lax-Milgram, by:

Theorem 11.5: Well-posedness in Continuous Case

If the following conditions are satisfied:

1. $a(\cdot, \cdot)$ and $b(\cdot, \cdot)$ are continuous (by definition).
2. $a(\cdot, \cdot)$ is coercive in X^0 :

$$\exists \alpha > 0 \quad \text{st} \quad a(v, v) \geq \alpha \|v\|_X^2 \quad \forall v \in X^0$$

3. The bilinear form $b(\cdot, \cdot)$ satisfies the compatibility condition:

$$\exists \beta > 0 \quad \text{st} \quad \forall \mu \in M \quad \exists v \in X, v \neq 0 \quad \text{st} \quad b(v, \mu) \geq \beta \|v\|_X \|\mu\|_M$$

This is to make sure that (SSPP) well-posedness coincides with the original problem posedness

Then, for every $l \in X'$ and $\sigma \in M'$ exists a unique solution $(\mu, \nu) \in X \times M$ to the saddle problem.

Moreover, the following a priori estimates hold:

$$\begin{aligned} \|u\|_X &\leq \frac{1}{\alpha} \left[\|l\|_{X'} + \frac{\alpha + \gamma}{\beta} \|\sigma\|_{M'} \right] \\ \|\eta\|_M &\leq \frac{1}{\beta} \left[\left(1 + \frac{\gamma}{\alpha} \right) \|l\|_{X'} + \frac{\gamma(\alpha + \gamma)}{\alpha\beta} \|\sigma\|_{M'} \right] \end{aligned}$$

Proof. Chapter 17 of the textbook □

Definition 11.6: Inf-sup condition (LBB)

Condition 3 is equivalent to:

$$\inf_{\mu \in M \neq 0} \sup_{v \in V \neq 0} \frac{b(v, \mu)}{\|v\|_X \|\mu\|_M} \geq \beta > 0$$

11.6 Well-posedness in Stokes problem

Theorem 11.7

If the data are regular enough ($\underline{f} \in [L^2(\Omega)]^d$). Then:

$$\exists! (\underline{u}, p) \in V \times Q \text{ solution of (WSE) and } \|\underline{u}\|_V + \|p\|_Q \leq C(\underline{f})$$

Proof. Recall:

$$V = [H_0^1(\Omega)]^d \text{ with } \|\underline{v}\|_V = \|\nabla v\|_{L^2(\Omega)} \quad (\text{or the sum of } v \text{ and } \nabla v, \text{ each in norm})$$

$$Q = L_0^2(\Omega) \text{ with } \|q\|_Q = \|q\|_{L^2(\Omega)}$$

Then:

1. a and b are continuous:

$$|a(\underline{u}, \underline{v})| = \left| \int_{\Omega} \nu \nabla \underline{u} \cdot \nabla \underline{v} \right| \lesssim \|\underline{u}\|_V \|\underline{v}\|_V \quad \forall \underline{u}, \underline{v} \in V$$

$$|b(\underline{v}, q)| = \left| - \int_{\Omega} \operatorname{div}(\underline{v}) q \right| \lesssim \|\underline{v}\|_V \|q\|_Q \quad \forall \underline{v} \in V$$

2. $a(\underline{u}, \underline{v}) \geq \alpha \|\underline{v}\|_V^2 \quad \forall \underline{v} \in V \quad \forall q \in Q$, so condition 2) holds in all V and then trivially holds on $V^0 \subseteq V$
3. It can be proven that $\exists \beta > 0$ such that:

$$\forall q \in Q \quad \exists \underline{v} \in V \quad \text{st} \quad b(\underline{v}, q) \geq B \|\underline{v}\|_V \|q\|_Q$$

$$\implies (\text{WSE}) \text{ is well posed}$$

This is proven using the following lemma.

□

Lemma 11.8

$\forall$ function q in $Q = L_0^2(\Omega) \quad \exists \underline{v} \in V = [H_0^1(\Omega)]^d$ such that:

$$q = \operatorname{div}(\underline{v}) \quad \wedge \quad \|\underline{v}\|_V \leq \|q\|_Q$$

Using this lemma, we can see that the compatibility condition is satisfied:

$$|b(\underline{v}, q)| = \left| - \int_{\Omega} \operatorname{div}(\underline{v}) q \right| = \left| - \int_{\Omega} q^2 \right| \quad (\operatorname{div}(\underline{v}) = q)$$

$$= \|q\|_{L^2(\Omega)}^2 \geq \|q\|_Q \|\underline{v}\|_V \quad (\|\underline{v}\|_V \leq \|q\|_Q)$$

This was the missing part to state that the stokes problem is well-posed.

11.7 Well-posedness of Discrete Abstract Saddle-Point Problems

Definition 11.9: Conforming Approximation of Abstract Saddle-Point Problem

Let be X_h, M_h two families of X, M , respectively, then the discretization of the problem is:

$$\text{Find } (u_h, \eta_h) \in X_h \times M_h \quad \text{st} \quad \begin{cases} a(\underline{u}_h, \underline{v}_h) + b(\underline{v}_h, \eta_h) = \langle l, v_h \rangle & \forall \underline{v}_h \in V_h \\ b(\underline{u}_h, \mu_h) = \langle \sigma, \mu_h \rangle & \forall \mu_h \in M_h \end{cases}$$

Theorem 11.10: Well-posedness in Discrete Case

If the following conditions are satisfied:

1. $a(\cdot, \cdot)$ and $b(\cdot, \cdot)$ are continuous (by definition).
2. $a(\cdot, \cdot)$ is coercive in X_h^0 :

$$\exists \alpha > 0 \quad \text{st} \quad a(v_h, v_h) \geq \alpha_h \|v_h\|_X^2 \quad \forall v_h \in X_h^0$$

3. The bilinear form $b(\cdot, \cdot)$ satisfies the compatibility condition:

$$\exists \beta_h > 0 \quad \text{st} \quad \forall \mu_h \in M_h \quad \exists v_h \in X_h, v_h \neq 0 \quad \text{st} \quad b(v_h, \mu_h) \geq \beta \|v_h\|_X \|\mu_h\|_M$$

Then, for every $l \in X'$ and $\sigma \in M'$ exists a unique solution $(\mu_h, \nu_h) \in X \times M$ to the saddle problem. Moreover, the solution satisfies the following stability conditions:

$$\begin{aligned} \|u_h\|_X &\leq \frac{1}{\alpha_h} \left[\|l\|_{X'} + \frac{\alpha_h + \gamma}{\beta_h} \|\sigma\|_{M'} \right] \\ \|\eta_h\|_M &\leq \frac{1}{\beta_h} \left[\left(1 + \frac{\gamma}{\alpha_h} \right) \|l\|_{X'} + \frac{\gamma(\alpha_h + \gamma)}{\alpha_h \beta_h} \|\sigma\|_{M'} \right] \end{aligned}$$

Remark.

Since in general $X_h^0 \not\subseteq X$, the continuous coercivity condition does not necessarily guarantee the discrete coercivity condition.

In the same way, the discrete compatibility condition is not inherited by the continuous one.

Theorem 11.11: Abstract Convergence

Let the assumptions of existence and uniqueness theorems in discrete and continuous cases be satisfied. Then, the solutions (u, η) and (u_h, η_h) of the abstract saddle point problems satisfy the following error estimates:

$$\begin{aligned} \|u - u_h\|_X &\lesssim \inf_{v_h \in X_h} \|u - v_h\|_X + \inf_{\mu_h \in M_h} \|\eta - \mu_h\|_M \\ \|\eta - \eta_h\|_X &\lesssim \inf_{v_h \in X_h} \|u - v_h\|_X + \inf_{\mu_h \in M_h} \|\eta - \mu_h\|_M \end{aligned}$$

Where the hidden constants depend on γ, δ, α_h and β_h

11.8 Well-posedness in Discrete Stokes problem

Let be:

$$V_h \subseteq V \quad \wedge \quad Q_h \subseteq Q$$

With this notation, the discrete weak problem is:

$$\text{Find } (u_h, p_h) \in V_h \times Q_h \quad \text{st} \quad \begin{cases} a(\underline{u}_h, \underline{v}_h) + b(\underline{v}_h, p_h) = F(\underline{v}_h) & \forall \underline{v}_h \in V_h \\ b(\underline{u}_h, q_h) = 0 & \forall q_h \in Q_h \end{cases} \quad (DS)$$

Now, with the previous theorem 11.10, we just need to prove:

1. Continuity of $a(\cdot, \cdot)$ and $b(\cdot, \cdot)$, trivial since $V_h \subseteq V$ and $Q_h \subseteq Q$
2. Recall that:

$$V_h^0 = \{\underline{v}_h \in V_h \quad \text{st} \quad b(\underline{v}_h, q_h) = 0 \quad \forall q_h \in Q_h\}$$

Clearly, $a(\underline{u}_h, \underline{u}_h) \gtrsim \alpha \|\underline{u}_h\|_V \quad \forall \underline{u}_h \in V_h$, since a is coercive in the whole space and $V_h^0 \subseteq V$

- 3.

$$\exists \beta_h > 0 \quad \text{st} \quad \forall q_h \in Q_h \quad \exists \underline{v}_h \in V_h, \quad \underline{v}_h \neq 0 \quad \text{st} \quad b(\underline{v}_h, q_h) \geq \|\underline{v}_h\|_V \|q_h\|_Q$$

This is more difficult to prove, since although $Q_h \subseteq Q$, the continuous relation can still find an element in V such that it is not in V_h .

In particular, if V_h and Q_h are compatible, then (DS) is well-posed.

And as we have seen in the abstract general problems, under the assumptions, the following results hold:

$$\begin{aligned} \|u - u_h\|_V &\lesssim \inf_{v_h \in V_h} \|u - v_h\|_V + \inf_{q_h \in Q_h} \|p - q_h\|_Q \\ \|p - q_h\|_Q &\lesssim \inf_{v_h \in V_h} \|u - v_h\|_V + \inf_{q_h \in Q_h} \|p - q_h\|_Q \end{aligned}$$

11.9 Algebraic Formulation

We can write (DS) as an algebraic system.

Let be:

$$\begin{aligned} \dim(V_h) = N \quad \wedge \quad V_h = \text{span}\{\underline{\varphi}_j, \quad j = 1, \dots, N\} &\implies \underline{u}_h = \sum_{j=1}^N \underline{u}_j \underline{\varphi}_j \\ \dim(Q_h) = M \quad \wedge \quad Q_h = \text{span}\{\phi_k, \quad k = 1, \dots, M\} &\implies p_h = \sum_{k=1}^M \underline{p}_k \phi_k \end{aligned}$$

Proceeding as usual, it can be got the following system:

$$\begin{cases} A\underline{u}_h + B^T \underline{p}_h = F \\ B\underline{u}_h = 0 \end{cases} \iff \underbrace{\begin{bmatrix} A & B^T \\ B & 0 \end{bmatrix}}_S \begin{bmatrix} \underline{u}_h \\ \underline{p}_h \end{bmatrix} = \begin{bmatrix} F \\ 0 \end{bmatrix}$$

where:

$$\underline{u}_h = \begin{bmatrix} u_1 \\ u_2 \\ \dots \\ u_N \end{bmatrix} \in \mathbb{R}^N \quad \wedge \quad \underline{p}_h = \begin{bmatrix} p_1 \\ p_2 \\ \dots \\ p_M \end{bmatrix} \in \mathbb{R}^M$$

and the matrices:

- $A \in \mathbb{R}^{N \times N}$ $A_{ij} = \int_{\Omega} \nabla \underline{\varphi}_j : \nabla \underline{\varphi}_i$ $(a(\underline{\varphi}_j, \underline{\varphi}_i))$
- $B \in \mathbb{R}^{M \times N}$ $A_{kj} = \int_{\Omega} \text{div}(\underline{\varphi}_j) \phi_k$ $(b(\underline{\varphi}_j, \phi_k))$
- $\underline{F} \in \mathbb{R}^N$ $F_i = \int_{\Omega} f \underline{\varphi}_i$

Remark.

The matrix S is block symmetric. A is the Laplacian, so it is SPD.

11.10 Algebraic Problem Well-posedness

The algebraic problem is well-posed if S is invertible.

First, S is block symmetric if A is symmetric. Actually, A is symmetric.

Now, S is invertible if and only if S has no null eigenvalues. This can be obtained from the inf-sup condition. The inf-sup preposition implies S is non-singular.

If we perform Gaussian Elimination by blocks in:

$$\begin{bmatrix} A & B^T \\ B & 0 \end{bmatrix} \begin{bmatrix} \underline{u}_h \\ \underline{p}_h \end{bmatrix} = \begin{bmatrix} \underline{F} \\ 0 \end{bmatrix}$$

From the first block of equations:

$$A\underline{u} + B^T \underline{p} = \underline{F}$$

As A is spd:

$$\underline{u} = A^{-1}[\underline{F} - B^T \underline{p}]$$

Note that we can do this only in theory, A is huge, so computationally is too costly.

Now, if we substitute this in the second block:

$$B\underline{u} = 0 \implies \underbrace{BA^{-1}B^T}_{=R \in \mathbb{R}^{M \times M}} \underline{p} = \underbrace{BA^{-1}\underline{F}}_{\underline{G}} \implies R\underline{p} = \underline{G}$$

R is invertible if and only if rank(R)=M. Note that R is a quadratic form, where A^{-1} is spd. Now, we have to analyze B.

Proposition 11.12

R is invertible $\iff \text{Ker}(B^T) = \{\underline{z} \neq 0 \text{ st } B^T \underline{z} = 0\} = \{0\}$.

Proof. Indeed, Given any $\underline{p} \in \mathbb{R}^m$ such that:

$$R\underline{p} = 0 \implies \underline{p} = 0$$

This is equivalent to: $\langle R\underline{p}, \underline{q} \rangle = \underline{q}^T R\underline{p} = 0 \quad \forall \underline{q} \neq 0$

Now, take $\underline{q} = \underline{p}$:

$$\implies \underline{p}^T R\underline{p} = \underline{p}^T (BA^{-1}B^T)\underline{p} = \langle A^{-1}B^T\underline{p}, B^T\underline{p} \rangle = 0$$

Now, let $\underline{w} = B^T\underline{p}$:

$$\langle A^{-1}\underline{w}, \underline{w} \rangle \implies \underline{w}^T A^{-1}\underline{w} = 0$$

So, since A is spd:

$$\underline{w} = 0 = B^T\underline{p}$$

So, an equivalent condition is:

$$\underline{p} \in \text{Ker}(B^T) \implies \underline{p} = 0$$

□

11.11 Choosing inf-sup stable dof's

Let be:

$$V = [H_0^1(\Omega)]^d \quad \wedge \quad Q = L_0^2(\Omega)$$

And a discretization:

$$V_h \subseteq V \quad \wedge \quad Q_h \subseteq Q$$

Where V_h is continuous FE and Q_h is discontinuous.

We would like to choose polynomial degree k in V and $k-1$ in Q , since L_0^2 gives one degree more of convergence.

For example, for linear elements in V , we choose functions of the pressure piecewise constant, since it would give us $H^{r+1} = H^1$ space of convergence, which matches with V .

Then, the error would be (it is incorrect):

$$\|u - u_h\|_V + \|p - p_h\|_Q \leq h^k + h^k$$

Unfortunately, this does not work, since it does not satisfy the inf-sup condition (It can be proven).

The next option could take continuous pressure. We use piecewise continuous polynomials of the same order for both velocity and pressure space. In this case, we 'lose' a degree of freedom:

$$\|u - u_h\|_V + \|p - p_h\|_Q \leq h^k + h^{k+1} \lesssim h^k$$

The advantage is that the degrees are the same in both, and this makes the implementation easier. Anyway, unfortunately, this does not satisfy the inf-sup condition either.

One of the correct options is to take continuous FE in V_h and discontinuous FE in Q_h . Choose for V_h $k \geq 2$ and for Q_h $k - 2$:

$$\mathbb{P}^2 - \mathbb{P}^0 \text{ pair in triangles} \quad \mathbb{Q}^2 - \mathbb{P}^0 \text{ pair in quads}$$

$\mathbb{P}_0 = \mathbb{Q}_0$.

There is a third setting called Crouzeix-Raviart elements, that also works (Right in the figure 4)

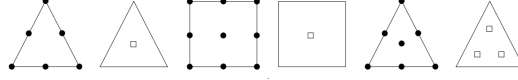


Figure 4: Degrees of Freedom of choices that satisfy inf-sup conditions. Triangles on left, quadrilaterals on center, Crouzeix-Raviart on right

All those 3 settings do pass the inf-sup condition.

There are other configurations that satisfy the conditions:

- $\mathbb{P}_1 - iso\mathbb{P}_2$
- Mini-element
- $\mathbb{P}_k - \mathbb{P}_{k-1} \quad k \geq 2$

In particular the smallest degree stable is $\mathbb{P}_2 - \mathbb{P}_1$

The stability proof can be found in (Brezzi & Fortin, 1991).

As a general rule:

- The larger the velocity space V_h , the more likely the inf-sup condition is satisfied.
- Otherwise, the V_h space must be rich enough compared with Q_h

11.12 Stabilization Techniques

We have seen that FE, which uses equal-degree polynomial for both velocity and pressure, does not fulfill the inf-sup condition and therefore is unstable.

However, stabilizing them is possible by SUPG or GLS techniques, like those encountered in the framework of the numerical approximation of advection-diffusion equations (see section 10.6).

Generally, we use stabilization techniques when we really want to use some spaces and they do not satisfy inf-sup condition.

We see the formulation for GLS stabilization for $\mathbb{P}_1 - \mathbb{P}_1$ space (adding the two previous equations):

$$\text{Find } (u_h, p_h) \in V_h \times Q_h \quad \text{st} \quad A_h(\underline{u}_h, p_h; \underline{v}_h, q_h) = F_h(\underline{v}_h, q_h) \quad (\text{GLSDS})$$

Where:

$$\begin{aligned} \bullet \quad A_h(\underline{u}_h, p_h; \underline{v}_h, q_h) &= a(\underline{u}_h, \underline{v}_h) + b(\underline{v}_h, p_h) - b(\underline{u}_h, q_h) \\ &+ \delta \sum_{K \in \mathcal{T}_h} h_K^2 \int_K (-\nu \Delta \underline{u}_h + \nabla p_h)(-\nu \Delta \underline{v}_h + \nabla q_h) \, dK \end{aligned}$$

- $F_h(\underline{v}_h, q_h) = \langle \underline{f}, \underline{v}_h \rangle + \delta \sum_{K \in \mathcal{T}_h} h_K^2 \int_K \underline{f} \cdot (-\nu \Delta \underline{v}_h + \nabla q_h) dK$
- δ a positive parameter chosen conveniently

This is stable since in the exact solution the original equation is satisfied:

$$(-\nu \Delta \underline{u}_h + \nabla p_h) = \underline{f} \quad \implies \quad (-\nu \Delta \underline{u}_h + \nabla p_h)(-\nu \Delta \underline{v}_h + \nabla q_h) = \underline{f} \cdot (-\nu \Delta \underline{v}_h + \nabla q_h)$$

It can be shown that:

$$\nu \|\nabla \underline{u}_h\|_{L^2(\Omega)}^2 + \delta \sum_{K \in \mathcal{T}_h} h_K^2 \|\nabla p_h\|_{L^2(K)}^2 \lesssim \|f\|_{L^2(\Omega)}^2$$

where the hidden constant is independent of h .

By applying Strang's lemma it can be shown:

$$\|\underline{u} - \underline{u}_h\|_{H^1(\Omega)} + \left(\delta \sum_{K \in \mathcal{T}_h} h_K^2 \|\nabla p - \nabla p_h\|_{L^2(K)}^2 \right)^{1/2} \lesssim h$$

With the stabilization, the formulation results as follows:

$$\begin{bmatrix} A & B^T \\ B & -C \end{bmatrix} \begin{bmatrix} \underline{u} \\ \underline{p} \end{bmatrix} = \begin{bmatrix} \underline{F} \\ 0 \end{bmatrix}$$

- $C = \{c_{km}\}_{k,m}$ spd:

$$c_{km} = \delta \sum_{K \in \mathcal{T}_h} h_K^2 \int_K \nabla \phi_m \cdot \nabla \phi_k dK$$

- $G = \{g_k\}_k$:

$$g_k = \delta \sum_{K \in \mathcal{T}_h} h_K^2 \int_K \underline{f} \cdot \nabla \phi_k dK$$

With C spd.

Doing Block Gaussian elimination (repeating steps):

$$R = BA^{-1}B^T + C$$

So this will help to the matrix to be spd.

12 Numerical Methods for Linear Elasticity

The PDE of this chapter are motivated by some problems where elastic bodies are involved. For example, given an elastic body, we want to apply a load and check the displacement of the body.

12.1 The problem

Let $\Omega \subseteq \mathbb{R}$ be an open bounded Lipschitz domain. Its closure $\bar{\Omega}$ is called the reference configuration.

We seek to characterize its shape upon a load via a mapping

$$\varphi : \bar{\Omega} \rightarrow \mathbb{R}^n \quad \tilde{\Omega} = \varphi(\bar{\Omega})$$

The simplest mathematical model for this elastic deformation is based on Hooke's law.

We are not interested in the exact mapping $\varphi(\cdot)$ since it depends on the position of the points x . On the other hand, the physical laws are frame invariance, so if the coordinate system changes, the laws only change with the coordinate transformation.

Definition 12.1: $\mathbb{I}$

The symbol $\mathbb{I}$ refers to all occurrence of identical mappings, for example I in the usual matrix setting.

Definition 12.2: $\epsilon(\underline{u})$

The Symmetric Deformation Tensor is defined as follows:

$$\epsilon(\underline{u}) = \frac{1}{2}(\nabla u + \nabla u^T)$$

It measures how a body deforms without considering rotations, since only the symmetric deformations produce elastic energy.

We introduce a displacement $u : \bar{\Omega} \rightarrow \mathbb{R}^d$ such that:

$$\varphi(x) = \mathbb{I}x + u(x)$$

The relation between the stress tensor σ and the force f is:

$$-\nabla \sigma = f$$

Hooke's law states that the stress σ depends linearly on the strain. Together with frame invariance, this implies the relation:

$$\sigma = 2\mu\epsilon(\underline{u}) + \lambda \text{trace}(\epsilon(\underline{u}))\mathbb{I}$$

Where $\lambda \geq 0$ and $\mu \geq 0$ are material properties called Lamé-Navier parameters.

The trace of the strain operator is equal to the gradient, so we have:

$$\text{trace}(\epsilon(\underline{u}))\mathbb{I} = \nabla \cdot \underline{u}$$

The final formulation is:

$$\begin{cases} -\nabla \cdot (2\mu \epsilon(\underline{u}) + \lambda \operatorname{trace}(\epsilon(\underline{u}))\mathbb{I}) = f & \text{in } \Omega \\ \underline{u} = 0 & \text{on } \Gamma_D \end{cases}$$

And the weak form is:

$$\int_{\Omega} [2\mu \epsilon(\underline{u}) : \nabla \underline{v} + \lambda (\nabla \cdot \underline{u}) \cdot \nabla \underline{v}] = \int_{\Omega} f \cdot \underline{v}$$

With the space $V = [H_0^1(\Omega)]^d$.

We note that (remembering : Frobenius norm defined in section 11.2):

$$\mathbb{I} : \nabla v = \sum_i \partial_i v_i = \nabla \cdot v$$

And also note that, as the symmetric space is orthogonal to the non-symmetric part of the same space:

$$\epsilon(u) : \nabla v = \epsilon(u) : \epsilon(v)$$

So we can rewrite the weak form:

$$\text{Find } u \in V \quad \text{st} \quad \mathcal{A}(u, v) = F(v) \quad \forall v \in V$$

where:

- $\mathcal{A}(u, v) = \int_{\Omega} [2\mu \epsilon(\underline{u}) : \epsilon(\underline{v}) + \lambda (\nabla \cdot \underline{u})(\nabla \cdot \underline{v})]$
- $F(v) = \int_{\Omega} f \cdot \underline{v}$

12.2 Well-Posedness

Lemma 12.3

In d dimensions, the bilinear form is continuous with continuity constant $M = 2\mu + d\lambda$. For $\partial\Omega = \Gamma_D$, the bilinear form is coercive with coercivity constant $\alpha = 2C_{\Omega}\mu$

Proof. Coercivity follows from Korn's inequality:

$$\|\epsilon(\underline{v})\|_{L^2(\Omega)} \geq C_{\Omega} \|v\|_V$$

Now, continuity of A and $\underline{f}$ is an exercise. □

Now, the Galerkin approximation over a conforming finite dimensional subspace $V_h \subseteq V$ is:

$$\text{Find } u_h \in V_h \quad \text{st} \quad \mathcal{A}(u_h, v_h) = F(v_h) \quad \forall v_h \in V_h$$

Now, we need $\mathcal{A}$ and F satisfy LM hypothesis (see theorem 3.6), so the Galerkin Approximation is well-posed for any closed subspace $V_h \subseteq V$. $\mathcal{A}$ is bounded and coercive as V_h ,

with the same continuity and coercivity constants as the original space V . Plus, F is linear and bounded. Then, the galerkin approximation is well-posed.

12.3 Locking Phenomenon

Anyway, in practice, we see that for some cases the FE method work, but for others don't. For a fixed μ , different values of λ could lead to good or bad results. In some cases, the FE approximation result gives a solution almost identical to the undeformed configuration. This is called Locking Phenomenon (See figure 5).

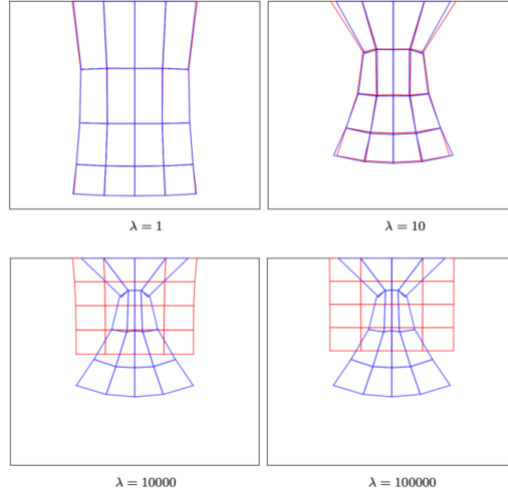


Figure 5: FE results for $\mu = 1$ and different λ s. Expected result in blue vs FE result in red.

It can be seen that if μ and λ are close in magnitude, the results are reasonable. If $\lambda \gg \mu$, the method does not work. This is due to the M and α related to the parameters set. $\sqrt{M/\alpha}$ becomes large as $\lambda \gg \mu$.

Remark.

- As $\lambda \rightarrow \infty$, the Galerkin approximation breaks down. The reason $M = M(\mu, \lambda)$ to $\alpha = \alpha(\mu)$ becomes bigger.
- Notice that λ penalizes $\|\nabla \cdot u\|_{L^2(\Omega)}^2$, therefore, if $\lambda \rightarrow \infty$, the displacement is not allowed to change the volume (nearly to the incompressible case).

Note that the method is converging as we refine the quadrature anyway, but it does it slowly.

12.4 Displacement Pressure Formulation

We can see that the problem appears due to the term:

$$\lambda(\nabla \cdot \underline{u})(\nabla \cdot \underline{v})$$

Now, introduce an auxiliary variable $p = -\lambda \nabla \cdot \underline{u}$, so setting $Q = L^2(\Omega)$ (this is important), the corresponding weak formulation corresponds to:

$$\text{Find } (u, p) \in V \times Q \quad \text{st} \quad \begin{cases} \int_{\Omega} [2\mu \epsilon(\underline{u}) : \epsilon(\underline{v}) - p(\nabla \cdot \underline{v})] = \int_{\Omega} \underline{f} \cdot \underline{v} & \forall \underline{v} \in V = [H_0^1(\Omega)]^d \\ - \int_{\Omega} \nabla \cdot \underline{u} q - \frac{1}{\lambda} \int_{\Omega} p q = 0 & \forall q \in Q = L^2(\Omega) \end{cases}$$

Where the last equation was obtained by integrating:

$$p = -\lambda \nabla \cdot \underline{u} \quad \implies \quad \frac{1}{\lambda} p = -\nabla \cdot \underline{u}$$

Theorem 12.4

Using the LBB theory, it can be shown that this problem is uniformly well-posed as $\lambda \rightarrow \infty$

Remark.

So we have moved from a simple-to-discretize but unstable formulation to a stable but harder-to-discretize one (stable in the sense of goodness of the approximation)

Now, the abstract formulation is:

$$\text{Find } (u, p) \quad \text{st} \quad \begin{cases} a(u, v) + b(v, p) = F(v) & \forall v \in V \\ b(u, q) - c(p, q) = 0 & \forall q \in Q \end{cases} \quad (\text{DISP})$$

Where:

- $a(\underline{u}, \underline{v}) = \int_{\Omega} 2\mu \epsilon(\underline{u}) : \epsilon(\underline{v})$
- $b(\underline{v}, p) = - \int_{\Omega} p \nabla \cdot \underline{v}$
- $c(p, q) = \frac{1}{\lambda} \int_{\Omega} p q$
- $F(\underline{v}) = \int_{\Omega} \underline{f} \cdot \underline{v}$

Remark.

Note that

$$c(p, q) = \frac{1}{\lambda} \int_{\Omega} p q \xrightarrow{\lambda \rightarrow \infty} 0$$

12.5 Displacement Pressure Formulation Well-Posedness

We start assuming that the case where $c(p, q) = 0$ is well-posed, and then we will find sufficient conditions over $c(\cdot, \cdot)$ to state well-posedness of the perturbed problem.

Assume that the assumptions of the general well-posedness theorem for saddle point problems hold:

- $a(\cdot, \cdot)$ is coercive on $K = \{v \in V \mid b(v, q) = 0 \quad \forall q \in Q\}$.
- $b(\cdot, \cdot): V \times Q \rightarrow \mathbb{R}$ satisfies the inf-sup condition.
- $c(\cdot, \cdot): Q \times Q \rightarrow \mathbb{R}$ is of the form $c(p, q) = \zeta \int_{\Omega} pq \quad \forall p, q \in Q, \quad \zeta > 0$

Then, for every $f \in V'$ and every $g \in Q'$, the problem (DISP) has a unique solution. Moreover, this solution satisfies:

$$\begin{aligned} \|u\|_V &\lesssim \|f\|_{V'} + \|g\|_{Q'} \\ \|p\|_V &\lesssim \|f\|_{V'} + \|g\|_{Q'} \end{aligned}$$

12.6 Final Remarks

There are two main strategies for overcoming the effects of locking:

- Use a mixed method, as we did in section 12.4. It is needed to choose an appropriate combination of finite spaces V_h and Q_h that lead to approximations for u and p that are indeed uniformly optimal.
- An alternative method is to retain the primal formulation, but in conjunction with higher-order finite element methods. It has its tradeoffs (e.g., higher computational cost, less flexibility for incompressibility)

There are plenty of other possible formulations for this problem. If we start from the mixed formulation:

1. We can sum the two equations, and the formulation becomes:

$$\underbrace{\int_{\Omega} [2\mu\epsilon(\underline{u}) : \epsilon(\underline{v}) - p(\nabla \cdot \underline{v})] - \int_{\Omega} \nabla \cdot \underline{u}q - \frac{1}{\lambda} \int_{\Omega} pq}_{=B((\underline{u}, p), (\underline{v}, q))} = \int_{\Omega} f\underline{v} \quad \forall (v \times q) \in V \times Q$$

This formulation satisfies symmetry.

The problem with this formulation is that coercivity does not hold. Indeed, take the specific case:

$$B((0, p), (0, p)) = -\frac{1}{\lambda} \|p\|_{L^2(\Omega)}^2 < 0$$

2. We can take the difference between the two equations, the formulation becomes:

$$\underbrace{\int_{\Omega} [2\mu\epsilon(\underline{u}) : \epsilon(\underline{v}) - p(\nabla \cdot \underline{v})] + \int_{\Omega} \nabla \cdot \underline{u}q + \frac{1}{\lambda} \int_{\Omega} pq}_{=B((\underline{u}, p), (\underline{v}, q))} = \int_{\Omega} f\underline{v} \quad \forall (v \times q) \in V \times Q$$

This formulation does not satisfy symmetry.

But in this case, coercivity holds:

$$B((\underline{u}, p), (\underline{u}, p)) = 2\mu \|\epsilon(\underline{u})\|_{L^2(\Omega)}^2 + \frac{1}{\lambda} \|p\|_{L^2(\Omega)}^2 > 0$$

13 Wave Propagation with Scalar Wave Equation

13.1 Introduction

The wave propagation is frequently encountered in many engineering applications (acoustics, electromagnetism, fluid dynamics, solid mechanics, meteorology, among others). There are different kinds of waves, such as electromagnetic waves, water waves, and ultrasound waves.

Acoustic waves are a type of mechanical wave that propagates through a medium (air, water, solid) due to an excitation. Sound waves are fundamentally longitudinal waves, where the medium's displacement is parallel to wave propagation.

Elastic waves are a type of mechanical wave that propagates in elastic or viscoelastic materials. In geological contexts, such as earthquakes, these waves are usually called seismic waves.

The aeroacoustics waves are a type of wave that generates noise via turbulent fluid motion or aerodynamic forces interacting with surfaces.

Some challenges of numerical solutions to wave propagation are:

- Correctly capture the wavefront.
- Boundary conditions have to be taken into consideration with care.
- Enough discretization points per wavelength.
- Low numerical dispersion and dissipation error.

13.2 Scalar wave equation

$$\text{Find } u(x, t) : \Omega \times (0, T] \rightarrow \mathbb{R} \quad st \quad \begin{cases} \ddot{u}(x, t) - c^2 \Delta u(x, t) = f(x, t) & \text{in } \Omega \times (0, T] \\ u(x, t) = 0 & \text{on } \partial\Omega \times (0, T] \\ u(x, 0) = u^0(x) & \text{in } \Omega \\ \dot{u}(x, 0) = v^0(x) & \text{in } \Omega \end{cases}$$

where:

- f, u^0, v^0 are given
- c is the wave speed

$\forall x, t \quad u(x, t)$ models the displacement from the rest configuration.

In this case, we can define the weak form.

Let $V = H_0^1(\Omega)$:

$$\text{Find } u(x, t) \in V \quad st \quad \begin{cases} \underbrace{\int_{\Omega} \ddot{u}v}_{M(\ddot{u}, v)} + c^2 \underbrace{\int_{\Omega} \nabla u \cdot \nabla v}_{A(u, v)} = \underbrace{\int_{\Omega} f(x, t)v}_{F(v)} & \forall v \in V \\ u(x, 0) = u^0(x) \\ \dot{u}(x, 0) = v^0(x) \end{cases} \quad \forall t \in (0, T]$$

Then, the abstract form of the problem is:

$$\text{Find } u(x, t) \in V \quad st \quad \begin{cases} M(\ddot{u}, v) + c^2 A(u, v) = F(v) & \forall v \in V \\ u(x, 0) = u^0(x) \\ \dot{u}(x, 0) = v^0(x) \end{cases} \quad \forall t \in (0, T]$$

13.3 Well-posedness

Definition 13.1: Energy Norm

$$\|u\|_\epsilon^2 = \|\dot{u}\|_{L^2(\Omega)}^2 + \|u\|_V^2$$

Theorem 13.2: Well-posedness Scalar Wave Equation

If the data are regular enough, then:

$$\exists! \text{ solution } u \in C^0(0, T; V) \cap C^1(0, T; L^2(\Omega))$$

which depends continuously on the data in the energy norm

13.4 Conforming Semidiscrete Approximations

If we want to apply:

- **FEM:** Take $V_h \subseteq V$ the space of conforming finite elements of degree $r \geq 1$ over a triangulation of $\mathcal{T}_h$, i.e.:

$$V_h = \{v_h \in V \quad st \quad v_h|_K \in \mathbb{P}^r(K) \quad \forall K \in \mathcal{T}_h\}$$

- **SEM:** Take $V_h \subseteq V$ the space of spectral elements of degree $p \geq 1$ over a triangulation of $\mathcal{T}_h$, i.e.:

$$V_h = \left\{ v_h \in V \quad st \quad v_h|_K = \hat{v} \circ F_K^{-1}, \quad \wedge \quad \hat{v} \in \mathbb{Q}^p(\hat{K}) \quad \forall K \in \mathcal{T}_h \right\}$$

As always:

$$\dim(V_h) = N_h \quad \wedge \quad V_h = \text{span}\{\varphi_j\}_{j=1}^{N_h}$$

Now we can reformulate the problem in V_h the conforming space:

$$\text{Find } u_h(x, t) \in V_h \quad st \quad \begin{cases} M(\ddot{u}_h, v_h) + c^2 A(u_h, v_h) = F(v_h) & \forall v_h \in V_h \\ u_h(x, 0) = u_h^0(x) \in V_h \\ \dot{u}_h(x, 0) = v_h^0(x) \in V_h \end{cases} \quad \forall t \in (0, T] \quad (\text{WPAPP})$$

The most important change is that $u_h^0(x)$ and $v_h^0(x)$ have to be suitable approximations of u^0 and v^0 in V_h .

In the case we want to add the numerical integration to SEM (SEM-IN), the main equation changes to:

$$M_h(\ddot{u}_h, v_h) + c^2 A_h(u_h, v_h) = F_h(v_h) \quad \forall v_h \in V_h$$

13.5 Non-Conforming Semidiscrete Approximation

We could also apply non conforming spaces $V_h \not\subseteq V$, for example Discontinuous Finite Elements (DG):

$$V_h = \{v_h \in L^2(\Omega) \quad st \quad v_h|_K \in \mathbb{P}^r(K) \quad \forall K \in \mathcal{T}_h\}$$

In this case, as we saw in section 7, taking $\theta = 1$

$$\begin{aligned} A_h(w_h, v_h) &= \int_{\Omega} \nabla_h w_h \cdot \nabla_h v_h - \sum_{F \in \mathcal{F}_h^I} \int_F \{\{\nabla_h w_h\}\} \cdot [[v_h]] - \sum_{F \in \mathcal{F}_h^I} \int_F [[w_h]] \cdot \{\{\nabla_h v_h\}\} + \sum_{F \in \mathcal{F}_h^I} \gamma [[w_h]] \cdot [[v_h]] \\ M_h(w_h, v_h) &= \int_{\Omega} w_h v_h \\ F_h(v_h) &= \int_{\Omega} f v_h \end{aligned}$$

This approximation is usually the most used, since it is highly parallelizable and the computation does not depend on the neighbors, due to it has independent DOFs.

13.6 Stabilization Analysis for DG

Let $u_h(t)$ be the semidiscrete approximation solution with a sufficiently large penalty parameter $\alpha > 0$. Then:

$$\|u_h(t)\|_{\epsilon_h} \lesssim \|u_h^0\|_{\epsilon_h} + \int_0^t \|f(\tau)\|_{L^2(\Omega)} d\tau \quad \forall t \in (0, T]$$

Now, assume that the exact solution is smooth enough:

$$\|u(t) - u_h(t)\|_{\epsilon_h} \leq C(T, u, \dot{u}, \ddot{u}, p) h^p \quad \forall t \in (0, T]$$

where: $\|u_h(t)\|_{\epsilon_h}^2 = \|\dot{u}_h(t)\|_{L^2(\Omega)}^2 + \|u_h(t)\|_{DG}^2$.

The same kind of bounds can be proven for the other formulations, defining suitable energy norms.

13.7 Algebraic Form

Let $V_h = \text{span}\{\varphi_j\}_{j=1}^{N_h}$, $N_h < \infty$, then:

$$u_h = \sum_{j=1}^{N_h} u_j(t) \varphi_j(x), \quad \underline{u}(t) = \begin{bmatrix} u_1(t) & u_2(t) & \dots & u_{N_h}(t) \end{bmatrix}^T \in \mathbb{R}^{N_h}$$

Let:

- $M_h \in \mathbb{R}^{N_h \times N_h}$: $\{M_h\}_{ij} = M_h(\varphi_j, \varphi_i)$
- $A_h \in \mathbb{R}^{N_h \times N_h}$: $\{A_h\}_{ij} = A_h(\varphi_j, \varphi_i)$
- $F_h \in \mathbb{R}^{N_h}$: $\{F_h\}_j = F_h(\varphi_j)$

Then (WPAPP) is equivalent to:

$$\text{Find } \underline{u} \in \mathbb{R}^{N_h} \quad \text{st} \quad \begin{cases} M_h \ddot{\underline{u}} + c^2 A_h \underline{u} = F_h \\ \underline{u}(0) = \underline{u}^0 \\ \dot{\underline{u}}(0) = \underline{v}^0 \end{cases} \quad \forall t \in (0, T] \quad (\text{WPALG})$$

Where:

$$\underline{u}^0 \in \mathbb{R}^{N_h} \quad \wedge \quad \underline{v}^0 \in \mathbb{R}^{N_h}$$

are the expansion coefficients of u_h^0 and $\dot{u}_h^0$ in the chosen basis.

In the case of DG M_h is either block-diagonal or diagonal. In SEM-NI M_h is diagonal. In these specific cases, integrating in time is parallelizable.

13.8 Time Integration

Let $\Delta t > 0$ fixed, and $t_0 = 0$. $t_n = t_0 + n\Delta t$, $n = 0, 1, \dots, N_T$. We have:

$$\underline{u}^n \approx \underline{u}(t_n) \quad \forall n = 0, 1, \dots, N_T$$

Now, we have two options:

13.8.1 Discretize (WPALG) in time with LeapFrog

Different time schemes can be used. Specifically, we will address the LeapFrog Scheme. It is an explicit method designed for second-order systems. It is conditionally stable, but the condition is not restrictive at all. Plus, it can be proven that it is the scheme of second order that keeps the disturbance and distortion error as low as possible.

First, we introduce:

$$\underline{v}(t) = \dot{\underline{u}}(t) \quad \wedge \quad \underline{G}(t, \underline{u}) = M_h^{-1}(\underline{F}(t) - c^2 A_h \underline{u}(t))$$

And approximate $\dot{\underline{u}}(t)$ and $\ddot{\underline{u}}(t)$ with those.

We update the displacements and velocities at each time snapshot t_{n+1} , starting from $(\underline{u}_0, \underline{v}_0)$. We can compute for $n = 0, 1, \dots, T$:

$$\begin{aligned} M_h \underline{u}^{n+1} &= \left[M_h - \frac{\Delta t^2}{2} c^2 A_h \right] \underline{u}^n + \Delta t M_h \underline{v}^n + \frac{\Delta t^2}{2} F_h(t_n) \\ M_h \underline{v}^{n+1} &= M_h \underline{v}^n - \frac{\Delta t}{2} c^2 A_h [\underline{u}^n + \underline{u}^{n+1}] + \frac{\Delta t}{2} [F_h(t_n) + F_h(t_{n+1})] \end{aligned}$$

The Courant-Friedrichs-Lewy (CFL) condition stands on Δt :

$$c \frac{\Delta t}{h} \lesssim 1$$

If the exact solution is smooth enough, and $u_h^n = \sum_{j=1}^N \underline{u}_j^n \varphi_j$ is the fully discrete approximation solution:

$$\|u(t_n) - u_h^n\|_\epsilon \lesssim h^p + (\Delta t)^2$$

Where the hidden constant depends on $p, T, u, \dot{u}, \ddot{u}, c$

Remark.

- This formulation (known as synchronized leapfrog) requires the time step to be constant.
- The asynchronous leapfrog updates the positions of the displacement and velocity at staggered time intervals, arranged so they “leapfrog” past one another

13.8.2 Rewrite (WPALG) as a first-order system

Defining:

$$\underline{v}(t) = \dot{\underline{u}}(t)$$

Then:

$$\text{Find } \underline{u} \in \mathbb{R}^{N_h} \quad st \quad \begin{cases} M_h \dot{\underline{u}} + c^2 A_h \underline{u} = F_h \\ \dot{\underline{u}}(t) - \underline{v}(t) = 0 \\ \underline{u}(0) = \underline{u}^0 \\ \dot{\underline{u}}(0) = \underline{v}^0 \end{cases} \quad \forall t \in (0, T]$$

Matricially:

$$\underbrace{\begin{bmatrix} M_h & 0 \\ 0 & I \end{bmatrix}}_{N_h} \underbrace{\begin{bmatrix} \dot{\underline{u}}(t) \\ \underline{u}(t) \end{bmatrix}}_{\underline{z}_h} + \underbrace{\begin{bmatrix} 0 & c^2 A_h \\ -I & 0 \end{bmatrix}}_{B_h} \underbrace{\begin{bmatrix} \underline{v}(t) \\ \underline{u}(t) \end{bmatrix}}_{\underline{z}_h} = \underbrace{\begin{bmatrix} F_h \\ 0 \end{bmatrix}}_{G_h}$$

So:

$$\begin{cases} N_h \dot{\underline{z}}_h + B_h \underline{z}_h = G_h \\ \underline{z}_h(0) = \underline{z}^0 \end{cases} \quad \forall t$$

$$\text{where: } \underline{z}^0 = \begin{bmatrix} \underline{v}^0 & \underline{u}^0 \end{bmatrix}^T$$

14 Wave Propagation with Elastodynamics Equation

This section analyzes the same problem presented in section 13.1, but with another formulation.

14.1 Fields of Applications

- Structural Engineering: To model vibrations on the beams that support structures
- Seismology: The Earth is mostly a solid body through which waves can propagate. Understanding how waves propagate is useful to predict future earthquakes, predict impact, diagnose explosions, and analyze the Earth's structure.

14.2 The Elastodynamics Equation

$$\rho \ddot{\underline{u}} - \nabla \cdot \sigma(\underline{u}) + G(\underline{u}) = \underline{f} \quad \text{in } \Omega \times (0, T]$$

where:

- $\underline{u}$ is the displacement
- ρ is the medium density
- λ and μ are the first and second Lamé elastic coefficients.
- The Hooke's law:

$$\sigma(\underline{u}) = 2\lambda \epsilon(\underline{u}) + \mu \text{trace}(\epsilon(\underline{u})) \iff \sigma(\underline{u}) = \mathbb{D} \epsilon(\underline{u})$$

With $\mathbb{D}$ elasticity tensor.

- $G(\underline{u}) = \begin{cases} 0 & \text{Fully elastic} \\ 2\rho\zeta\dot{\underline{u}} + \rho\zeta^2\underline{u} & \text{Viscoelastic} \end{cases}$
With ζ =damping coefficient (s^{-1})

14.3 Semidiscrete DG formulation

Let $V_h = \{v_h \in [L^2(\Omega)]^d \quad \text{st} \quad v_h|_K \in [\mathbb{P}^p(K)]^d \quad \forall K \in \mathcal{T}_h\}$:

$$\text{Find } u_h(t) \in V_h \quad \text{st} \quad \begin{cases} \int_{\Omega} \ddot{u}_h \cdot v_h + \mathcal{A}_h(u_h, v_h) + \int_{\Omega} \mathcal{G}(u_h) \cdot v_h = \mathcal{F}_h(v_h) \\ u_h(0) = u_h^0 \in V_h^p \\ \dot{u}_h(0) = v_h^0 \in V_h^p \end{cases} \quad \forall t \in (0, T]$$

where:

- $\mathcal{A}_h(u_h, v_h) = \int_{\Omega} \sigma(u_h) : \epsilon(v_h) - \int_{\mathcal{F}_h} \{\{\sigma(u_h)\}\} : [[v_h]] - \int_{\mathcal{F}_h} \{\{\sigma(v_h)\}\} : [[u_h]] + \int_{\mathcal{F}} \gamma[[u_h]][[v_h]]$

$$\gamma \approx \{\{\mathbb{D}\}\} \alpha \frac{p^2}{h}$$

- $\mathcal{F}_h(v_h) = \int_{\Omega} \underline{f} \cdot v_h$

Definition 14.1: $\|\cdot\|_{\mathcal{E}_h}$

$$\|v_h(t)\|_{\mathcal{E}_h}^2 := \|\rho^{1/2}\dot{v}_h(t)\|_{L^2(\Omega)}^2 + \|\rho^{1/2}\zeta v_h(t)\|_{L^2(\Omega)}^2 + \|\mathbb{D}^{1/2}\epsilon(v_h(t))\|_{L^2(\Omega)}^2 + \|\gamma^{1/2}[[v_h(t)]]\|_{L^2(\mathcal{F}_h)}^2$$

14.4 Stability and Convergence

Theorem 14.2

For any time $t \in (0, T]$, let $\underline{u}_h(t) \in V_h$ be the semidiscrete DG approximation (with a sufficiently large penalty parameter α), then:

$$\|\underline{u}_h(t)\|_{\mathcal{E}_h}^2 \lesssim \|\underline{u}_h(0)\|_{\mathcal{E}_h}^2 + \int_0^t \|\underline{u}(\tau)\|_{L^2(\Omega)}^2 d\tau \quad 0 < t \leq T$$

Theorem 14.3

For any time $t \in (0, T]$, assume that the exact solution $\underline{u}(t)$ is sufficiently regular. Let $\underline{u}_h(t) \in V_h$ be the semidiscrete DG approximation (with a sufficiently large penalty parameter α), then:

$$\|\underline{u}(t) - \underline{u}_h(t)\|_{\mathcal{E}_h} \leq C(T, \underline{u}, \dot{\underline{u}}, \ddot{\underline{u}}) \left(\frac{h}{p}\right)^p p^{1/2} \quad \forall t \in (0, T]$$

Where the constant depends on the material properties

14.5 Fully Discrete Scheme

The semidiscrete formulation in algebraic form is:

$$\begin{cases} M\ddot{\underline{u}}(t) + 2\rho\zeta M\dot{\underline{u}}(t) + (A + \rho\zeta^2 I)\underline{u}(t) = \underline{F}(t) \\ \underline{u}(0) = \underline{u}^0 \\ \dot{\underline{u}}(0) = \underline{v}^0 \end{cases} \quad \forall t \in (0, T]$$

By exercise, the leapfrog method can be stated.

15 Domain Decomposition Problems

15.1 Alternating Schwaz Method for Overlapping subdomains

Consider (an elliptic) partial differential equation of the form:

$$Lu = f \quad \text{in } \Omega = \Omega_1 \cup \Omega_2$$

With boundary condition $u = g$ on $\partial\Omega$.

Define Γ_1 and Γ_2 as the boundaries of one of the domains, inserted in the other one:

$$\Gamma_1 = \partial\Omega_1 \cap \Omega_2 \quad \wedge \quad \Gamma_2 = \partial\Omega_2 \cap \Omega_1$$

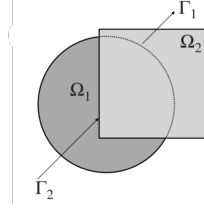


Figure 6: Compounded Boundary example: Circle and Square

Given $u^{(0)}$:

1. On Ω_1 , solve:

$$\begin{cases} Lu_1^{(k+\frac{1}{2})} = f & \text{in } \Omega_1 \\ u_1^{(k+\frac{1}{2})} = g & \text{in } \partial\Omega_1 \setminus \Gamma_1 \\ u_1^{(k+\frac{1}{2})} = u_2^{(k)} & \text{in } \Gamma_1 \end{cases}$$

2. On Ω_2 , solve:

$$\begin{cases} Lu_2^{(k+1)} = f & \text{in } \Omega_2 \\ u_2^{(k+1)} = g & \text{in } \partial\Omega_2 \setminus \Gamma_2 \\ u_2^{(k+1)} = u_1^{(k+\frac{1}{2})} & \text{in } \Gamma_2 \end{cases}$$

3. Define:

$$u^{(k+1)} \begin{cases} u_1^{(k+1)} & \text{in } \Omega \setminus \Omega_2 \\ u_2^{(k+1)} & \text{in } \Omega_2 \end{cases}$$

15.2 Discretized Schwarz Methods

- Discretization yields a $n \times n$ symmetric positive definite linear algebraic system of the form $A\underline{x} = \underline{b}$.
- For $i = 1, 2$, let S_i be set on n_i indices of grid points in the interior of Ω_i , where $n_i = |S_i|$.
- Because subdomains overlap, $S_1 \cap S_2 \neq \emptyset$, and $n_1 + n_2 > n$.
- For $i = 1, 2$, let R_i be $n_i \times n$ the boolean restriction matrix such that for any vector $\underline{v} \in \mathbb{R}^n$, $\underline{v}_i = R_i \underline{v} \in \mathbb{R}^{n_i}$ contains precisely those components of $\underline{v}$ corresponding to indices in S_i . This is also called the Restriction Operator.
- Let $R_i^T \in \mathbb{R}^{n \times n_i}$ be the extension matrix that expands the vector $\underline{v}_i \in \mathbb{R}^{n_i}$ into a vector $\underline{v} = R_i^T \underline{v}_i \in \mathbb{R}^n$. It keeps the components in Ω_i the same as $\underline{v}_i$, and the rest in 0. This is also called the Prolongation Operator.

- Note that for any i :

$$A_i v_i = R_i A (R_i^T v_i)$$

So the principal submatrices $A_i \in \mathbb{R}^{n_i \times n_i}$, $i = 1, 2$ of A correspond to two subdomains given by:

$$\begin{aligned} - A_1 &= R_1 A R_1^T \\ - A_2 &= R_2 A R_2^T \end{aligned}$$

15.3 Multiplicative Schwarz Method

For the discretized problem, the alternating Schwarz iteration is called Multiplicative Schwarz Method, and takes the following form:

$$\begin{aligned} \underline{x}^{(k+\frac{1}{2})} &= \underline{x}^{(k)} + R_1^T A_1^{-1} R_1 (\underline{b} - A \underline{x}^{(k)}) \\ \underline{x}^{(k+1)} &= \underline{x}^{(k+\frac{1}{2})} + R_2^T A_2^{-1} R_2 (\underline{b} - A \underline{x}^{(k+\frac{1}{2})}) \end{aligned}$$

The overall error $\underline{e}^{(k)} = \underline{x} - \underline{x}^{(k)}$ is updated as:

$$\underline{e}^{(k+1)} = B_{MS} \underline{e}^{(k)} \quad B_{MS} = (I - R_2^T A_2^{-1} R_2 A)(I - R_1^T A_1^{-1} R_1 A)$$

15.4 Additive Schwarz Method

Anyway, with the multiplicative method, we have no parallelism, since the problems must be solved sequentially. Instead, we can use the block Jacobi approach, also called Additive Schwarz Method:

$$\begin{aligned} \underline{x}^{(k+\frac{1}{2})} &= \underline{x}^{(k)} + R_1^T A_1^{-1} R_1 (\underline{b} - A \underline{x}^{(k)}) \\ \underline{x}^{(k+1)} &= \underline{x}^{(k)} + R_2^T A_2^{-1} R_2 (\underline{b} - A \underline{x}^{(k)}) \end{aligned}$$

The overall error $\underline{e}^{(k)} = \underline{x} - \underline{x}^{(k)}$ is updated as:

$$\underline{e}^{(k+1)} = B_{AS} \underline{e}^{(k)} \quad B_{AS} = (R_2^T A_2^{-1} R_2 A + R_1^T A_1^{-1} R_1) A$$

If we replace $\underline{x}^{(k+\frac{1}{2})}$ in $\underline{x}^{(k)}$ equation, we get:

$$x^{(k+1)} = x^{(k)} + P_{ad}^{-1} \underline{r}^{(k)}, \quad \underline{r}^{(k)} = (\underline{b} - A \underline{x}^{(k)})$$

where P_{ad}^{-1} is called the preconditioner:

$$P_{ad}^{-1} = R_1^T A_1^{-1} R_1 + R_2^T A_2^{-1} R_2$$

We can also use accelerated methods for convergence:

$$x^{(k+1)} = x^{(k)} + \alpha_k P_{ad}^{-1} \underline{r}^{(k)}$$

15.5 Many Overlapping Subdomains

To achieve a higher degree of parallelism with Schwarz method we can apply a two-domain algorithm recursively or using more subdomains.

In the case of p subdomains, the additive schwarz preconditioner will be:

$$P_{ad}^{-1} = \sum_{i=1}^p R_i^T A_i^{-1} R_i$$

This is highly parallel, but not algorithmically scalable, since the convergence rate degrades as p grows.

To deal with this degradation, a coarse correction can be done, adding a subproblem with a coarse grid solving the global problem, giving a global perspective of the problem:

$$P_{ad}^{-1} = \sum_{i=0}^p R_i^T A_i^{-1} R_i = R_0^T A_0^{-1} R_0 + \sum_{i=1}^p R_i^T A_i^{-1} R_i$$

15.6 Colouring techniques

The multiplicative Schwarz preconditioner is inherently serial. We can use a colouring technique to identify the sets of subdomains that can be processed concurrently.

This helps us to identify the limit of the degree of parallelism if there is a low number of subdomains per color.

In general, the multiplicative schwarz method converges faster than the additive one, but the last one have better parallel speedup.

15.7 Non-Overlapping Subdomains

We now consider subdomains that have a common boundary Γ , but there is no overlapping between them. Then, we can consider three mutually disjoint sets:

- S_1 interior nodes in Ω_1
- S_2 interior nodes in Ω_2
- S_Γ nodes in Γ

Now, we can obtain a symmetric block linear system:

$$\begin{bmatrix} A_{11} & 0 & A_{1\Gamma} \\ 0 & A_{22} & A_{2\Gamma} \\ A_{1\Gamma}^T & A_{2\Gamma}^T & A_{\Gamma\Gamma} \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_\Gamma \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_\Gamma \end{pmatrix}$$

We can do Block LU factorization:

$$\begin{bmatrix} A_{11} & 0 & A_{1\Gamma} \\ 0 & A_{22} & A_{2\Gamma} \\ A_{1\Gamma}^T & A_{2\Gamma}^T & A_{\Gamma\Gamma} \end{bmatrix} = \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ A_{1\Gamma}^T A_{11}^{-1} & A_{2\Gamma}^T A_{22}^{-1} & I \end{bmatrix} \begin{bmatrix} A_{11} & 0 & A_{1\Gamma} \\ 0 & A_{22} & A_{2\Gamma} \\ 0 & 0 & S \end{bmatrix}$$

Where S is called the Shur Complement:

$$S = A_{\Gamma\Gamma} - A_{1\Gamma}^T A_{11}^{-1} A_{1\Gamma} - A_{2\Gamma}^T A_{22}^{-1} A_{2\Gamma}$$

In particular this last equation is:

$$S \underline{x}_\Gamma = \tilde{b}_\Gamma$$

And we can recover b_Γ from:

$$\tilde{b}_\Gamma = b_\Gamma - A_{1\Gamma}^T A_{11}^{-1} b_1 - A_{2\Gamma}^T A_{22}^{-1} b_2$$

And the remaining unknowns can be computed in parallel with:

$$\begin{aligned} \underline{x}_1 &= A_{11}^{-1} (b_1 - A_{1\Gamma} \underline{x}_\Gamma) \\ \underline{x}_2 &= A_{22}^{-1} (b_2 - A_{2\Gamma} \underline{x}_\Gamma) \end{aligned}$$

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